

Chapter 10

Reset and Clock

10.1 Reset

10.1.1 Overview

ESP32-P4 provides four types of reset that occur at different levels, namely CPU Reset, Core Reset, System Reset, and Chip Reset. All reset types mentioned above (except Chip Reset) preserve the data stored in internal memory. Figure 10.1-1 shows the scopes of affected subsystems by each type of reset.

10.1.2 Architectural Overview

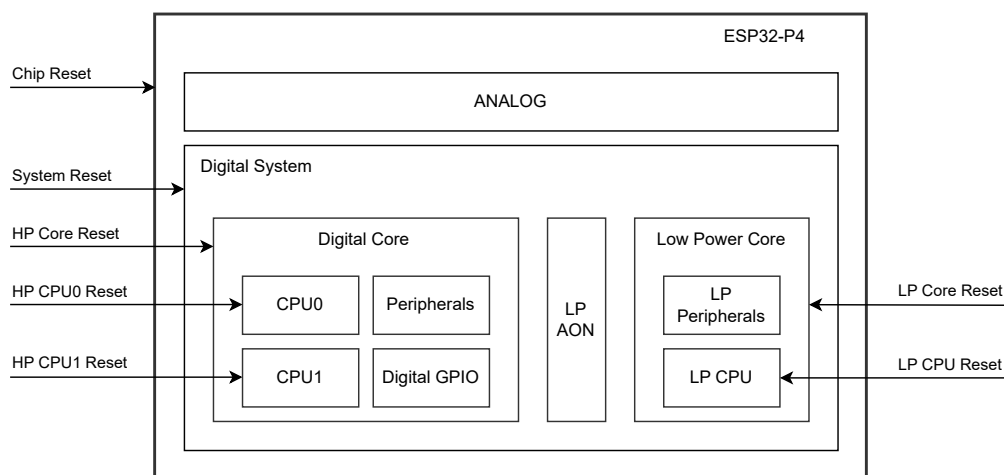


Figure 10.1-1. Reset Types

ESP32-P4's Digital System consists of [High Performance System \(HP system\)](#) that includes Digital Core and HP memory, and [Low Power System \(LP system\)](#) that includes the Low Power Always On (LP AON) system, LP Core, and LP memory. See Figure 10.1-1 for details (note that HP memory and LP memory would not be reset, so they are not shown in the figure).

10.1.3 Features

- Four reset types:
 - CPU Reset: resets CPU core. HP CPU0, HP CPU1, and LP CPU can be reset independently:
 - * HP CPU0 will be automatically released from reset after chip power-up, and start execution from CPU Reset Vector (0x30100000).

- * HP CPU1 is at reset by default after chip power-up, and needs to be manually released from reset by writing 0 to [HP_SYS_CLKRST_RST_EN_CORE1_GLOBAL](#). Once released from reset, HP CPU1 will start execution from CPU Reset Vector (0x30100000).
- * LP CPU is at reset after chip power-up, and needs to be manually released from reset by configuring the power management unit (PMU). Once released from reset, LP CPU will start execution from LP_SYSTEM_LP_CPU_BOOT_ADDR (0x50100000 by default). For details, see Chapter 14 [Low-Power Management](#).
- Core Reset: resets the whole digital system except for LP AON. HP core and LP core can be reset independently: HP Core Reset resets HP CPU0, HP CPU1, HP peripherals, HP GPIO, etc., and LP Core Reset resets LP CPU and LP peripherals.
- System Reset: resets the whole digital system, including the LP system.
- Chip Reset: resets the whole chip.
- Software reset and hardware reset:
 - Software Reset: triggered via software by configuring the corresponding registers of CPU, see Chapter 14 [Low-Power Management](#).
 - Hardware Reset: triggered directly by the hardware.

10.1.4 Functional Description

CPU will be reset immediately when any type of reset above occurs. After the reset is released, users can retrieve reset source codes by reading the following fields:

- HP CPU0: [LP_CLKRST_HPCORE0_RESET_CAUSE](#)
- HP CPU1: [LP_CLKRST_HPCORE1_RESET_CAUSE](#)
- LP CPU: [LP_CLKRST_LPCORE_RESET_CAUSE](#)

Table 10.1-1 and Table 10.1-2 list possible reset sources and the types of reset they trigger.

Table 10.1-1. HP CPU Reset Source

Code	Source	Reset Type	Note
0x01	Chip reset ¹	Chip Reset	—
0x03	Software system reset	HP Core Reset	Triggered by configuring LP_SYSTEM_SYS_SW_RST
0x05	PMU core reset	HP Core Reset	See Chapter 14 Low-Power Management
0x07	MWDT core reset ²	HP Core Reset	See Chapter 17 Watchdog Timers (WDT)
0x09	RWDT core reset	HP Core Reset	See Chapter 17 Watchdog Timers (WDT)
0x0B	MWDT CPU reset	HP CPU0/1 Reset	See Chapter 17 Watchdog Timers (WDT)
0x0C	Software CPU reset	HP CPU0/1 Reset	Triggered by configuring LP_CLKRST_HPCORE0_SW_RESET or LP_CLKRST_HPCORE1_SW_RESET
0x0D	RWDT CPU reset	HP CPU0/1 Reset	See Chapter 17 Watchdog Timers (WDT)

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Table 10.1-1 – cont'd from previous page

Code	Source	Reset Type	Note
0x0F	Brown-out system reset	Chip Reset or System Reset	Triggered by the brown-out detector. Once brown-out status is detected, the detector will trigger System Reset or Chip Reset, depending on the register configuration. See Chapter 14 Low-Power Management
0x10	RWDT system reset	System Reset	See Chapter 17 Watchdog Timers (WDT)
0x12	Super Watchdog reset	System Reset	See Chapter 17 Watchdog Timers (WDT)
0x13	Power glitch reset	System Reset	See Chapter 23 Brown-out Detector
0x1B	PSDET reset	System Reset	See Chapter 24 Power Supply Detector (PSDET)
0x14	eFuse reset	HP Core Reset	Triggered by eFuse CRC error
0x16	USB (JTAG) reset	HP Core Reset	Triggered when external USB host sends a specific command to the JTAG interface of USB Serial/JTAG Controller. See Chapter 51 USB Serial/JTAG Controller (USB_SERIAL_JTAG)
0x17	USB (UART) reset	HP Core Reset	Triggered when external USB host sends a specific command to the serial interface of USB Serial/JTAG Controller. See Chapter 51 USB Serial/JTAG Controller (USB_SERIAL_JTAG)
0x18	JTAG CPU reset	HP CPU0/1 Reset	Triggered when a “GDB Resetting CPU” instruction is received
0x1A	Lockup reset	HP CPU0/1 Reset	Triggered when the CPU enters lockup. HP CPU0 and HP CPU1 can be reset in lockup state by configuring LP_CLKRST_HPCORE_x_LOCKUP_RESET_EN

- Chip Reset can be triggered by the following sources:
 - Chip power-up
 - Brown-out detector
- The HP system contains two watchdog timers called Main System Watchdog Timers (MWDT), and the MWDT that can trigger reset can be configured via [HP_SYS_CLKRST_HPCORE0_WDT_RESET_SOURCE_SEL](#) and [HP_SYS_CLKRST_HPCORE0_WDT_RESET_SOURCE_SEL](#) for HP CPU0 and HP CPU1 respectively.

Table 10.1-2. LP CPU Reset Source

Code	Source	Reset Type	Note
0x01	Chip reset	Chip Reset	—
0x09	PMU LP Peripheral reset	LP Core Reset	See Chapter 14 Low-Power Management
0x0A	PMU LP CPU reset ¹	LP CPU Reset	See Chapter 14 Low-Power Management
0x0F	Brown-out system reset	Chip Reset or System Reset	Triggered by the brown-out detector
0x10	RWDT system reset	System Reset	See Chapter 17 Watchdog Timers (WDT)
0x12	Super Watchdog reset	System Reset	See Chapter 17 Watchdog Timers (WDT)

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Table 10.1-2 – cont'd from previous page

Code	Source	Reset Type	Note
0x13	Power glitch reset	System Reset	See Chapter 23 Brown-out Detector
0x14	Software LP CPU reset	LP CPU Reset	Triggered by configuring LPPERI_RST_EN_LP_CORE

¹ LP CPU is reset by PMU after chip power-up. To retrieve the correct reset source code (0x01) when the LP CPU runs for the first time, the source code for PMU LP CPU reset (0x0A) will be masked. After the CPU's first run, clear [LP_CLKRST_LPCORE_RESET_CAUSE_PMU_LP_CPU_MASK](#) to unmask the 0x0A source code.

10.1.5 Peripheral Reset

Peripherals can be reset individually by configuring corresponding registers, or globally by Core Reset, System Reset, or Chip Reset.

ESP32-P4 has three groups of peripheral reset registers, prefixed with HP_SYS_CLKRST, LP_CLKRST, and LPPERI. See Section [10.4 Register Summary](#) for detailed information.

10.2 Clock

10.2.1 Overview

ESP32-P4 clocks are mainly sourced from oscillator (OSC, including Resistor-Capacitor circuit), crystal (XTAL), and PLL circuit, and then processed by the dividers or selectors, which allows most functional modules to select their working clock according to their power consumption and performance requirements. Figure [10.2-1](#) shows the HP and LP system clock structure, including the main clock path and typical peripheral clock generator circuit. ESP32-P4 has a large number of peripherals, and limited by space, not all peripheral clocks are shown in the figure.

10.2.2 Architectural Overview

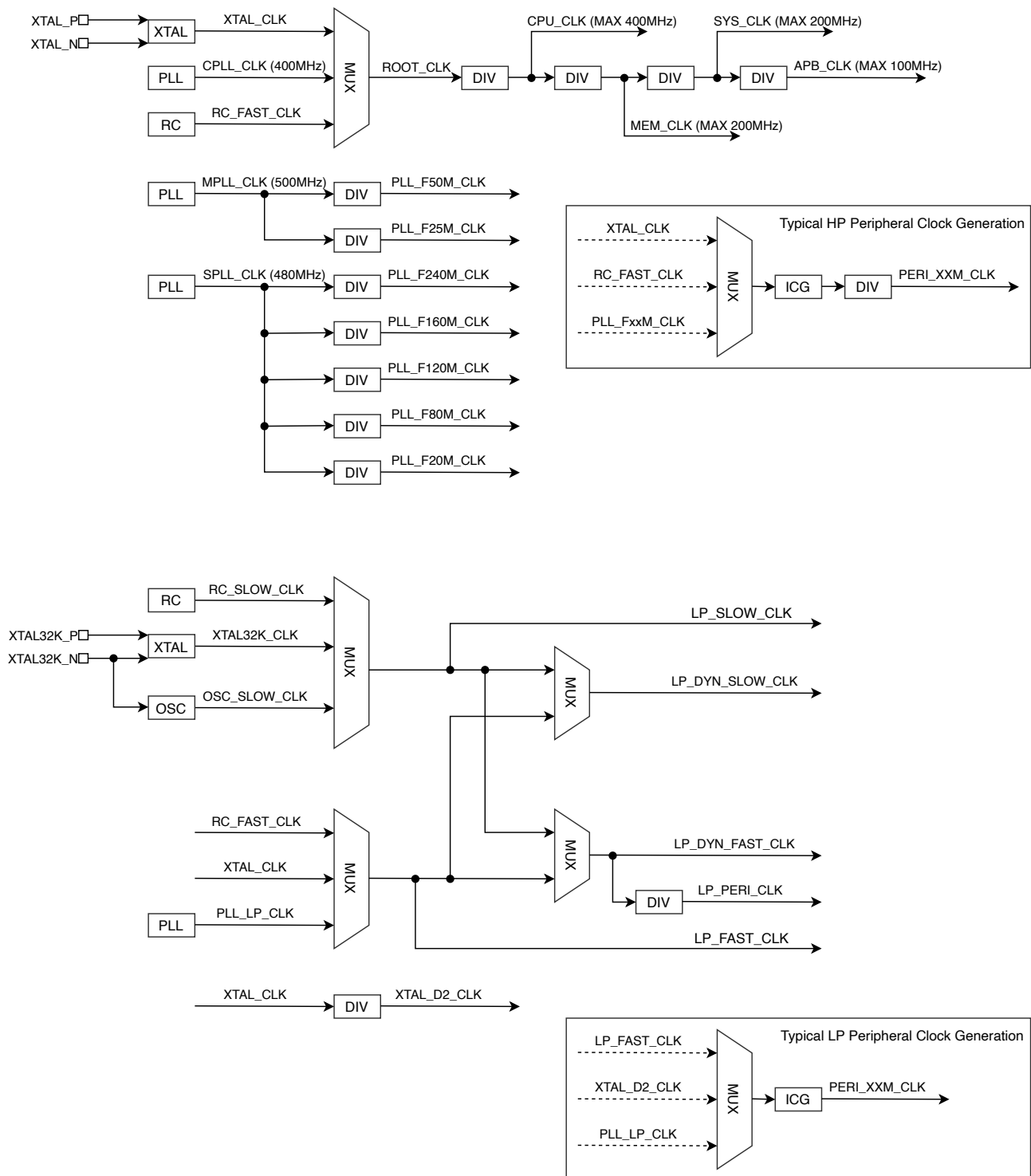


Figure 10.2-1. HP and LP System Clock

10.2.3 Features

ESP32-P4 clocks can be classified into two types depending on their frequencies:

- High speed clocks for devices working at a higher frequency, such as HP CPU0/1 and digital peripherals

- CPLL_CLK: internal 360 MHz PLL clock. Its reference clock is XTAL_CLK
- MPLL_CLK: internal 500 MHz PLL clock. Its reference clock is XTAL_CLK
- SPLL_CLK: internal 480 MHz PLL clock. Its reference clock is XTAL_CLK
- Slow speed clocks for LP system and some peripherals working in low-power mode
 - XTAL32K_CLK: external 32 kHz crystal clock
 - RC_SLOW_CLK: internal 150 kHz slow RC oscillator
 - OSC_SLOW_CLK: external slow clock input through [XTAL_32K_N](#), with a frequency of 32 kHz by default. After configuring this GPIO, also configure the Hold function (see Chapter 9 [GPIO Matrix and IO MUX > 9.9 Pin Hold Feature](#))
 - XTAL_CLK: 40 MHz external crystal clock
 - RC_FAST_CLK: internal fast RC oscillator with adjustable frequency (20 MHz by default)
 - PLL_LP_CLK: internal PLL clock with a frequency of 8 MHz by default. Its reference clock can be XTAL32K_CLK

10.2.4 Functional Description

10.2.4.1 HP System Clock

As Figure 10.2-1 shows, XTAL_CLK, CPLL_CLK, and RC_FAST_CLK are the multiplexer inputs to generate ROOT_CLK. For selecting which input, see Table 10.2-1.

Table 10.2-1. ROOT_CLK Clock Source

LP_CLKRST_HP_ROOT_CLK_SRC_SEL	Clock Source
0	XTAL_CLK
1	CPLL_CLK
2	RC_FAST_CLK

ROOT_CLK is then divided by a series of dividers, and sequentially derives:

- CPU_CLK, which drives HP CPUs and their logic
- MEM_CLK, which drives the internal memories (L2 Cache, L2MEM, ROM) and their logic
- SYS_CLK, which is the HP high-speed bus clock that drives the AXI and AHB bus logic
- APB_CLK, which is the HP low-speed bus clock that drives the APB bus logic

The above four derived clocks are synchronous SoC clocks, asynchronous to peripheral clocks described in Section 10.2.4.3 [Peripheral Clocks](#). Peripheral clocks are asynchronous with each other unless otherwise stated.

The maximum frequencies allowed for these SoC clocks are as follows. Software should ensure that the frequencies of these clocks do not exceed the maximum frequency.

- CPU_CLK: 360 MHz
- MEM_CLK: 200 MHz

- SYS_CLK: 200 MHz
- APB_CLK: 100 MHz

Each module has an individual clock gate for the SoC clock it uses. For example, users can enable the clock for HP CPU0 and disable the clock for HP CPU1 by setting the [HP_SYS_CLKRST_CORE0_CPU_CLK_EN](#) bit and clearing the [HP_SYS_CLKRST_CORE1_CPU_CLK_EN](#) bit. “CPU_CLK” in bit names indicates that the corresponding clock belongs to the CPU_CLK group shown in Figure 10.2-1.

The internal PLL clocks of ESP32-P4, namely MPLL_CLK and SPLL_CLK, generate reference clocks for peripherals through a set of parallel clock dividers.

Besides these two PLL clocks, there are two additional PLL clocks dedicated to specific modules:

- SDIO_PLL, which generates three clocks:
 - SDIO_PLLO_CLK, used for SDIO’s high-speed SDIO_SLF_CLK (internal signal clock) and the source of SDIO’s low-speed clock
 - SDIO_PLL1_CLK, users for SDIO’s high-speed SDIO_DRV_CLK (output signal driving clock)
 - SDIO_PLL2_CLK, used for SDIO’s high-speed SDIO_SAM_CLK (input signal sampling clock)
- APLL_CLK, which is dedicated to I2S

A typical peripheral clock generation circuit consists of a clock selector, a clock gating unit, and a divider. In high-performance mode, the peripheral can select a high-speed clock source without division for higher processing speed. In low-power mode, the peripheral can select a low-speed clock source or divide it by a large divisor to lower power consumption. When the peripheral is disabled, its clock can be directly disabled.

Note:

Updates to [HP_SYS_CLKRST_ROOT_CLK_CTRL0/1/2/3_REG](#) will take effect only after [HP_SYS_CLKRST_SOC_CLK_DIV_UPDATE](#) is set.

10.2.4.2 LP System Clock

The LP system can operate when most other clocks are disabled. LP system clocks include LP_SLOW_CLK, LP_FAST_CLK, LP_DYN_SLOW_CLK, LP_DYN_FAST_CLK, and XTAL_D2_CLK.

The clock sources for LP_SLOW_CLK and LP_FAST_CLK are low-frequency clocks:

- LP_SLOW_CLK has four clock sources, see Table 10.2-2:

Table 10.2-2. LP_SLOW_CLK Clock Source Selection

LP_CLKRST_SLOW_CLK_SEL	Clock Source
0	RC_SLOW_CLK
1	XTAL32K_CLK
2	Invalid
3	OSC_SLOW_CLK

- LP_FAST_CLK has three clock sources, see Table 10.2-3:

Table 10.2-3. LP_FAST_CLK Clock Source Selection

LP_CLKRST_FAST_CLK_SEL	Clock Source
0	RC_FAST_CLK
1	XTAL_CLK
2	PLL_LP_CLK

LP_DYN_SLOW_CLK and LP_DYN_FAST_CLK can be derived from LP_SLOW_CLK or LP_FAST_CLK depending on the chip's power mode (see Chapter [14 Low-Power Management](#)), and should be able to operate in all power modes to drive LP AON modules. These two clocks are always synchronous with each other.

- In Active mode, select LP_FAST_CLK as the clock source. In this case, LP_DYN_SLOW_CLK has the same frequency with LP_SLOW_CLK, and the same phase with LP_FAST_CLK;
- In Deep-sleep mode, select the clock source according to PMU configurations. If the LP CPU is off, LP_SLOW_CLK is the clock source, and has the same frequency and phase as LP_DYN_FAST_CLK in this case; if the LP CPU is on, LP_FAST_CLK is the clock source.

LP_PERI_CLK is derived from LP_DYN_FAST_CLK and drives all the buses for LP peripherals.

XTAL_D2_CLK is derived from XTAL_CLK divided by 2.

10.2.4.3 Peripheral Clocks

Table [10.2-4](#), Table [10.2-5](#), Table [10.2-6](#), and Table [10.2-7](#) list the derived HP/LP clock sources and HP clocks/LP clocks for each peripheral.

Table 10.2-4. Derived HP Clock sources

Derived Clock	Source Clock							Derived Clock						
	RC_FAST_CLK	XTAL_CLK	CPLL_CLK	MPLL_CLK	SPLL_CLK	APLL_CLK	SDIO_PLL0_CLK	ROOT_CLK	CPU_CLK	MEM_CLK	SYS_CLK	PLL_F80M_CLK	PLL_F160M_CLK	PLL_F240M_CLK
ROOT_CLK	Y	Y	Y						Y					
CPU_CLK										Y				
MEM_CLK											Y			
SYS_CLK												Y		
APB_CLK													Y	
PLL_F50M_CLK					Y									
PLL_F25M_CLK					Y									
PLL_F240M_CLK						Y								
PLL_F160M_CLK						Y								
PLL_F120M_CLK						Y								
PLL_F80M_CLK						Y								
PLL_F20M_CLK						Y								
PLL_F12M_CLK						Y								
PLL_F48M_CLK						Y								
FLASH_CLK			Y	Y		Y								
PSRAM_CLK			Y	Y	Y	Y	Y							
GPSPi_CLK	Y	Y				Y	Y	Y						
ADC_CLK	Y	Y										Y		
UART_CLK	Y	Y										Y		
CRYPTO_CLK	Y	Y											Y	Y

Table 10.2-5. HP Peripheral Clocks

Peripheral	Clock	Clock Source															Derived Clock						Clock from IO
		RC_FAST_CLK	XTAL_CLK	APLL_CLK	CPLL_CLK	SPLL_CLK	MPLL_CLK	SDIO_PLL0_CLK	PLL_F12M_CLK	PLL_F20M_CLK	PLL_F25M_CLK	PLL_F48M_CLK	PLL_F80M_CLK	PLL_F120M_CLK	PLL_F160M_CLK	PLL_F240M_CLK	FLASH_CLK	PSRAM_CLK	GPSPi_CLK	ADC_CLK	UART_CLK	CRYPTO_CLK	
SPI0/1 (flash)	FLASH_PLL_CLK																Y						
	FLASH_CORE_CLK																Y						
SPI0/1 (PSRAM)	PSRAM_PLL_CLK																	Y					
	PSRAM_CORE_CLK																	Y					
I2S	I2S_RX_CLK		Y	Y											Y								PAD_I2S_MCLK
	I2S_TX_CLK		Y	Y											Y								PAD_I2S_MCLK
I2C	I2C_CLK	Y	Y																				
MCPWM	MCPWM_CLK	Y	Y												Y								
TIMG	TIMERGRP_TO_CLK	Y	Y										Y										
	TIMERGRP_T1_CLK	Y	Y										Y										
MWDT	TIMERGRP_WDT_CLK	Y	Y										Y										
TWAI	TWAI_CLK	Y	Y																				
GP-SPI2/GP-SPI3	GPSPi_HS_CLK																		Y				
	GPSPi_MST_CLK																		Y				
EMAC	EMAC_RMII_CLK																						PAD_EMAC_TXRX_CLK PAD_EMAC_RX_CLK PAD_EMAC_TX_CLK
	EMAC_RX_CLK																						PAD_EMAC_TXRX_CLK PAD_EMAC_RX_CLK
	EMAC_TX_CLK																						PAD_EMAC_TXRX_CLK PAD_EMAC_TXRX_CLK PAD_EMAC_TX_CLK
	EMAC_PTP_REF_CLK		Y										Y										
CAM OUT	CAM_CLK		Y	Y											Y								
IO MUX	IOMUX_CLK		Y										Y										
LCD	LCD_CLK		Y	Y											Y								

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Table 10.2-5 – cont'd from previous page

Peripheral	Clock	Clock Source															Derived Clock						Clock from IO
		RC_FAST_CLK	XTAL_CLK	APLL_CLK	CPLL_CLK	SPLL_CLK	MPLL_CLK	SDIO_PLL0_CLK	PLL_F12M_CLK	PLL_F20M_CLK	PLL_F25M_CLK	PLL_F48M_CLK	PLL_F80M_CLK	PLL_F120M_CLK	PLL_F160M_CLK	PLL_F240M_CLK	FLASH_CLK	PSRAM_CLK	GPSPi_CLK	ADC_CLK	UART_CLK	CRYPTO_CLK	
LEDC	LEDC_CLK	Y	Y									Y											
ADC	HPADC_CLK																			Y			
	HPADC1_SARCLK																			Y			
	HPADC2_SARCLK																			Y			
AES	AES_CLK																					Y	
SHA	SHA_CLK																						
RSA	RSA_CLK																						
ECC	ECC_CLK																						
RSA_DS	DSA_CLK																						
HMAC	HMAC_CLK																						
Key Manager	KM_CLK																						
ECDSA_DS	ECDSA_DS_CLK																						
	SEC_CLK																						
RMT	RMT_CLK	Y	Y									Y											
System Timer	SYSTIMER_CLK	Y	Y																				
UART	UART_FCLK																				Y		
	UART_SCLK																				Y		
	UART_SLP_CLK																					PAD_UART_SLP_CLK	
PARLIO	PARLIO_RX_CLK	Y	Y												Y							PAD_PARLIO_RX_CLK	
	PARLIO_TX_CLK	Y	Y												Y							PAD_PARLIO_TX_CLK	
I3S	I3C_MST_CLK		Y								Y			Y	Y								
ISP	ISP_CLK		Y												Y	Y							
MIPI DSI	DSI_DPHY_CFG_CLK	Y								Y	Y												
	DSI_DPHY_PLL_REFCLK		Y	Y	Y	Y	Y																
	DSI_DPICKL		Y	Y											Y	Y							
MIPI CSI	CSI_DPHY_CFG_CLK	Y								Y	Y												
H264	H264_CLK		Y													Y							
USB_SERIAL_JTAG	USB_SERIAL_JTAG_CLK										Y												
USB 2.0 Full-Speed OTG	USB_OTG11_CLK										Y												
USB 2.0 High-Speed OTG	USB_HS_PHY_REFCLK								Y	Y												PAD_USB_HSPHY_REFCLK	
SD/MMC Host Controller (SDHOST)	SDIO_DRV_CLK (low-speed)							Y							Y								
	SDIO_SLF_CLK (low-speed)																						
	SDIO_SAM_CLK (low-speed)																						

Table 10.2-6. Derived LP Clock Sources

Derived clock	Source clock						Derived clock					Clock from IO
	RC_SLOW_CLK 150 kHz	XTAL32K_CLK 32 kHz	OSC_SLOW_CLK 32 kHz	RC_FAST_CLK 20 MHz	XTAL_CLK 40 MHz	PLL_LP_CLK 8 MHz	LP_SLOW_CLK	LP_FAST_CLK	LP_DYN_SLOW_CLK	LP_DYN_FAST_CLK	XTAL_D2_CLK	
LP_FAST_CLK				Y	Y	Y						
LP_SLOW_CLK	Y	Y	Y									
LP_DYN_FAST_CLK							Y	Y				
LP_DYN_SLOW_CLK							Y	Y				
LP_PERI_CLK										Y		
XTAL_D2_CLK					Y							

Table 10.2-7. LP Peripheral Clocks

Derived clock	Source clock						Derived clock						Clock from IO
	RC_SLOW_CLK 150 kHz	XTAL32K_CLK 32 kHz	OSC_SLOW_CLK 32 kHz	RC_FAST_CLK 20 MHz	XTAL_CLK 40 MHz	PLL_LP_CLK 8 MHz	LP_SLOW_CLK	LP_FAST_CLK	LP_DYN_SLOW_CLK	LP_DYN_FAST_CLK	LP_PERI_CLK	XTAL_D2_CLK	
eFuse Controller (EFUSE)												Y	
RTC Watchdog Timer (RWDT)	Y								Y	Y			
RTC Timer									Y	Y			
Brown-out Detector										Y			
Power Management Unit (PMU)	Y							Y	Y	Y			
LP UART						Y				Y		Y	PAD_LP_UART_SLP_CLK
LP SPI													PAD_LP_SPI_CLK
LP I2C						Y				Y		Y	
Analog I2C Controller								Y					
LP I2S Controller						Y				Y		Y	
LP ADC Controller											Y		

Note:

To protect the encryption and decryption peripherals from DPA (Differential Power Analysis) attacks, a random divider strategy is implemented for its function clock. Four security levels are available, depending on the range of the random divider. Users can select the security level by configuring [HP_SYS_CLKRST_SEC_DPA_CFG_SEL](#).

If [HP_SYS_CLKRST_SEC_DPA_CFG_SEL](#) is set to 1, the security level is determined by the configuration of [HP_SYS_CLKRST_SEC_DPA_LEVEL](#). Otherwise, the security level is determined by the value of [EFUSE_SEC_DPA_LEVEL](#).

10.3 Programming Procedures

10.3.1 HP System Clock Configuration

- Configure the clock divisor for CPU_CLK: [HP_SYS_CLKRST_CPU_CLK_DIV_NUM](#) for the integral part, [HP_SYS_CLKRST_CPU_CLK_DIV_NUMERATOR](#) for the numerator of the fractional part, and [HP_SYS_CLKRST_CPU_CLK_DIV_DENOMINATOR](#) for the denominator of the fractional part. Write 1 to [HP_SYS_CLKRST_SOC_CLK_DIV_UPDATE](#) to apply the frequency divisor configurations for CPU_CLK.
- Configure the clock divisor for MEM_CLK: [HP_SYS_CLKRST_MEM_CLK_DIV_NUM](#) for the integral part, [HP_SYS_CLKRST_MEM_CLK_DIV_NUMERATOR](#) for the numerator of the fractional part, and [HP_SYS_CLKRST_MEM_CLK_DIV_DENOMINATOR](#) for the denominator of the fractional part. Write 1 to [HP_SYS_CLKRST_SOC_CLK_DIV_UPDATE](#) to apply the frequency divisor configurations for MEM_CLK.
- Configure the clock divisor for SYS_CLK: [HP_SYS_CLKRST_SYS_CLK_DIV_NUM](#) for the integral part, [HP_SYS_CLKRST_SYS_CLK_DIV_NUMERATOR](#) for the numerator of the fractional part, and [HP_SYS_CLKRST_SYS_CLK_DIV_DENOMINATOR](#) for the denominator of the fractional part. Write 1 to [HP_SYS_CLKRST_SOC_CLK_DIV_UPDATE](#) to apply the frequency divisor configurations for SYS_CLK.
- Configure the clock divisor for APB_CLK: [HP_SYS_CLKRST_APB_CLK_DIV_NUM](#) for the integral part, [HP_SYS_CLKRST_APB_CLK_DIV_NUMERATOR](#) for the numerator of the fractional part, and [HP_SYS_CLKRST_APB_CLK_DIV_DENOMINATOR](#) for the denominator of the fractional part. Write 1 to [HP_SYS_CLKRST_SOC_CLK_DIV_UPDATE](#) to apply the frequency divisor configurations for APB_CLK.
- Configure the clock source for ROOT_CLK via [LP_CLKRST_HP_ROOT_CLK_SRC_SEL](#).

Notice:

1. It is recommended to first configure the frequency divisors for the SoC clocks and apply the configurations, and then configure the clock source for ROOT_CLK.
2. Before changing the clock source for ROOT_CLK, software needs to ensure that the configured clock divisors do not result in clock frequencies exceeding the allowed maximum frequencies.
3. Writing 1 to [HP_SYS_CLKRST_SOC_CLK_DIV_UPDATE](#) will apply the frequency divisor configurations for CPU_CLK, MEM_CLK, SYS_CLK, and APB_CLK at one go, but it is not recommended to do so. It would be better to apply the configuration for each clock individually.
4. When switching from a high divisor to a low divisor, the recommended configuration order is APB_CLK

→ SYS_CLK → MEM_CLK → CPU_CLK. When switching from a low divisor to a high divisor, the recommended configuration order is CPU_CLK → MEM_CLK → SYS_CLK → APB_CLK. This helps to prevent high-frequency clock glitches.

10.3.2 LP System Clock Configuration

- Configure the clock source of LP_SLOW_CLK via [LP_CLKRST_SLOW_CLK_SEL](#)
- Configure the clock source of LP_FAST_CLK via [LP_CLKRST_FAST_CLK_SEL](#)
- Configure the clock divisor of LP_PERI_CLK via [LP_CLKRST_LP_PERI_DIV_NUM](#)

10.3.3 Peripheral Clock Reset and Configuration

Notice:

ESP32-P4 features low power consumption. This is why some peripheral clocks are gated (disabled) by default. Before using any of these peripherals, it is mandatory to enable the clock for the given peripheral by setting the corresponding CLK_EN bit to 1, and release the peripheral from the reset state to make it operational by setting the RST_EN bit to 0.

The clocks of most peripherals can be classified into two types:

- Bus clock: used to configure peripheral registers.
- Function clock: such as UART's reference clock, used by peripherals to operate.

The function clock of most peripherals can be selected from multiple clock sources. In the description of the gating registers, it will be stated whether a register is a bus clock (SYS_CLK, APB_CLK) gating register or a function clock gating register.

Bus clock enable/disable registers, function clock enable/disable registers, clock source selection registers, and clock frequency division registers are grouped into the HP_SYS_CLKRST, LP_CLKRST, and LPPERI modules. Except for EMA, SDMMAC, USB Serial/JTAG, Full-Speed USB 2.0 OTG, and High-Speed USB 2.0 OTG, whose reset configuration registers are in the LP_CLKRST module, all the other HP clock and reset registers are in the HP_SYS_CLKRST module. For more information, see Section [10.4 Register Summary](#).

When changing the clock configuration, it is recommended to turn off the corresponding clock gate first, update the clock divider configurations, clock sources, and other configurations, and then turn it on again.

When a peripheral is not working, users can turn off its function clock by configuring related registers. Turning off the peripheral's function clock does not affect the rest of the system.

Take I2C clock configuration as an example.

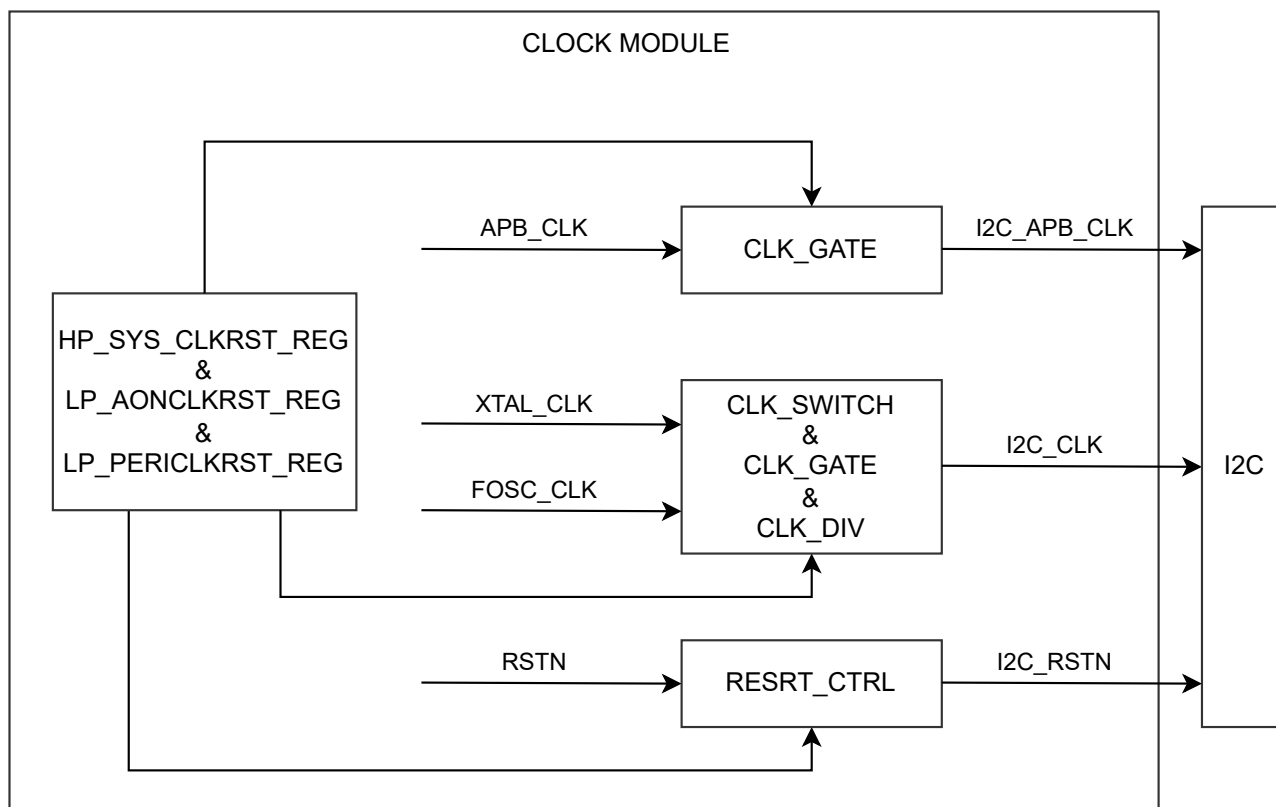


Figure 10.3-1. Clock Configuration Example

Figure 10.3-1 shows the clock structure of I2C. The clock structure of other peripherals is similar to this one. CLK_SWITCH is used to select a clock output and CLK_GATE to turn on/off the clock.

In scenarios that require low power consumption, when the peripheral is not in use, in addition to turning off the function clock, the bus clock of the peripheral can also be turned off to further lower power consumption.

Note that if you turn off the bus clock first, the functional clock may continue working. Therefore, when turning off clocks, it is recommended to turn off the functional clock first and then the bus clock; when turning on clocks, it is recommended to turn on the bus clock first and then the functional clock.

Note:

In this chapter, all divisor configuration registers are configured with the actual divisor minus 1.

10.4 Register Summary

10.4.1 Reset and Clock (HP_SYS_CLKRST) Register Summary

The addresses in this section are relative to HP_SYS_CLKRST base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

Name	Description	Address	Access
Clock gate enable register			
HP_SYS_CLKRST_CLK_ENO_REG	Register clock gate enable register	0x0000	R/W
HP_SYS_CLKRST_CLK_FORCE_ON_CTRL0_REG	System and peripheral clock gate configuration register	0x00B4	R/W
Root clock configuration register			
HP_SYS_CLKRST_ROOT_CLK_CTRL0_REG	Root clock configuration register 0	0x0004	varies
HP_SYS_CLKRST_ROOT_CLK_CTRL1_REG	Root clock configuration register 1	0x0008	R/W
HP_SYS_CLKRST_ROOT_CLK_CTRL2_REG	Root clock configuration register 2	0x000C	R/W
HP_SYS_CLKRST_ROOT_CLK_CTRL3_REG	Root clock configuration register 3	0x0010	R/W
Bus clock configuration register			
HP_SYS_CLKRST_SOC_CLK_CTRL0_REG	Bus clock configuration register 0	0x0014	R/W
HP_SYS_CLKRST_SOC_CLK_CTRL1_REG	Bus clock configuration register 1	0x0018	R/W
HP_SYS_CLKRST_SOC_CLK_CTRL2_REG	Bus clock configuration register 2	0x001C	R/W
HP_SYS_CLKRST_SOC_CLK_CTRL3_REG	Bus clock configuration register 3	0x0020	R/W
Reference clock configuration register			
HP_SYS_CLKRST_REF_CLK_CTRL0_REG	Reference clock configuration register 0	0x0024	R/W
HP_SYS_CLKRST_REF_CLK_CTRL1_REG	Reference clock configuration register 1	0x0028	R/W
HP_SYS_CLKRST_REF_CLK_CTRL2_REG	Reference clock configuration register 2	0x002C	R/W
Peripheral clock configuration register			
HP_SYS_CLKRST_PERI_CLK_CTRL00_REG	Peripheral clock configuration register	0x0030	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL01_REG	Peripheral clock configuration register	0x0034	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL02_REG	Peripheral clock configuration register	0x0038	varies
HP_SYS_CLKRST_PERI_CLK_CTRL03_REG	Peripheral clock configuration register	0x003C	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL10_REG	Peripheral clock configuration register	0x0040	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL11_REG	Peripheral clock configuration register	0x0044	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL12_REG	Peripheral clock configuration register	0x0048	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL13_REG	Peripheral clock configuration register	0x004C	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL14_REG	Peripheral clock configuration register	0x0050	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL15_REG	Peripheral clock configuration register	0x0054	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL16_REG	Peripheral clock configuration register	0x0058	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL17_REG	Peripheral clock configuration register	0x005C	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL18_REG	Peripheral clock configuration register	0x0060	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL19_REG	Peripheral clock configuration register	0x0064	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL110_REG	Peripheral clock configuration register	0x0068	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL111_REG	Peripheral clock configuration register	0x006C	R/W

Name	Description	Address	Access
HP_SYS_CLKRST_PERI_CLK_CTRL112_REG	Peripheral clock configuration register	0x0070	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL113_REG	Peripheral clock configuration register	0x0074	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL114_REG	Peripheral clock configuration register	0x0078	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL115_REG	Peripheral clock configuration register	0x007C	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL116_REG	Peripheral clock configuration register	0x0080	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL117_REG	Peripheral clock configuration register	0x0084	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL118_REG	Peripheral clock configuration register	0x0088	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL119_REG	Peripheral clock configuration register	0x008C	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL120_REG	Peripheral clock configuration register	0x0090	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL20_REG	Peripheral clock configuration register	0x0094	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL21_REG	Peripheral clock configuration register	0x0098	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL22_REG	Peripheral clock configuration register	0x009C	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL23_REG	Peripheral clock configuration register	0x00A0	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL24_REG	Peripheral clock configuration register	0x00A4	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL25_REG	Peripheral clock configuration register	0x00A8	R/W
HP_SYS_CLKRST_PERI_CLK_CTRL26_REG	Peripheral clock configuration register	0x00AC	R/W
DPA clock configuration register			
HP_SYS_CLKRST_DPA_CTRL0_REG	DPA clock configuration register	0x00B8	R/W
Analog PLL calibration configuration register			
HP_SYS_CLKRST_ANA_PLL_CTRL0_REG	Analog PLL calibration configuration register	0x00BC	varies
Peripheral reset register			
HP_SYS_CLKRST_HP_RST_EN0_REG	Peripheral reset register 0	0x00C0	R/W
HP_SYS_CLKRST_HP_RST_EN1_REG	Peripheral reset register 1	0x00C4	R/W
HP_SYS_CLKRST_HP_RST_EN2_REG	Peripheral reset register 2	0x00C8	R/W
HP_SYS_CLKRST_HP_FORCE_NORST0_REG	Peripheral reset disable register register 0	0x00CC	R/W
HP_SYS_CLKRST_HP_FORCE_NORST1_REG	Peripheral reset disable register register 1	0x00D0	R/W
HP CPU WDT reset configuration register			
HP_SYS_CLKRST_HPWDT_CORE0_RST_CTRL0_REG	HP CPU0 WDT reset configuration register	0x00D4	R/W
HP_SYS_CLKRST_HPWDT_CORE1_RST_CTRL0_REG	HP CPU1 WDT reset configuration register	0x00D8	R/W
HP_SYS_CLKRST_HPCORE_WDT_RESET_SOURCE0_REG	HP CPU WDT reset source configuration register	0x00EC	R/W
CPU clock status register			
HP_SYS_CLKRST_CPU_SRC_FREQ0_REG	CPU source clock frequency register	0x00DC	RO
HP_SYS_CLKRST_CPU_CLK_STATUS0_REG	CPU clock status register	0x00E0	RO
WAITI Signal Control Register			
HP_SYS_CLKRST_CPU_WAITI_CTRL0_REG	WAITI signal control register	0x00F4	R/W

10.4.2 LP Always on Clock and Reset (LP_CLKRST) Register Summary

The addresses in this section are relative to LP_CLKRST base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Configuration registers			
LP_CLKRST_LP_CLK_CONF_REG	LP root clock configuration register	0x0000	R/W
LP_CLKRST_LP_CLK_EN_REG	LP always on peripheral clock configuration register	0x0008	R/W
LP_CLKRST_LP_RST_EN_REG	LP always on peripheral reset configuration register	0x000C	R/W
LP_CLKRST_RESET_CAUSE_REG	HP CPU and LP CPU reset cause register	0x0010	varies
LP_CLKRST_HPCPU_RESET_CTRL0_REG	HP CPU reset configuration register 0	0x0014	varies
LP_CLKRST_HPCPU_RESET_CTRL1_REG	HP CPU reset configuration register 1	0x0018	R/W
LP_CLKRST_FOSC_CNTL_REG	RC_FAST_CLK frequency configuration register	0x001C	R/W
LP_CLKRST_XTAL32K_REG	XTAL32K_CLK configuration register	0x0030	R/W
LP_CLKRST_HP_CLK_CTRL_REG	HP clock control register	0x0040	R/W
LP_CLKRST_HP_USB_CLKRST_CTRL0_REG	USB clock and reset control register 0	0x0044	R/W
LP_CLKRST_HP_USB_CLKRST_CTRL1_REG	USB clock and reset control register 1	0x0048	R/W
LP_CLKRST_HP_SDMMC_EMAC_RST_CTRL_REG	SDMMC and EMAC reset control register	0x004C	R/W
Version register			
LP_CLKRST_DATE_REG	Version control register	0x03FC	R/W

10.4.3 LP Peripheral Clock and Reset (LPPERI) Register Summary

The addresses in this section are relative to LPPERI base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Configuration registers			
LPPERI_CLK_EN_REG	LP peripheral bus clock configuration register	0x0000	R/W
LPPERI_CORE_CLK_SEL_REG	LP peripheral function clock configuration register	0x0004	R/W
LPPERI_RESET_EN_REG	LP peripheral reset configuration register	0x0008	varies
LPPERI_CPU_REG	LP CPU access configuration register	0x000C	R/W
LPPERI_MEM_CTRL_REG	LP UART miscellaneous control register	0x0028	varies
LPPERI_ADC_CTRL_REG	LP ADC clock configuration register	0x002C	R/W
LPPERI_LP_I2S_RXCLK_DIV_NUM_REG	LP I2S RX clock configuration register	0x0030	R/W
LPPERI_LP_I2S_RXCLK_DIV_XYZ_REG	LP I2S RX clock configuration register	0x0034	R/W
Version register			

Name	Description	Address	Access
LPPERI_DATE_REG	Version control register	0x03FC	R/W

10.5 Registers

10.5.1 Reset and Clock (HP_SYS_CLKRST) Registers

The addresses in this section are relative to HP_SYS_CLKRST base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 10.1. HP_SYS_CLKRST_CLK_EN0_REG (0x0000)

(reserved)																HP_SYS_CLKRST_CLK_EN															
31																														1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	Reset

HP_SYS_CLKRST_CLK_EN Configures resister clock gating.
0: Support clock only when application writes registers
1: Force on clock gating for registers
(R/W)

Register 10.2. HP_SYS_CLKRST_ROOT_CLK_CTRL0_REG (0x0004)

(reserved)			HP_SYS_CLKRST_CPU_CLK_DIV_DENOMINATOR																	HP_SYS_CLKRST_CPU_CLK_DIV_NUMERATOR																	HP_SYS_CLKRST_CPU_CLK_DIV_NUM																	HP_SYS_CLKRST_SOC_CLK_DIV_UPDATE																	HP_SYS_CLKRST_CPUICM_DELAY_NUM																
31			29			28			21												20			13												12			5												4			3			0																														
0			0			0			0												0			0												0			0			0			0			Reset																																							

HP_SYS_CLKRST_CPUICM_DELAY_NUM Configures the time required from entering WFI mode to the actual shutdown of the CPU clock.
Measurement unit: clock cycles. (R/W)

HP_SYS_CLKRST_SOC_CLK_DIV_UPDATE Configures whether to update the divisors for CPU_CLK, MEM_CLK, SYS_CLK, APB_CLK.

0: Not update

1: Update

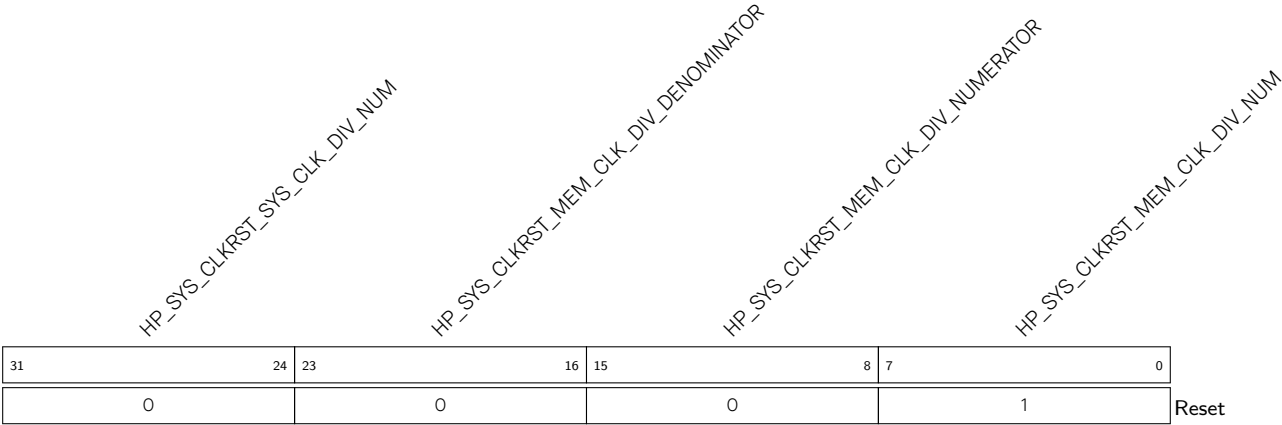
(WT)

HP_SYS_CLKRST_CPU_CLK_DIV_NUM Configures the integer part of the CPU_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_CPU_CLK_DIV_NUMERATOR Configures the numerator of the divisor's fractional part for CPU_CLK. (R/W)

HP_SYS_CLKRST_CPU_CLK_DIV_DENOMINATOR Configures the denominator of the divisor's fractional part for CPU_CLK. (R/W)

Register 10.3. HP_SYS_CLKRST_ROOT_CLK_CTRL1_REG (0x0008)



HP_SYS_CLKRST_MEM_CLK_DIV_NUM Configures the integer part of the MEM_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_MEM_CLK_DIV_NUMERATOR Configures the numerator of the divisor’s fractional part for MEM_CLK. (R/W)

HP_SYS_CLKRST_MEM_CLK_DIV_DENOMINATOR Configures the denominator of the divisor’s fractional part for MEM_CLK . (R/W)

HP_SYS_CLKRST_SYS_CLK_DIV_NUM Configures the integer part of the SYS_CLK clock divisor.
(R/W)

Register 10.4. HP_SYS_CLKRST_ROOT_CLK_CTRL2_REG (0x000C)

HP_SYS_CLKRST_APB_CLK_DIV_NUMERATOR							
HP_SYS_CLKRST_APB_CLK_DIV_NUM							
HP_SYS_CLKRST_SYS_CLK_DIV_DENOMINATOR							
HP_SYS_CLKRST_SYS_CLK_DIV_NUMERATOR							
31	24	23	16	15	8	7	0
0		1		0		0	
Reset							

- HP_SYS_CLKRST_SYS_CLK_DIV_NUMERATOR Configures the numerator of the divisor’s fractional part for SYS_CLK. (R/W)
- HP_SYS_CLKRST_SYS_CLK_DIV_DENOMINATOR Configures the denominator of the divisor’s fractional part for SYS_CLK. (R/W)
- HP_SYS_CLKRST_APB_CLK_DIV_NUM Configures the integer part of the APB_CLK fclock divisor. (R/W)
- HP_SYS_CLKRST_APB_CLK_DIV_NUMERATOR Configures the numerator of the divisor’s fractional part for APB_CLK. (R/W)

Register 10.5. HP_SYS_CLKRST_ROOT_CLK_CTRL3_REG (0x0010)

(reserved)																HP_SYS_CLKRST_APB_CLK_DIV_DENOMINATOR	
31							8	7									0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Reset																	

- HP_SYS_CLKRST_APB_CLK_DIV_DENOMINATOR Configures the denominator of the divisor’s fractional part for APB_CLK. (R/W)

Register 10.6. HP_SYS_CLKRST_SOC_CLK_CTRL0_REG (0x0014)

HP_SYS_CLKRST_PSRAM_SYS_CLK_EN																																HP_SYS_CLKRST_FLASH_SYS_CLK_EN																																HP_SYS_CLKRST_ICM_SYS_CLK_EN																																HP_SYS_CLKRST_SPMON_SYS_CLK_EN																																HP_SYS_CLKRST_L2MEMMON_SYS_CLK_EN																																HP_SYS_CLKRST_L2MEM_SYS_CLK_EN																																HP_SYS_CLKRST_L2CACHE_SYS_CLK_EN																																HP_SYS_CLKRST_TRACE_SYS_CLK_EN																																HP_SYS_CLKRST_MISC_SYS_CLK_EN																																HP_SYS_CLKRST_ICM_MEM_CLK_EN																																HP_SYS_CLKRST_L2MEMMON_MEM_CLK_EN																																HP_SYS_CLKRST_L2MEM_MEM_CLK_EN																																HP_SYS_CLKRST_L1CACHE_I1_MEM_CLK_EN																																HP_SYS_CLKRST_L1CACHE_I0_MEM_CLK_EN																																HP_SYS_CLKRST_L1CACHE_D_MEM_CLK_EN																																HP_SYS_CLKRST_L1CACHE_MEM_CLK_EN																																HP_SYS_CLKRST_GDMA_CPU_CLK_EN																																HP_SYS_CLKRST_ICM_CPU_CLK_EN																																HP_SYS_CLKRST_TRACE_CPU_CLK_EN																																HP_SYS_CLKRST_L1CACHE_I1_CPU_CLK_EN																																HP_SYS_CLKRST_L1CACHE_I0_CPU_CLK_EN																																HP_SYS_CLKRST_L1CACHE_D_CPU_CLK_EN																																HP_SYS_CLKRST_BUSMON_CPU_CLK_EN																																HP_SYS_CLKRST_SPM_CPU_CLK_EN																																HP_SYS_CLKRST_CORE1_CPU_CLK_EN																																HP_SYS_CLKRST_CORE0_CPU_CLK_EN																																HP_SYS_CLKRST_MISC_CPU_CLK_EN																																HP_SYS_CLKRST_CORE1_CLIC_CLK_EN																																HP_SYS_CLKRST_CORE0_CLIC_CLK_EN																																																																																																																																																											
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Reset																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
1	1	1	0	0	1	1	0	1	1	0	1	1	1	1	1	1	0	0	1	0	1	1	1	1	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Reset

HP_SYS_CLKRST_CORE0_CLIC_CLK_EN Configures whether to enable the bus clock of HP CPU0 CLIC.
 0: Disable
 1: Enable
 (R/W)

HP_SYS_CLKRST_CORE1_CLIC_CLK_EN Configures whether to enable the bus clock of HP CPU1 CLIC.
 0: Disable
 1: Enable
 (R/W)

HP_SYS_CLKRST_MISC_CPU_CLK_EN Configures whether to enable the bus clock of miscellaneous modules (CPU_CLK clock domain).
 0: Disable
 1: Enable
 (R/W)

HP_SYS_CLKRST_CORE0_CPU_CLK_EN Configures whether to enable the bus clock of HP CPU0 (CPU_CLK clock domain).
 0: Disable
 1: Enable
 (R/W)

HP_SYS_CLKRST_CORE1_CPU_CLK_EN Configures whether to enable the bus clock of HP CPU1 (CPU_CLK clock domain).
 0: Disable
 1: Enable
 (R/W)

HP_SYS_CLKRST_SPM_CPU_CLK_EN Configures whether to enable the bus clock of SPM monitor (CPU_CLK clock domain).
 0: Disable
 1: Enable
 (R/W)

Continued on the next page...

Register 10.6. HP_SYS_CLKRST_SOC_CLK_CTRL0_REG (0x0014)

Continued from the previous page...

HP_SYS_CLKRST_BUSMON_CPU_CLK_EN Configures whether to enable the bus clock of bus monitor (CPU_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L1CACHE_CPU_CLK_EN Configures whether to enable the bus clock of L1 Cache (CPU_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L1CACHE_D_CPU_CLK_EN Configures whether to enable the bus clock of L1 D-Cache (CPU_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L1CACHE_IO_CPU_CLK_EN Configures whether to enable the bus clock of L1 I-Cache 0 (CPU_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L1CACHE_I1_CPU_CLK_EN Configures whether to enable the bus clock of L1 I-Cache 1 (CPU_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TRACE_CPU_CLK_EN Configures whether to enable the bus clock of RISC-V Trace Encoder (CPU_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_ICM_CPU_CLK_EN Configures the bus clock of the bus module (CPU_CLK clock domain).

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.6. HP_SYS_CLKRST_SOC_CLK_CTRL0_REG (0x0014)

Continued from the previous page...

HP_SYS_CLKRST_GDMA_CPU_CLK_EN Configures whether to enable the bus clock of VDMA (CPU_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L1CACHE_MEM_CLK_EN Configures whether to enable the bus clock of L1 Cache (MEM_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L1CACHE_D_MEM_CLK_EN Configures whether to enable the bus clock of L1 D-Cache (MEM_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L1CACHE_IO_MEM_CLK_EN Configures whether to enable the bus clock of L1 I-Cache0 (MEM_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L1CACHE_I1_MEM_CLK_EN Configures whether to enable the bus clock of L1 I-Cache1 (MEM_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L2CACHE_MEM_CLK_EN Configures whether to enable the bus clock of L2 Cache (MEM_CLK clock domain).

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.6. HP_SYS_CLKRST_SOC_CLK_CTRL0_REG (0x0014)

Continued from the previous page...

HP_SYS_CLKRST_L2MEM_MEM_CLK_EN Configures whether to enable the bus clock of L2 memory (MEM_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L2MEMMON_MEM_CLK_EN Configures whether to enable the bus clock of L1 memory monitor (MEM_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_ICM_MEM_CLK_EN Configures the bus clock of the bus module (MEM_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_MISC_SYS_CLK_EN Configures whether to enable the bus clock of miscellaneous modules (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TRACE_SYS_CLK_EN Configures whether to enable the bus clock of RISC-V Trace Encoder (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L2CACHE_SYS_CLK_EN Configures whether to enable the bus clock of L2 Cache (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_L2MEM_SYS_CLK_EN Configures whether to enable the bus clock of L2 memory (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.6. HP_SYS_CLKRST_SOC_CLK_CTRL0_REG (0x0014)

Continued from the previous page...

HP_SYS_CLKRST_L2MEMMON_SYS_CLK_EN Configures whether to enable the bus clock of L2 memory monitor (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_SPMMON_SYS_CLK_EN Configures whether to enable the bus clock of L2 SPM monitor (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_ICM_SYS_CLK_EN Configures the bus clock of the bus module (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_FLASH_SYS_CLK_EN Configures whether to enable the bus clock of flash (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_PSRAM_SYS_CLK_EN Configures whether to enable the bus clock of PSRAM (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

Register 10.7. HP_SYS_CLKRST_SOC_CLK_CTRL1_REG (0x0018)

HP_SYS_CLKRST_H264_SYS_CLK_EN																																HP_SYS_CLKRST_BITRAMBLER_TX_SYS_CLK_EN																																HP_SYS_CLKRST_BITRAMBLER_RX_SYS_CLK_EN																																HP_SYS_CLKRST_BITRAMBLER_SYS_CLK_EN																																HP_SYS_CLKRST_KEY_MANAGER_SYS_CLK_EN																																HP_SYS_CLKRST_CRYPTO_SYS_CLK_EN																																(reserved)																																HP_SYS_CLKRST_ETM_SYS_CLK_EN																																HP_SYS_CLKRST_PARLIO_SYS_CLK_EN																																HP_SYS_CLKRST_UART4_SYS_CLK_EN																																HP_SYS_CLKRST_UART3_SYS_CLK_EN																																HP_SYS_CLKRST_UART2_SYS_CLK_EN																																HP_SYS_CLKRST_UART1_SYS_CLK_EN																																HP_SYS_CLKRST_UHCI_SYS_CLK_EN																																HP_SYS_CLKRST_USB_SYS_CLK_EN																																HP_SYS_CLKRST_USB_OTG20_SYS_CLK_EN																																HP_SYS_CLKRST_OTG11_SYS_CLK_EN																																HP_SYS_CLKRST_SDMMC_SYS_CLK_EN																																HP_SYS_CLKRST_EMAC_SYS_CLK_EN																																HP_SYS_CLKRST_DSI_SYS_CLK_EN																																HP_SYS_CLKRST_CSI_SYS_CLK_EN																																HP_SYS_CLKRST_CSI_HOST_SYS_CLK_EN																																(reserved)																																HP_SYS_CLKRST_PPA_BRG_SYS_CLK_EN																																HP_SYS_CLKRST_JPEG_SYS_CLK_EN																																HP_SYS_CLKRST_DMA2D_SYS_CLK_EN																																HP_SYS_CLKRST_GDMA_SYS_CLK_EN																																HP_SYS_CLKRST_AXI_PDMA_SYS_CLK_EN																																HP_SYS_CLKRST_AHB_PDMA_SYS_CLK_EN																																HP_SYS_CLKRST_GSP3_SYS_CLK_EN																																HP_SYS_CLKRST_GSP2_SYS_CLK_EN																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
0	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	1	Reset																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														

HP_SYS_CLKRST_GSP2_SYS_CLK_EN Configures whether to enable the bus clock of GP-SPI2 (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_GSP3_SYS_CLK_EN Configures whether to enable the bus clock of GP-SPI3 (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_AHB_PDMA_SYS_CLK_EN Configures whether to enable the bus clock of GDMA-AHB (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_AXI_PDMA_SYS_CLK_EN Configures whether to enable the bus clock of GDMA-AXI (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_GDMA_SYS_CLK_EN Configures whether to enable the bus clock of VDMA (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.7. HP_SYS_CLKRST_SOC_CLK_CTRL1_REG (0x0018)

Continued from the previous page...

HP_SYS_CLKRST_DMA2D_SYS_CLK_EN Configures whether to enable the bus clock of 2DDMA (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_JPEG_SYS_CLK_EN Configures whether to enable the bus clock of JPEG (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_PPA_SYS_CLK_EN Configures whether to enable the bus clock of PPA (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_CSI_BRG_SYS_CLK_EN Configures whether to enable the bus clock of CSI Bridge and ISP (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_CSI_HOST_SYS_CLK_EN Configures whether to enable the bus clock of CSI Host (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_DSI_SYS_CLK_EN Configures whether to enable the bus clock of DSI (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.7. HP_SYS_CLKRST_SOC_CLK_CTRL1_REG (0x0018)

Continued from the previous page...

HP_SYS_CLKRST_EMAC_SYS_CLK_EN Configures whether to enable the bus clock of EMAC (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_SDMMC_SYS_CLK_EN Configures whether to enable the bus clock of SDMMC (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_USB_OTG11_SYS_CLK_EN Configures whether to enable the bus clock of Full-Speed USB 2.0 OTG (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_USB_OTG20_SYS_CLK_EN Configures whether to enable the bus clock of High-Speed USB 2.0 OTG (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_UHCI_SYS_CLK_EN Configures whether to enable the bus clock of UHCI (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_UART0_SYS_CLK_EN Configures whether to enable the bus clock of UART0 (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_UART1_SYS_CLK_EN Configures whether to enable the bus clock of UART1 (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.7. HP_SYS_CLKRST_SOC_CLK_CTRL1_REG (0x0018)

Continued from the previous page...

HP_SYS_CLKRST_UART2_SYS_CLK_EN Configures whether to enable the bus clock of UART2 (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_UART3_SYS_CLK_EN Configures whether to enable the bus clock of UART3 (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_UART4_SYS_CLK_EN Configures whether to enable the bus clock of UART4 (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_PARLIO_SYS_CLK_EN Configures whether to enable the bus clock of PARLIO (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_ETM_SYS_CLK_EN Configures whether to enable the bus clock of ETM (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W) 0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_CRYPTOSYS_CLK_EN Configures whether to enable the bus clock of the crypto modules (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.7. HP_SYS_CLKRST_SOC_CLK_CTRL1_REG (0x0018)

Continued from the previous page...

HP_SYS_CLKRST_KEY_MANAGER_SYS_CLK_EN Configures whether to enable the bus clock of the Key Manager (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_BITSRAMBLER_SYS_CLK_EN Configures whether to enable the bus clock of the Bit Scrambler (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_BITSRAMBLER_RX_SYS_CLK_EN Configures whether to enable the bus clock of the Bit Scrambler RX (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_BITSRAMBLER_TX_SYS_CLK_EN Configures whether to enable the bus clock of the Bit Scrambler TX (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_H264_SYS_CLK_EN Configures whether to enable the bus clock of H264 (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

Register 10.8. HP_SYS_CLKRST_SOC_CLK_CTRL2_REG (0x001C)

HP_SYS_CLKRST_PARLO_APB_CLK_EN	HP_SYS_CLKRST_PONT_APB_CLK_EN	HP_SYS_CLKRST_USB_DEVICE_APB_CLK_EN	HP_SYS_CLKRST_MCPWM1_APB_CLK_EN	HP_SYS_CLKRST_MCPWMO_APB_CLK_EN	HP_SYS_CLKRST_TWA12_APB_CLK_EN	HP_SYS_CLKRST_TWA11_APB_CLK_EN	HP_SYS_CLKRST_TWAIO_APB_CLK_EN	HP_SYS_CLKRST_SYSTIMER_APB_CLK_EN	HP_SYS_CLKRST_TIMERGRP1_APB_CLK_EN	HP_SYS_CLKRST_TIMERGRP0_APB_CLK_EN	HP_SYS_CLKRST_GSP13_APB_CLK_EN	HP_SYS_CLKRST_GSP12_APB_CLK_EN	HP_SYS_CLKRST_I3C_S1V_APB_CLK_EN	HP_SYS_CLKRST_I3C_MST_APB_CLK_EN	HP_SYS_CLKRST_I2S2_APB_CLK_EN	HP_SYS_CLKRST_I2S1_APB_CLK_EN	HP_SYS_CLKRST_I2S0_APB_CLK_EN	HP_SYS_CLKRST_I2C1_APB_CLK_EN	HP_SYS_CLKRST_I2CO_APB_CLK_EN	HP_SYS_CLKRST_UART4_APB_CLK_EN	HP_SYS_CLKRST_UART3_APB_CLK_EN	HP_SYS_CLKRST_UART2_APB_CLK_EN	HP_SYS_CLKRST_UART1_APB_CLK_EN	HP_SYS_CLKRST_UART0_APB_CLK_EN	HP_SYS_CLKRST_UH01_APB_CLK_EN	HP_SYS_CLKRST_ADC_APB_CLK_EN	HP_SYS_CLKRST_INTRMTX_APB_CLK_EN	HP_SYS_CLKRST_ICM_APB_CLK_EN	HP_SYS_CLKRST_SYSREG_APB_CLK_EN	HP_SYS_CLKRST_HP_CLKRST_APB_CLK_EN	HP_SYS_CLKRST_RMT_SYS_CLK_EN
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	1	0	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	1	1	1	1	0	Reset

HP_SYS_CLKRST_RMT_SYS_CLK_EN Configures whether to enable the bus clock of RMT (SYS_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_HP_CLKRST_APB_CLK_EN Configures whether to enable the bus clock of the Reset and Clock module (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_SYSREG_APB_CLK_EN Configures whether to enable the bus clock of the HP System Registers (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_ICM_APB_CLK_EN Configures the bus clock of the bus module (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_INTRMTX_APB_CLK_EN Configures whether to enable the bus clock of the Interrupt Matrix (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_ADC_APB_CLK_EN Configures whether to enable the bus clock of the ADC (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.8. HP_SYS_CLKRST_SOC_CLK_CTRL2_REG (0x001C)

Continued from the previous page...

HP_SYS_CLKRST_UHCI_APB_CLK_EN Configures whether to enable the bus clock of the UHCI (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_UART0_APB_CLK_EN Configures whether to enable the bus clock of UART0 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_UART1_APB_CLK_EN Configures whether to enable the bus clock of UART1 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_UART2_APB_CLK_EN Configures whether to enable the bus clock of UART2 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_UART3_APB_CLK_EN Configures whether to enable the bus clock of UART3 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_UART4_APB_CLK_EN Configures whether to enable the bus clock of UART4 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_I2CO_APB_CLK_EN Configures whether to enable the bus clock of I2CO (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.8. HP_SYS_CLKRST_SOC_CLK_CTRL2_REG (0x001C)

Continued from the previous page...

HP_SYS_CLKRST_I2C1_APB_CLK_EN Configures whether to enable the bus clock of I2C1 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_I2S0_APB_CLK_EN Configures whether to enable the bus clock of I2S0 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_I2S1_APB_CLK_EN Configures whether to enable the bus clock of I2S1 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_I2S2_APB_CLK_EN Configures whether to enable the bus clock of I2S2 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_I3C_MST_APB_CLK_EN Configures whether to enable the bus clock of the I3C Master (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_I3C_SLV_APB_CLK_EN Configures whether to enable the bus clock of the I3C Slave (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_GPSPI2_APB_CLK_EN Configures whether to enable the bus clock of GP-SPI2 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.8. HP_SYS_CLKRST_SOC_CLK_CTRL2_REG (0x001C)

Continued from the previous page...

HP_SYS_CLKRST_GPSPI3_APB_CLK_EN Configures whether to enable the bus clock of GP-SPI3 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TIMERGRPO_APB_CLK_EN Configures whether to enable the bus clock of Timer Group 0 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TIMERGRP1_APB_CLK_EN Configures whether to enable the bus clock of Timer Group 1 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_SYSTIMER_APB_CLK_EN Configures whether to enable the bus clock of the System Timer (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TWAIO_APB_CLK_EN Configures whether to enable the bus clock of TWAIO (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TWA11_APB_CLK_EN Configures whether to enable the bus clock of TWA11 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TWA12_APB_CLK_EN Configures whether to enable the bus clock of TWA12 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.8. HP_SYS_CLKRST_SOC_CLK_CTRL2_REG (0x001C)

Continued from the previous page...

HP_SYS_CLKRST_MCPWM0_APB_CLK_EN Configures whether to enable the bus clock of MCPWM0 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_MCPWM1_APB_CLK_EN Configures whether to enable the bus clock of MCPWM1 (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_USB_DEVICE_APB_CLK_EN Configures whether to enable the bus clock of the USB Serial/JTAG (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_PCNT_APB_CLK_EN Configures whether to enable the bus clock of the PCNT (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_PARLIO_APB_CLK_EN Configures whether to enable the bus clock of the PARLIO (APB_CLK clock domain).

0: Disable

1: Enable

(R/W)

Register 10.9. HP_SYS_CLKRST_SOC_CLK_CTRL3_REG (0x0020)

(reserved)																								HP_SYS_CLKRST_IOMUX_APB_CLK_EN			
																								HP_SYS_CLKRST_ETM_APB_CLK_EN			
																								HP_SYS_CLKRST_LCDCAM_APB_CLK_EN			
																								HP_SYS_CLKRST_LEDC_APB_CLK_EN			
31																					4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	Reset	

HP_SYS_CLKRST_LEDC_APB_CLK_EN Configures whether to enable the bus clock of LEDC (APB_CLK clock domain).
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_LCDCAM_APB_CLK_EN Configures whether to enable the bus clock of the LCD_CAM (APB_CLK clock domain).
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_ETM_APB_CLK_EN Configures whether to enable the bus clock of ETM (APB_CLK clock domain).
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_IOMUX_APB_CLK_EN Configures whether to enable the bus clock of IO MUX (APB_CLK clock domain).
0: Disable
1: Enable
(R/W)

Register 10.10. HP_SYS_CLKRST_REF_CLK_CTRL0_REG (0x0024)

HP_SYS_CLKRST_REF_160M_CLK_DIV_NUM							
HP_SYS_CLKRST_REF_240M_CLK_DIV_NUM							
HP_SYS_CLKRST_REF_25M_CLK_DIV_NUM							
HP_SYS_CLKRST_REF_50M_CLK_DIV_NUM							
31	24	23	16	15	8	7	0
2		1		19		9	
Reset							

- HP_SYS_CLKRST_REF_50M_CLK_DIV_NUM Configures the divisor of PLL_F50M_CLK. It is not recommended for users to change. (R/W)
- HP_SYS_CLKRST_REF_25M_CLK_DIV_NUM Configures the divisor of PLL_F25M_CLK. It is not recommended for users to change. (R/W)
- HP_SYS_CLKRST_REF_240M_CLK_DIV_NUM Configures the divisor of PLL_F240M_CLK. It is not recommended for users to change. (R/W)
- HP_SYS_CLKRST_REF_160M_CLK_DIV_NUM Configures the divisor of PLL_F160M_CLK. It is not recommended for users to change. (R/W)

Register 10.11. HP_SYS_CLKRST_REF_CLK_CTRL1_REG (0x0028)

(reserved)				HP_SYS_CLKRST_REF_240M_CLK_EN				(reserved)				HP_SYS_CLKRST_REF_25M_CLK_EN				HP_SYS_CLKRST_REF_50M_CLK_EN				(reserved)				HP_SYS_CLKRST_REF_20M_CLK_DIV_NUM				HP_SYS_CLKRST_REF_80M_CLK_DIV_NUM				HP_SYS_CLKRST_REF_120M_CLK_DIV_NUM			
31	30	29	28	27	26	24			23	16			15	8			7	0																	
0	1	0	1	1	0	0	0	23			5			3			Reset																		

Reset

HP_SYS_CLKRST_REF_120M_CLK_DIV_NUM Configures the divisor of PLL_F120M_CLK. (R/W)

HP_SYS_CLKRST_REF_80M_CLK_DIV_NUM Configures the divisor of PLL_F80M_CLK. (R/W)

HP_SYS_CLKRST_REF_20M_CLK_DIV_NUM Configures the divisor of PLL_F20M_CLK. (R/W)

HP_SYS_CLKRST_REF_50M_CLK_EN Configures whether to enable PLL_F50M_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_REF_25M_CLK_EN Configures whether to enable PLL_F25M_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_REF_240M_CLK_EN Configures whether to enable PLL_F240M_CLK.

0: Disable

1: Enable

(R/W)

Register 10.12. HP_SYS_CLKRST_REF_CLK_CTRL2_REG (0x002C)

(reserved)																				HP_SYS_CLKRST_REF_20M_CLK_EN				(reserved)				HP_SYS_CLKRST_REF_80M_CLK_EN				(reserved)				HP_SYS_CLKRST_REF_120M_CLK_EN				(reserved)				HP_SYS_CLKRST_REF_160M_CLK_EN				
31																				9	8	7	5	4	3	2	1	0																				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	1	Reset																		

HP_SYS_CLKRST_REF_160M_CLK_EN Configures whether to enable PLL_F160M_CLK.
0: Disable
1: Enable
User modification is not recommended. (R/W)

HP_SYS_CLKRST_REF_120M_CLK_EN Configures whether to enable PLL_F120M_CLK.
0: Disable
1: Enable
User modification is not recommended. (R/W)

HP_SYS_CLKRST_REF_80M_CLK_EN Configures whether to enable PLL_F80M_CLK.
0: Disable
1: Enable
User modification is not recommended. (R/W)

HP_SYS_CLKRST_REF_20M_CLK_EN Configures whether to enable PLL_F20M_CLK.
0: Disable
1: Enable
User modification is not recommended. (R/W)

Register 10.13. HP_SYS_CLKRST_PERI_CLK_CTRL00_REG (0x0030)

(reserved) HP_SYS_CLKRST_EMAC_RX_CLK_EN HP_SYS_CLKRST_EMAC_RX_CLK_SRC_SEL HP_SYS_CLKRST_EMAC_RMII_CLK_EN HP_SYS_CLKRST_EMAC_RMII_CLK_SRC_SEL HP_SYS_CLKRST_PAD_EMAC_REF_CLK_EN HP_SYS_CLKRST_PSRAM_CORE_CLK_DIV_NUM HP_SYS_CLKRST_PSRAM_CORE_CLK_EN HP_SYS_CLKRST_PSRAM_PLL_CLK_EN HP_SYS_CLKRST_PSRAM_CLK_SRC_SEL HP_SYS_CLKRST_FLASH_CORE_CLK_DIV_NUM HP_SYS_CLKRST_FLASH_CORE_CLK_EN HP_SYS_CLKRST_FLASH_PLL_CLK_EN HP_SYS_CLKRST_FLASH_CLK_SRC_SEL																											
31	30	29	28	27	26	25	24	23					16	15	14	13	12	11					4	3	2	1	0
0	0	0	0	0	0	0	0	0				1	1	0	3				1	1	0	Reset					

HP_SYS_CLKRST_FLASH_CLK_SRC_SEL Configures the clock source for FLASH_CLK.

- 0: XTAL_CLK
 - 1: SPLL_CLK (480 MHz)
 - 2: CPLL_CLK (360 MHz)
 - 3: Invalid
- (R/W)

HP_SYS_CLKRST_FLASH_PLL_CLK_EN Configures whether to enable FLASH_PLL_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_FLASH_CORE_CLK_EN Configures whether to enable FLASH_CORE_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_FLASH_CORE_CLK_DIV_NUM Configures the clock divisor of FLASH_CORE_CLK. (R/W)

HP_SYS_CLKRST_PSRAM_CLK_SRC_SEL Configures the clock source for PSRAM_CLK.

- 0: XTAL_CLK
 - 1: MPLL_CLK
 - 2: SPLL_CLK (480 MHz)
 - 3: CPLL_CLK (360 MHz)
- (R/W)

HP_SYS_CLKRST_PSRAM_PLL_CLK_EN Configures whether to enable PSRAM_PLL_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_PSRAM_CORE_CLK_EN Configures whether to enable PSRAM_CORE_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

Continued on the next page...

Register 10.13. HP_SYS_CLKRST_PERI_CLK_CTRL00_REG (0x0030)

Continued from the previous page...

HP_SYS_CLKRST_PSRAM_CORE_CLK_DIV_NUM Configures the clock divisor of PSRAM_CORE_CLK. (R/W)

HP_SYS_CLKRST_PAD_EMAC_REF_CLK_EN Configures whether to enable PAD_EMAC_REF_CLK.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_EMAC_RMII_CLK_SRC_SEL Configures the clock source for EMAC_RMII_CLK.
0: PAD_EMAC_TXRX_CLK
1: PAD_EMAC_RX_CLK
2: PAD_EMAC_TX_CLK
3: Invalid
(R/W)

HP_SYS_CLKRST_EMAC_RMII_CLK_EN Configures whether to enable EMAC_RMII_CLK.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_EMAC_RX_CLK_SRC_SEL Configures the clock source for EMAC_RX_CLK.
0: PAD_EMAC_TXRX_CLK
1: PAD_EMAC_RX_CLK
(R/W)

HP_SYS_CLKRST_EMAC_RX_CLK_EN Configures whether to enable EMAC_RX_CLK.
0: Disable
1: Enable
(R/W)

Register 10.14. HP_SYS_CLKRST_PERI_CLK_CTRL01_REG (0x0034)

(reserved)								HP_SYS_CLKRST_SDIO_LS_CLK_EN								HP_SYS_CLKRST_SDIO_LS_CLK_SRC_SEL								(reserved)								HP_SYS_CLKRST_SDIO_HS_MODE								HP_SYS_CLKRST_EMAC_PTP_REF_CLK_EN								HP_SYS_CLKRST_EMAC_PTP_REF_CLK_SRC_SEL								HP_SYS_CLKRST_EMAC_TX_CLK_DIV_NUM								HP_SYS_CLKRST_EMAC_TX_CLK_EN								HP_SYS_CLKRST_EMAC_TX_CLK_SRC_SEL								HP_SYS_CLKRST_EMAC_RX_CLK_DIV_NUM							
31								25								24	23	22	21	20	19	18	17								10								9	8	7	0																																													
0								0								0	0	0	0	0	0	0	1								0								0	1								Reset																																							

HP_SYS_CLKRST_EMAC_RX_CLK_DIV_NUM Configures the clock divisor for EMAC_RX_CLK.
(R/W)

HP_SYS_CLKRST_EMAC_TX_CLK_SRC_SEL Configures the clock source for EMAC_TX_CLK.
0: PAD_EMAC_TXRX_CLK
1: PAD_EMAC_TX_CLK
(R/W)

HP_SYS_CLKRST_EMAC_TX_CLK_EN Configures whether to enable EMAC_TX_CLK.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_EMAC_TX_CLK_DIV_NUM Configures the clock divisor for EMAC_TX_CLK. (R/W)

HP_SYS_CLKRST_EMAC_PTP_REF_CLK_SRC_SEL Configures the clock source for EMAC_PTP_REF_CLK.
0: XTAL_CLK
1: PLL_F80M_CLK
(R/W)

HP_SYS_CLKRST_EMAC_PTP_REF_CLK_EN Configures whether to enable EMAC_PTP_REF_CLK.
0: Disable
1: Enable
(R/W)

Continued on the next page...

Register 10.14. HP_SYS_CLKRST_PERI_CLK_CTRL01_REG (0x0034)

Continued from the previous page...

HP_SYS_CLKRST_SDIO_HS_MODE Configures the mode of SDIO clock.

0: Low-speed mode

1: High-speed mode

(R/W)

HP_SYS_CLKRST_SDIO_LS_CLK_SRC_SEL Configures the low-speed clock source for SDIO.

0: PLL_F160M_CLK

1: APLL_CLK

(R/W)

HP_SYS_CLKRST_SDIO_LS_CLK_EN Configures whether to enable SDIO low-speed clock.

0: Disable

1: Enable

(R/W)

Register 10.15. HP_SYS_CLKRST_PERI_CLK_CTRL02_REG (0x0038)

HP_SYS_CLKRST_MIP1_DSI_DPHY_CLK_SRC_SEL		HP_SYS_CLKRST_SDIO_LS_SAM_CLK_EN		HP_SYS_CLKRST_SDIO_LS_DRV_CLK_EN		HP_SYS_CLKRST_SDIO_LS_SLF_CLK_EN		HP_SYS_CLKRST_SDIO_LS_SAM_CLK_EDGE_SEL		HP_SYS_CLKRST_SDIO_LS_DRV_CLK_EDGE_SEL		HP_SYS_CLKRST_SDIO_LS_SLF_CLK_EDGE_SEL		HP_SYS_CLKRST_SDIO_LS_CLK_EDGE_N		HP_SYS_CLKRST_SDIO_LS_CLK_EDGE_H		HP_SYS_CLKRST_SDIO_LS_CLK_EDGE_L		HP_SYS_CLKRST_SDIO_LS_CLK_EDGE_CFG_UPDATE		HP_SYS_CLKRST_SDIO_LS_CLK_DIV_NUM	
31	30	29	28	27	26	25	24	23	22	21	20	17	16	13	12	9	8	7	0			Reset	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			

HP_SYS_CLKRST_SDIO_LS_CLK_DIV_NUM Configures the divisor of the SDIO low-speed clock.
(R/W)

HP_SYS_CLKRST_SDIO_LS_CLK_EDGE_CFG_UPDATE Configures whether to update the SDIO low-speed clock phase configuration.
0: Not update
1: Update (WT)

HP_SYS_CLKRST_SDIO_LS_CLK_EDGE_L Configures the threshold for the SDIO low-speed clock signal to switch to the low level. (R/W)

HP_SYS_CLKRST_SDIO_LS_CLK_EDGE_H Configures the threshold for the SDIO low-speed clock signal to switch to the high level. (R/W)

HP_SYS_CLKRST_SDIO_LS_CLK_EDGE_N Configures the threshold for clearing the count value of the SDIO low-speed clock. (R/W)

HP_SYS_CLKRST_SDIO_LS_SLF_CLK_EDGE_SEL Configures SDIO_SLF_CLK to select the SDIO low-speed clock phase. (R/W)

HP_SYS_CLKRST_SDIO_LS_DRV_CLK_EDGE_SEL Configures SDIO_DRV_CLK to select the SDIO low-speed clock phase. (R/W)

HP_SYS_CLKRST_SDIO_LS_SAM_CLK_EDGE_SEL Configures SDIO_SAM_CLK to select the SDIO low-speed clock phase. (R/W)

HP_SYS_CLKRST_SDIO_LS_SLF_CLK_EN Configures whether to enable SDIO_SLF_CLK.
0: Disable
1: Enable
(R/W)

Continued on the next page...

Register 10.15. HP_SYS_CLKRST_PERI_CLK_CTRL02_REG (0x0038)

Continued from the previous page...

HP_SYS_CLKRST_SDIO_LS_DRV_CLK_EN Configures whether to enable SDIO_DRV_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_SDIO_LS_SAM_CLK_EN Configures whether to enable SDIO_SAM_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_MIPI_DSI_DPHY_CLK_SRC_SEL Configures the clock source for DSI_DPHY_CFG_CLK and DSI_DPHY_PLL_REFCLK.

0: PLL_F20M_CLK

1: RC_FAST_CLK

2: PLL_F25M_CLK

3: Invalid

(R/W)

Register 10.16. HP_SYS_CLKRST_PERI_CLK_CTRL03_REG (0x003C)

(reserved)																HP_SYS_CLKRST_MIPI_DSI_DPICLK_DIV_NUM								HP_SYS_CLKRST_MIPI_DSI_DPICLK_EN				HP_SYS_CLKRST_MIPI_CSI_DPHY_CFG_CLK_SEL				HP_SYS_CLKRST_MIPI_DSI_DPHY_CLK_EN				HP_SYS_CLKRST_MIPI_DSI_DPHY_PLL_REFCLK_EN			
31																16	15								8	7	6	5	4	3	2	1	0						
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset									

HP_SYS_CLKRST_MIPI_DSI_DPHY_CFG_CLK_EN Configures whether to enable DSI_DPHY_CFG_CLK clock.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_MIPI_DSI_DPHY_PLL_REFCLK_EN Configures whether to enable DSI_DPHY_PLL_REFCLK clock.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_MIPI_CSI_DPHY_CLK_SRC_SEL Configures the clock source for CSI_DPHY_CFG_CLK.
0: PLL_F20M_CLK
1: RC_FAST_CLK
2: PLL_F25M_CLK
3: Invalid
(R/W)

HP_SYS_CLKRST_MIPI_CSI_DPHY_CFG_CLK_EN Configures whether to enable CSI_DPHY_CFG_CLK clock.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_MIPI_DSI_DPICLK_SRC_SEL Configures the clock source for DSI_DPI_CLK.
0: XTAL_CLK
1: PLL_F240M_CLK
2: PLL_F160M_CLK
3: Invalid
(R/W)

Register 10.16. HP_SYS_CLKRST_PERI_CLK_CTRL03_REG (0x003C)

Continued from the previous page...

HP_SYS_CLKRST_MIPI_DSI_DPICKL_EN Configures whether to enable DSI_DPI_CLK clock.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_MIPI_DSI_DPICKL_DIV_NUM Configures the clock divisor of DSI_DPI_CLK.

(R/W)

Register 10.17. HP_SYS_CLKRST_PERI_CLK_CTRL10_REG (0x0040)

(reserved)				HP_SYS_CLKRST_I2C1_CLK_EN HP_SYS_CLKRST_I2C1_CLK_SRC_SEL				HP_SYS_CLKRST_I2CO_CLK_DIV_DENOMINATOR				HP_SYS_CLKRST_I2CO_CLK_DIV_NUMERATOR				HP_SYS_CLKRST_I2CO_CLK_DIV_NUM				HP_SYS_CLKRST_I2CO_CLK_EN HP_SYS_CLKRST_I2CO_CLK_SRC_SEL			
31	28	27	26	25	18	17	10	9	2	1	0	Reset											
0	0	0	0	0	0	0	0	0	0	0													

HP_SYS_CLKRST_I2CO_CLK_SRC_SEL Configures the clock source for I2CO_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

(R/W)

HP_SYS_CLKRST_I2CO_CLK_EN Configures whether to enable I2CO_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_I2CO_CLK_DIV_NUM Configures the integer part of the clock divisor of I2CO_CLK. (R/W)

HP_SYS_CLKRST_I2CO_CLK_DIV_NUMERATOR Configures the numerator of the divisor's fractional part for clock divisor of I2CO_CLK. (R/W)

HP_SYS_CLKRST_I2CO_CLK_DIV_DENOMINATOR Configures the denominator of the divisor's fractional part for clock divisor of I2CO_CLK. (R/W)

HP_SYS_CLKRST_I2C1_CLK_SRC_SEL Configures the clock source for I2C1_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

(R/W)

HP_SYS_CLKRST_I2C1_CLK_EN Configures whether to enable I2C1_CLK.

0: Disable

1: Enable

(R/W)

Register 10.18. HP_SYS_CLKRST_PERI_CLK_CTRL11_REG (0x0044)

(reserved)					HP_SYS_CLKRST_I2SO_RX_CLK_SRC_SEL					HP_SYS_CLKRST_I2SO_RX_CLK_EN					HP_SYS_CLKRST_I2C1_CLK_DIV_DENOMINATOR					HP_SYS_CLKRST_I2C1_CLK_DIV_NUMERATOR					HP_SYS_CLKRST_I2C1_CLK_DIV_NUM				
31					27	26	25	24	23					16	15					8	7					0			
0	0	0	0	0	0	0	0	0				0				0				0				Reset					

Reset

HP_SYS_CLKRST_I2C1_CLK_DIV_NUM Configures the integer part of the I2C1_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2C1_CLK_DIV_NUMERATOR Configures the numerator of the divisor's fractional part for I2C1_CLK clock divisor. (R/W)

HP_SYS_CLKRST_I2C1_CLK_DIV_DENOMINATOR Configures the denominator of the divisor's fractional part for I2C1_CLK clock divisor. (R/W)

HP_SYS_CLKRST_I2SO_RX_CLK_EN Configures whether to enable I2SO_RX_CLK.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_I2SO_RX_CLK_SRC_SEL Configures the clock source for I2SO_RX_CLK.
0: XTAL_CLK
1: APLL_CLK
2: PAD_I2SO_MCLK
3: Invalid
(R/W)

Register 10.19. HP_SYS_CLKRST_PERI_CLK_CTRL12_REG (0x0048)

(reserved)						HP_SYS_CLKRST_I2SO_RX_DIV_Y											HP_SYS_CLKRST_I2SO_RX_DIV_X													HP_SYS_CLKRST_I2SO_RX_DIV_N												
31	26					25	17										16	8								7	0															
0	0	0	0	0	0	0	0										0								0							Reset										

- HP_SYS_CLKRST_I2SO_RX_DIV_N Configures the integer part of the I2SO_RX_CLK clock divisor.
(R/W)
- HP_SYS_CLKRST_I2SO_RX_DIV_X Configures the coefficient x of the I2SO_RX_CLK clock divisor.
(R/W)
- HP_SYS_CLKRST_I2SO_RX_DIV_Y Configures the coefficient y of the I2SO_RX_CLK clock divisor.
(R/W)

Register 10.20. HP_SYS_CLKRST_PERI_CLK_CTRL13_REG (0x004C)

31	30	29	21	20	13	12	11	10	9	8	0
0	0		0		0	0	0	0		0	Reset

HP_SYS_CLKRST_I2SO_RX_DIV_Z Configures the coefficient z of I2SO_RX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2SO_RX_DIV_YN1 Configures the coefficient yn1 of I2SO_RX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S0_TX_CLK_EN Configures whether to enable I2S0_TX_CLK.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_I2SO_TX_CLK_SRC_SEL Configures the clock source for I2SO_TX_CLK.

- 0: XTAL_CLK
- 1: APLL_CLK
- 2: PAD_I2SO_MCLK
- 3: Invalid

(R/W)

HP_SYS_CLKRST_I2S0_TX_DIV_N Configures the integer part of the clock divisor of I2S0_TX_CLK.
(R/W)

HP_SYS_CLKRST_I2S0_TX_DIV_X Configures the coefficient x of I2S0_TX_CLK clock divisor. (R/W)

Register 10.21. HP_SYS_CLKRST_PERI_CLK_CTRL14_REG (0x0050)

(reserved)																HP_SYS_CLKRST_I2S1_RX_DIV_N						HP_SYS_CLKRST_I2S1_RX_CLK_SRC_SEL						HP_SYS_CLKRST_I2S1_RX_CLK_EN						HP_SYS_CLKRST_I2S0_MST_CLK_SEL						HP_SYS_CLKRST_I2S0_TX_DIV_YN1						HP_SYS_CLKRST_I2S0_TX_DIV_Z						HP_SYS_CLKRST_I2S0_TX_DIV_Y											
31	30																							23	22	21	20	19	18	17	9																8	0															
0	0																							0	0	0	0	0																0																Reset			

HP_SYS_CLKRST_I2S0_TX_DIV_Y Configures the coefficient y for I2S0_TX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S0_TX_DIV_Z Configures the coefficient z for I2S0_TX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S0_TX_DIV_YN1 Configures the coefficient yn1 for I2S0_TX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S0_MST_CLK_SEL Configures the clock source for the output clock PAD_I2S0_MST_CLK.
0: I2S0_RX_CLK
1: I2S0_TX_CLK
(R/W)

HP_SYS_CLKRST_I2S1_RX_CLK_EN Configures whether to enable I2S1_RX_CLK.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_I2S1_RX_CLK_SRC_SEL Configures the clock source for I2S1_RX_CLK.
0: XTAL_CLK
1: APLL_CLK
2: PAD_I2S1_MCLK
3: Invalid
(R/W)

HP_SYS_CLKRST_I2S1_RX_DIV_N Configures the integer part of the clock divisor for I2S1_RX_CLK.
(R/W)

Register 10.22. HP_SYS_CLKRST_PERI_CLK_CTRL15_REG (0x0054)

(reserved)																HP_SYS_CLKRST_I2S1_TX_CLK_SRC_SEL																HP_SYS_CLKRST_I2S1_TX_CLK_EN																HP_SYS_CLKRST_I2S1_RX_DIV_YN1																HP_SYS_CLKRST_I2S1_RX_DIV_Z																HP_SYS_CLKRST_I2S1_RX_DIV_Y																HP_SYS_CLKRST_I2S1_RX_DIV_X															
31	30	29	28	27	26											18	17											9	8											0																																																																							
0	0	0	0	0	0										0										0										Reset																																																																												

HP_SYS_CLKRST_I2S1_RX_DIV_X Configures the coefficient x for I2S1_RX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S1_RX_DIV_Y Configures the coefficient y for I2S1_RX_CLK clock divisor. (R/W)

HP_SYS_CLKRST_I2S1_RX_DIV_Z Configures the coefficient z for I2S1_RX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S1_RX_DIV_YN1 Configures the coefficient yn1 for I2S1_RX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S1_TX_CLK_EN Configures whether to enable PI2S1_TX_CLK.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_I2S1_TX_CLK_SRC_SEL Configures the clock source for I2S1_TX_CLK.

- 0: XTAL_CLK
- 1: APLL_CLK
- 2: PAD_I2S1_MCLK
- 3: Invalid

(R/W)

Register 10.23. HP_SYS_CLKRST_PERI_CLK_CTRL16_REG (0x0058)

(reserved)						HP_SYS_CLKRST_I2S1_TX_DIV_Y											HP_SYS_CLKRST_I2S1_TX_DIV_X											HP_SYS_CLKRST_I2S1_TX_DIV_N										
31						26	25									17	16								8	7												0
0	0	0	0	0	0	0		0										0									0											Reset

HP_SYS_CLKRST_I2S1_TX_DIV_N Configures the integer part of the clock divisor for I2S1_TX_CLK.
(R/W)

HP_SYS_CLKRST_I2S1_TX_DIV_X Configures the coefficient x for I2S1_TX_CLK clock divisor. (R/W)

HP_SYS_CLKRST_I2S1_TX_DIV_Y Configures the coefficient y for I2S1_TX_CLK clock divisor. (R/W)

Register 10.24. HP_SYS_CLKRST_PERI_CLK_CTRL17_REG (0x005C)

(reserved)												HP_SYS_CLKRST_I2S2_RX_DIV_X												HP_SYS_CLKRST_I2S2_RX_DIV_N												HP_SYS_CLKRST_I2S2_RX_CLK_SRC_SEL												HP_SYS_CLKRST_I2S2_RX_CLK_EN												HP_SYS_CLKRST_I2S1_MST_CLK_SEL												HP_SYS_CLKRST_I2S1_TX_DIV_YN1												HP_SYS_CLKRST_I2S1_TX_DIV_Z											
31	30																													22	21	14										13	12	11	10	9	8	0																																															
0	0																													0										0	0	0	0	0																							Reset																												

HP_SYS_CLKRST_I2S1_TX_DIV_Z Configures the coefficient z for I2S1_TX_CLK clock divisor. (R/W)

HP_SYS_CLKRST_I2S1_TX_DIV_YN1 Configures the coefficient yn1 for I2S1_TX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S1_MST_CLK_SEL Configures the clock source for the output clock PAD_I2S1_MST_CLK.

0: I2S1_RX_CLK

1: I2S1_TX_CLK

(R/W)

HP_SYS_CLKRST_I2S2_RX_CLK_EN Configures whether to enable I2S2_RX_CLK.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_I2S2_RX_CLK_SRC_SEL Configures the clock source for I2S2_RX_CLK.

- 0: XTAL_CLK
- 1: APLL_CLK
- 2: PAD_I2S2_MCLK
- 3: Invalid

(R/W)

HP_SYS_CLKRST_I2S2_RX_DIV_N Configures the integer part of the clock divisor for I2S2_RX_CLK. (R/W)

HP_SYS_CLKRST_I2S2_RX_DIV_X Configures the coefficient x for I2S2_RX_CLK clock divisor.
(R/W)

Register 10.25. HP_SYS_CLKRST_PERI_CLK_CTRL18_REG (0x0060)

(reserved)			HP_SYS_CLKRST_I2S2_TX_DIV_N				HP_SYS_CLKRST_I2S2_TX_CLK_SRC_SEL				HP_SYS_CLKRST_I2S2_TX_CLK_EN				HP_SYS_CLKRST_I2S2_RX_DIV_YN1				HP_SYS_CLKRST_I2S2_RX_DIV_Z				HP_SYS_CLKRST_I2S2_RX_DIV_Y			
31	30	29					22	21	20	19	18	17					9	8					0			
0	0	0					0	0	0	0					0											
																					Reset					

HP_SYS_CLKRST_I2S2_RX_DIV_Y Configures the coefficient y for I2S2_RX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S2_RX_DIV_Z Configures the coefficient z for I2S2_RX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S2_RX_DIV_YN1 Configures the coefficient yn1 for I2S2_RX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S2_TX_CLK_EN Configures whether to enable I2S2_TX_CLK.
0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_I2S2_TX_CLK_SRC_SEL Configures the clock source for I2S2_TX_CLK.

- 0: XTAL_CLK
- 1: APLL_CLK
- 2: PAD_I2S2_MCLK
- 3: Invalid

(R/W)

HP_SYS_CLKRST_I2S2_TX_DIV_N Configures the integer part for I2S2_TX_CLK clock divisor. (R/W)

Register 10.26. HP_SYS_CLKRST_PERI_CLK_CTRL19_REG (0x0064)

HP_SYS_CLKRST_LCD_CLK_EN																HP_SYS_CLKRST_I2S2_TX_DIV_Z																HP_SYS_CLKRST_I2S2_TX_DIV_Y																HP_SYS_CLKRST_I2S2_TX_DIV_X																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
HP_SYS_CLKRST_LCD_CLK_SEL				HP_SYS_CLKRST_I2S2_TX_DIV_YN1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															

HP_SYS_CLKRST_I2S2_TX_DIV_X Configures the coefficient x of I2S2_TX_CLK clock divisor. (R/W)

HP_SYS_CLKRST_I2S2_TX_DIV_Y Configures the coefficient y of I2S2_TX_CLK clock divisor. (R/W)

HP_SYS_CLKRST_I2S2_TX_DIV_Z Configures the coefficient z of I2S2_TX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S2_TX_DIV_YN1 Configures the coefficient yn1 of I2S2_TX_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_I2S2_MST_CLK_SEL Configures the clock source for output clock PAD_I2S2_MST_CLK.
0: I2S2_RX_CLK
1: I2S2_TX_CLK
(R/W)

HP_SYS_CLKRST_LCD_CLK_SRC_SEL Configures the clock source for LCD_CLK.
0: XTAL_CLK
1: PLL_F160M_CLK
2: APLL_CLK
3: Invalid
(R/W)

HP_SYS_CLKRST_LCD_CLK_EN Configures whether to enable LCD_CLK.
0: Disable
1: Enable
(R/W)

Register 10.27. HP_SYS_CLKRST_PERI_CLK_CTRL110_REG (0x0068)

(reserved)					HP_SYS_CLKRST_UART0_CLK_EN					HP_SYS_CLKRST_UART0_CLK_SRC_SEL					HP_SYS_CLKRST_LCD_CLK_DIV_DENOMINATOR					HP_SYS_CLKRST_LCD_CLK_DIV_NUMERATOR					HP_SYS_CLKRST_LCD_CLK_DIV_NUM				
31					27	26	25	24	23					16	15					8	7					0			
0	0	0	0	0	0	1	0	0				0				0				0				Reset					

Reset

HP_SYS_CLKRST_LCD_CLK_DIV_NUM Configures the integer part of the LCD_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_LCD_CLK_DIV_NUMERATOR Configures the numerator of the divisor's fractional part for LCD_CLK. (R/W)

HP_SYS_CLKRST_LCD_CLK_DIV_DENOMINATOR Configures the denominator of the divisor's fractional part for LCD_CLK. (R/W)

HP_SYS_CLKRST_UART0_CLK_SRC_SEL Configures the clock source for UART0_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

2: PLL_F80M_CLK

3: Invalid

(R/W)

HP_SYS_CLKRST_UART0_CLK_EN Configures whether to enable UART0_CLK.

0: Disable

1: Enable

(R/W)

Register 10.28. HP_SYS_CLKRST_PERI_CLK_CTRL111_REG (0x006C)

(reserved)						HP_SYS_CLKRST_UART1_CLK_EN						HP_SYS_CLKRST_UART1_CLK_SRC_SEL						HP_SYS_CLKRST_UART0_SCLK_DIV_DENOMINATOR						HP_SYS_CLKRST_UART0_SCLK_DIV_NUMERATOR						HP_SYS_CLKRST_UART0_SCLK_DIV_NUM					
31						27	26	25	24	23						16	15						8	7						0					
0	0	0	0	0	0	1	0		0						0						0						Reset								

HP_SYS_CLKRST_UART0_SCLK_DIV_NUM Configures the integer part of the UART0_SCLK clock divisor. (R/W)

HP_SYS_CLKRST_UART0_SCLK_DIV_NUMERATOR Configures the numerator of the divisor's fractional part for UART0_SCLK. (R/W)

HP_SYS_CLKRST_UART0_SCLK_DIV_DENOMINATOR Configures the denominator of the divisor's fractional part for UART0_SCLK. (R/W)

HP_SYS_CLKRST_UART1_CLK_SRC_SEL Configures the clock source for UART1_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

2: PLL_F80M_CLK

3: Invalid

(R/W)

HP_SYS_CLKRST_UART1_CLK_EN Configures whether to enable UART1_CLK.

0: Disable

1: Enable

(R/W)

Register 10.29. HP_SYS_CLKRST_PERI_CLK_CTRL112_REG (0x0070)

(reserved)					HP_SYS_CLKRST_UART2_CLK_EN					HP_SYS_CLKRST_UART2_CLK_SRC_SEL					HP_SYS_CLKRST_UART1_SCLK_DIV_DENOMINATOR					HP_SYS_CLKRST_UART1_SCLK_DIV_NUMERATOR					HP_SYS_CLKRST_UART1_SCLK_DIV_NUM				
31					27	26	25	24	23					16	15					8	7					0			
0	0	0	0	0	1	0			0					0					0					Reset					

Reset

HP_SYS_CLKRST_UART1_SCLK_DIV_NUM Configures the integer part of the UART1_SCLK clock divisor. (R/W)

HP_SYS_CLKRST_UART1_SCLK_DIV_NUMERATOR Configures the numerator of the divisor's fractional part for UART1_SCLK. (R/W)

HP_SYS_CLKRST_UART1_SCLK_DIV_DENOMINATOR Configures the denominator of the divisor's fractional part for UART1_SCLK. (R/W)

HP_SYS_CLKRST_UART2_CLK_SRC_SEL Configures the clock source for UART2_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

2: PLL_F80M_CLK

3: Invalid

(R/W)

HP_SYS_CLKRST_UART2_CLK_EN Configures whether to enable the UART2_CLK clock.

0: Disable

1: Enable

(R/W)

Register 10.30. HP_SYS_CLKRST_PERI_CLK_CTRL113_REG (0x0074)

(reserved)						HP_SYS_CLKRST_UART3_CLK_EN						HP_SYS_CLKRST_UART3_CLK_SRC_SEL						HP_SYS_CLKRST_UART2_SCLK_DIV_DENOMINATOR						HP_SYS_CLKRST_UART2_SCLK_DIV_NUMERATOR						HP_SYS_CLKRST_UART2_SCLK_DIV_NUM					
31		27				26	25	24	23				16				15	8				7	0												
0		0				0		0		0		1				0		0				0				Reset									

HP_SYS_CLKRST_UART2_SCLK_DIV_NUM Configures the integer part of the UART2_SCLK clock divisor. (R/W)

HP_SYS_CLKRST_UART2_SCLK_DIV_NUMERATOR Configures the numerator of the fractional part of the UART2_SCLK. (R/W)

HP_SYS_CLKRST_UART2_SCLK_DIV_DENOMINATOR Configures the denominator of the fractional part of the UART2_SCLK. (R/W)

HP_SYS_CLKRST_UART3_CLK_SRC_SEL Configures the clock source for UART3_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

2: PLL_F80M_CLK

3: Invalid

(R/W)

HP_SYS_CLKRST_UART3_CLK_EN Configures whether to enable the UART3_CLK clock.

0: Disable

1: Enable

(R/W)

Register 10.31. HP_SYS_CLKRST_PERI_CLK_CTRL114_REG (0x0078)

(reserved)						HP_SYS_CLKRST_UART4_CLK_EN						HP_SYS_CLKRST_UART4_CLK_SRC_SEL						HP_SYS_CLKRST_UART3_SCLK_DIV_DENOMINATOR						HP_SYS_CLKRST_UART3_SCLK_DIV_NUMERATOR						HP_SYS_CLKRST_UART3_SCLK_DIV_NUM					
31						27	26	25	24	23						16	15						8	7						0					
0	0	0	0	0	0	1	0	0					0					0					0					Reset							

HP_SYS_CLKRST_UART3_SCLK_DIV_NUM Configures the integer part of the UART3_SCLK clock divisor. (R/W)

HP_SYS_CLKRST_UART3_SCLK_DIV_NUMERATOR Configures the numerator of the fractional part of the UART3_SCLK. (R/W)

HP_SYS_CLKRST_UART3_SCLK_DIV_DENOMINATOR Configures the denominator of the fractional part of the UART3_SCLK. (R/W)

HP_SYS_CLKRST_UART4_CLK_SRC_SEL Configures the clock source for UART4_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

2: PLL_F80M_CLK

3: Invalid

(R/W)

HP_SYS_CLKRST_UART4_CLK_EN Configures whether to enable the UART4_CLK clock.

0: Disable

1: Enable

(R/W)

Register 10.32. HP_SYS_CLKRST_PERI_CLK_CTRL115_REG (0x007C)

(reserved)																HP_SYS_CLKRST_TWA12_CLK_EN																HP_SYS_CLKRST_TWA12_CLK_SRC_SEL																HP_SYS_CLKRST_TWA11_CLK_EN																HP_SYS_CLKRST_TWA11_CLK_SRC_SEL																HP_SYS_CLKRST_TWA10_CLK_EN																HP_SYS_CLKRST_TWA10_CLK_SRC_SEL																HP_SYS_CLKRST_UART4_SCLK_DIV_DENOMINATOR																HP_SYS_CLKRST_UART4_SCLK_DIV_NUMERATOR																HP_SYS_CLKRST_UART4_SCLK_DIV_NUM															
31	30	29	28	27	26	25	24	23																	16	15																	8	7																	0																																																																																																		
0	0	0	0	0	0	0	0	0	0																																0																																0																Reset																																																																						

Reset

HP_SYS_CLKRST_UART4_SCLK_DIV_NUM Configures the integer part of the UART4_SCLK clock divisor. (R/W)

HP_SYS_CLKRST_UART4_SCLK_DIV_NUMERATOR Configures the numerator of the fractional part of the UART4_SCLK clock divisor. (R/W)

HP_SYS_CLKRST_UART4_SCLK_DIV_DENOMINATOR Configures the denominator of the fractional part of the UART4_SCLK clock divisor. (R/W)

HP_SYS_CLKRST_TWAIO_CLK_SRC_SEL Configures the clock source for TWAIO_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

(R/W)

HP_SYS_CLKRST_TWAIO_CLK_EN Configures whether to enable TWAIO_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TWA1_CLK_SRC_SEL Configures the clock source for TWA1_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

(R/W)

HP_SYS_CLKRST_TWA1_CLK_EN Configures whether to enable TWA1_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TWA2_CLK_SRC_SEL Configures the clock source for TWA2_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

(R/W)

HP_SYS_CLKRST_TWA2_CLK_EN Configures whether to enable TWA2_CLK.

0: Disable

1: Enable

(R/W)

Register 10.33. HP_SYS_CLKRST_PERI_CLK_CTRL116_REG (0x0080)

(reserved)								HP_SYS_CLKRST_GPSPi3_HS_CLK_EN				HP_SYS_CLKRST_GPSPi3_CLK_SRC_SEL				HP_SYS_CLKRST_GPSPi2_MST_CLK_EN				HP_SYS_CLKRST_GPSPi2_MST_CLK_DIV_NUM				HP_SYS_CLKRST_GPSPi2_HS_CLK_DIV_NUM				HP_SYS_CLKRST_GPSPi2_HS_CLK_EN				HP_SYS_CLKRST_GPSPi2_CLK_SRC_SEL			
31								25	24	23	21		20	19								12	11								4	3	2	0	
0 0 0 0 0 0 0								1	0			1	0							0							1	0			Reset				

HP_SYS_CLKRST_GPSPI2_CLK_SRC_SEL Configures the clock source for GPSPI2_CLK.

- 0: XTAL_CLK
 - 1: RC_FAST_CLK
 - 2: SDIO_PLLO_CLK
 - 3: APLL_CLK
 - 4: SPLL_CLK (480 MHz)
 - 5-7: Invalid
- (R/W)

HP_SYS_CLKRST_GPSPI2_HS_CLK_EN Configures whether to enable GPSPI2_HS_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_GPSPI2_HS_CLK_DIV_NUM Configures the integer part of the clock divisor of GPSPI2_HS_CLK. (R/W)

HP_SYS_CLKRST_GPSPI2_MST_CLK_DIV_NUM Configures the integer part of the clock divisor of GPSPI2_MST_CLK. (R/W)

HP_SYS_CLKRST_GPSPI2_MST_CLK_EN Configures whether to enable GPSPI2_MST_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_GPSPI3_CLK_SRC_SEL Configures the clock source for GPSPI3_CLK.

- 0: XTAL_CLK
 - 1: RC_FAST_CLK
 - 2: SDIO_PLLO_CLK
 - 3: APLL_CLK
 - 4: SPLL_CLK (480 MHz)
 - 5-7: Invalid
- (R/W)

HP_SYS_CLKRST_GPSPI3_HS_CLK_EN Configures whether to enable GPSPI3_HS_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

Register 10.34. HP_SYS_CLKRST_PERI_CLK_CTRL117_REG (0x0084)

(reserved)				HP_SYS_CLKRST_PARLIO_RX_CLK_DIV_NUM				HP_SYS_CLKRST_PARLIO_RX_CLK_EN				HP_SYS_CLKRST_PARLIO_RX_CLK_SRC_SEL				HP_SYS_CLKRST_GPSPI3_MST_CLK_EN				HP_SYS_CLKRST_GPSPI3_MST_CLK_DIV_NUM				HP_SYS_CLKRST_GPSPI3_HS_CLK_DIV_NUM			
31	28	27		20	19	18	17	16	15		8	7															
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

HP_SYS_CLKRST_GPSPI3_HS_CLK_DIV_NUM Configures the integer part of the clock divisor for GPSPI3_HS_CLK. (R/W)

HP_SYS_CLKRST_GPSPI3_MST_CLK_DIV_NUM Configures the integer part of the clock divisor for GPSPI3_MST_CLK. (R/W)

HP_SYS_CLKRST_GPSPI3_MST_CLK_EN Configures whether to enable GPSPI3_MST_CLK.
 0: Disable
 1: Enable
 (R/W)

HP_SYS_CLKRST_PARLIO_RX_CLK_SRC_SEL Configures the clock source for PARLIO_RX_CLK.
 0: XTAL_CLK
 1: RC_FAST_CLK
 2: PLL_F160M_CLK
 3: PAD_PARLIO_RX_CLK
 (R/W)

HP_SYS_CLKRST_PARLIO_RX_CLK_EN Configures whether to enable PARLIO_RX_CLK.
 0: Disable
 1: Enable
 (R/W)

HP_SYS_CLKRST_PARLIO_RX_CLK_DIV_NUM Configures the integer part of the clock divisor for PARLIO_RX_CLK. (R/W)

Register 10.35. HP_SYS_CLKRST_PERI_CLK_CTRL118_REG (0x0088)

(reserved)					HP_SYS_CLKRST_PARLIO_TX_CLK_DIV_NUM				HP_SYS_CLKRST_PARLIO_TX_CLK_EN				HP_SYS_CLKRST_PARLIO_TX_CLK_SRC_SEL				HP_SYS_CLKRST_PARLIO_RX_CLK_DIV_DENOMINATOR				HP_SYS_CLKRST_PARLIO_RX_CLK_DIV_NUMERATOR			
31	27				26	19				18	17	16	15	8				7	0					
0	0	0	0	0	0				0	0	0				0				Reset					

Reset

HP_SYS_CLKRST_PARLIO_RX_CLK_DIV_NUMERATOR Configures the numerator of the divisor's fractional part for clock divisor for PARLIO_RX_CLK. (R/W)

HP_SYS_CLKRST_PARLIO_RX_CLK_DIV_DENOMINATOR Configures the denominator of the divisor's fractional part for clock divisor for PARLIO_RX_CLK. (R/W)

HP_SYS_CLKRST_PARLIO_TX_CLK_SRC_SEL Configures the clock source for PARLIO_TX_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

2: PLL_F160M_CLK

3: PAD_PARLIO_TX_CLK

(R/W)

HP_SYS_CLKRST_PARLIO_TX_CLK_EN Configures whether to enable PARLIO_TX_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_PARLIO_TX_CLK_DIV_NUM Configures the integer part of the clock divisor for PARLIO_TX_CLK. (R/W)

Register 10.36. HP_SYS_CLKRST_PERI_CLK_CTRL119_REG (0x008C)

(reserved)		HP_SYS_CLKRST_CAM_CLK_EN		HP_SYS_CLKRST_CAM_CLK_SRC_SEL		HP_SYS_CLKRST_I3C_MST_CLK_DIV_NUM		HP_SYS_CLKRST_I3C_MST_CLK_EN		HP_SYS_CLKRST_I3C_MST_CLK_SRC_SEL		HP_SYS_CLKRST_PARLIO_TX_CLK_DIV_DENOMINATOR		HP_SYS_CLKRST_PARLIO_TX_CLK_DIV_NUMERATOR	
31	30	29	28	27	26	19	18	17	16	15	8	7			0
0	0	0	0	0	0	0	0	0	0	0	0	0			Reset

HP_SYS_CLKRST_PARLIO_TX_CLK_DIV_NUMERATOR Configures the numerator of the divisor's fractional part for PARLIO_TX_CLK. (R/W)

HP_SYS_CLKRST_PARLIO_TX_CLK_DIV_DENOMINATOR Configures the denominator of the divisor's fractional part for PARLIO_TX_CLK. (R/W)

HP_SYS_CLKRST_I3C_MST_CLK_SRC_SEL Configures the clock source for I3C_MST_CLK.

0: XTAL_CLK

1: PLL_F160M_CLK

2: PLL_F120M_CLK

3: Invalid

(R/W)

HP_SYS_CLKRST_I3C_MST_CLK_EN Configures whether to enable I3C_MST_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_I3C_MST_CLK_DIV_NUM Configures the integer part of the I3C_MST_CLK clock divisor. (R/W)

HP_SYS_CLKRST_CAM_CLK_SRC_SEL Configures the clock source for the output clock CAM_CLK.

0: XTAL_CLK

1: PLL_F160M_CLK

2: APLL_CLK

3: Invalid

(R/W)

HP_SYS_CLKRST_CAM_CLK_EN Configures whether to enable the output clock CAM_CLK.

0: Disable

1: Enable

(R/W)

Register 10.37. HP_SYS_CLKRST_PERI_CLK_CTRL120_REG (0x0090)

(reserved)								HP_SYS_CLKRST_CAM_CLK_DIV_DENOMINATOR								HP_SYS_CLKRST_CAM_CLK_DIV_NUMERATOR								HP_SYS_CLKRST_CAM_CLK_DIV_NUM																																							
31								24								23								16								15								8								7								0							
0 0 0 0 0 0 0 0								0								0								0								Reset																															

HP_SYS_CLKRST_CAM_CLK_DIV_NUM Configures the integer part of the CAM_CLK clock divisor.
(R/W)

HP_SYS_CLKRST_CAM_CLK_DIV_NUMERATOR Configures the numerator of the divisor's fractional part for CAM_CLK. (R/W)

HP_SYS_CLKRST_CAM_CLK_DIV_DENOMINATOR Configures the denominator of the divisor's fractional part for CAM_CLK. (R/W)

Register 10.38. HP_SYS_CLKRST_PERI_CLK_CTRL20_REG (0x0094)

31	30	29	28	27	26	25	24	23	22	21	14	13	12	11	10	3	2	1	0
1	1	0	1	0	1	0				0		0	0		0		0	0	Reset

HP_SYS_CLKRST_MCPWMO_CLK_SRC_SEL Configures the clock source for MCPWMO_CLK.

- 0: XTAL_CLK
1: RC_FAST_CLK
2: PLL_F160_CLK
3: Invalid
(R/W)

HP_SYS_CLKRST_MCPWMO_CLK_EN Configures whether to enable MCPWMO_CLK.

- 0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_MCPWMO_CLK_DIV_NUM Configures the clock divisor of MCPWMO_CLK. (R/W)

HP_SYS_CLKRST_MCPWM1_CLK_SRC_SEL Configures the clock source for MCPWM1_CLK.

- 0: XTAL_CLK
1: RC_FAST_CLK
2: PLL_F160_CLK
3: Invalid
(R/W)

HP_SYS_CLKRST_MCPWM1_CLK_EN Configures whether to enable MCPWM1_CLK.

- 0: Disable
1: Enable
(R/W)

HP_SYS_CLKRST_MCPWM1_CLK_DIV_NUM Configures the clock divisor of MCPWM1_CLK. (R/W)

HP_SYS_CLKRST_TIMERGRPO_TO_SRC_SEL Configures the clock source for TIMER-GRPO TO CLK.

- 0: XTAL_CLK
1: RC_FAST_CLK
2: PLL_F80M_CLK
3: Invalid
(R/W)

Register 10.38. HP_SYS_CLKRST_PERI_CLK_CTRL20_REG (0x0094)

Continued from the previous page...

HP_SYS_CLKRST_TIMERGRPO_TO_CLK_EN Configures whether to enable TIMERGRPO_TO_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TIMERGRPO_T1_SRC_SEL Configures the clock source for TIMERGRPO_T1_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

2: PLL_F80M_CLK

3: Invalid

(R/W)

HP_SYS_CLKRST_TIMERGRPO_T1_CLK_EN Configures whether to enable TIMERGRPO_T1_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TIMERGRPO_WDT_SRC_SEL Configures the clock source for TIMER-GRPO_WDT_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

2: PLL_F80M_CLK

3: Invalid

(R/W)

HP_SYS_CLKRST_TIMERGRPO_WDT_CLK_EN Configures whether to enable TIMER-GRPO_WDT_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_TIMERGRPO_TGRT_CLK_EN Configures whether to enable TIMER-GRPO_TGRT_CLK.

0: Disable

1: Enable

(R/W)

Register 10.39. HP_SYS_CLKRST_PERI_CLK_CTRL21_REG (0x0098)

<i>(reserved)</i>													<i>HP_SYS_CLKRST_SYSTIMER_CLK_EN</i>													<i>HP_SYS_CLKRST_TIMERGRP1_WDT_CLK_EN</i>													<i>HP_SYS_CLKRST_TIMERGRP1_WDT_SRC_SEL</i>													<i>HP_SYS_CLKRST_TIMERGRP1_T1_CLK_EN</i>													<i>HP_SYS_CLKRST_TIMERGRP1_T1_SRC_SEL</i>													<i>HP_SYS_CLKRST_TIMERGRP1_TO_CLK_EN</i>													<i>HP_SYS_CLKRST_TIMERGRP1_TO_SRC_SEL</i>													<i>HP_SYS_CLKRST_TIMERGRP0_TGRT_CLK_DIV_NUM</i>													<i>HP_SYS_CLKRST_TIMERGRP0_TGRT_CLK_SRC_SEL</i>												
31	30	29	28	27	26	25	24	23	22	21	20	19											4	3											0																																																																																														
0	1	0	1	0	1	0	1	0	1	0	0										0										Reset																																																																																																		

Reset

HP_SYS_CLKRST_TIMERGRPO_TGRT_CLK_SRC_SEL Configures the clock source for TIMER-GRPO_TGRT_CLK.

0: MPLL_CLK (500 MHz)

1: SPLL_CLK (480 MHz)

2: CPLL_CLK (360 MHz)

3: APLL_CLK

4: SDIO_PLLO_CLK

5: SDIO_PLL1_CLK

6: SDIO_PLL2_CLK

7: RC_FAST_CLK

8: RC_SLOW_CLK

9: Invalid

10: XTAL32K_CLK

11: PLL_LP_CLK

12-15: Clock tie 0

(R/W)

HP_SYS_CLKRST_TIMERGRPO_TGRT_CLK_DIV_NUM Configures the clock divisor of TIMER-GRPO_TGRT_CLK. (R/W)

HP_SYS_CLKRST_TIMERGRP1_TO_SRC_SEL Configures the clock source for TIMERGRP1_TO_CLK.

0: XTAL_CLK

1: RC_FAST_CLK

2: PLL_F80M_CLK

3: Invalid

(R/W)

HP_SYS_CLKRST_TIMERGRP1_TO_CLK_EN Configures whether to enable TIMERGRP1_TO_CLK.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.39. HP_SYS_CLKRST_PERI_CLK_CTRL21_REG (0x0098)

Continued from the previous page...

HP_SYS_CLKRST_TIMERGRP1_T1_SRC_SEL Configures the clock source for TIMERGRP1_T1_CLK.

- 0: XTAL_CLK
 - 1: RC_FAST_CLK
 - 2: PLL_F80M_CLK
 - 3: Invalid
- (R/W)

HP_SYS_CLKRST_TIMERGRP1_T1_CLK_EN Configures whether to enable TIMERGRP1_T1_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_TIMERGRP1_WDT_SRC_SEL Configures the clock source for TIMERGRP1_WDT_CLK.

- 0: XTAL_CLK
 - 1: RC_FAST_CLK
 - 2: PLL_F80M_CLK
 - 3: Invalid
- (R/W)

HP_SYS_CLKRST_TIMERGRP1_WDT_CLK_EN Configures whether to enable TIMERGRP1_WDT_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_SYSTIMER_CLK_SRC_SEL Configures the clock source for SYSTIMER_CLK.

- 0: XTAL_CLK
 - 1: RC_FAST_CLK
- (R/W)

HP_SYS_CLKRST_SYSTIMER_CLK_EN Configures whether to enable SYSTIMER_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

Register 10.40. HP_SYS_CLKRST_PERI_CLK_CTRL22_REG (0x009C)

HP_SYS_CLKRST_ADC_CLK_SRC_SEL				HP_SYS_CLKRST_RMT_CLK_DIV_DENOMINATOR				HP_SYS_CLKRST_RMT_CLK_DIV_NUMERATOR				HP_SYS_CLKRST_RMT_CLK_DIV_NUM				HP_SYS_CLKRST_RMT_CLK_EN				HP_SYS_CLKRST_RMT_CLK_SRC_SEL				HP_SYS_CLKRST_LEDC_CLK_EN				HP_SYS_CLKRST_LEDC_CLK_SRC_SEL			
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

HP_SYS_CLKRST_LEDC_CLK_SRC_SEL Configures the clock source for LEDC_CLK.

- 0: XTAL_CLK
 - 1: RC_FAST_CLK
 - 2: PLL_F80M_CLK
 - 3: Invalid
- (R/W)

HP_SYS_CLKRST_LEDC_CLK_EN Configures whether to enable LEDC_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_RMT_CLK_SRC_SEL Configures the clock source for RMT_CLK.

- 0: XTAL_CLK
 - 1: RC_FAST_CLK
 - 2: PLL_F80M_CLK
 - 3: Invalid
- (R/W)

HP_SYS_CLKRST_RMT_CLK_EN Configures whether to enable RMT_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_RMT_CLK_DIV_NUM Configures the integer part of the RMT_CLK clock divisor.
(R/W)**HP_SYS_CLKRST_RMT_CLK_DIV_NUMERATOR** Configures the numerator of the divisor's fractional part for RMT_CLK. (R/W)**HP_SYS_CLKRST_RMT_CLK_DIV_DENOMINATOR** Configures the denominator of the divisor's fractional part for RMT_CLK. (R/W)

Continued on the next page...

Register 10.40. HP_SYS_CLKRST_PERI_CLK_CTRL22_REG (0x009C)

Continued from the previous page...

HP_SYS_CLKRST_ADC_CLK_SRC_SEL Configures the clock source for HPADC_CLK.

- 0: XTAL_CLK
- 1: RC_FAST_CLK
- 2: PLL_F80M_CLK
- 3: Invalid

(R/W)

Register 10.41. HP_SYS_CLKRST_PERI_CLK_CTRL23_REG (0x00A0)

(reserved)								HP_SYS_CLKRST_ADC_CLK_DIV_DENOMINATOR								HP_SYS_CLKRST_ADC_CLK_DIV_NUMERATOR								HP_SYS_CLKRST_ADC_CLK_DIV_NUM								HP_SYS_CLKRST_ADC_CLK_EN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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HP_SYS_CLKRST_ADC_CLK_EN Configures whether to enable HPADC_CLK.

- 0: Disable
- 1: Enable

(R/W)

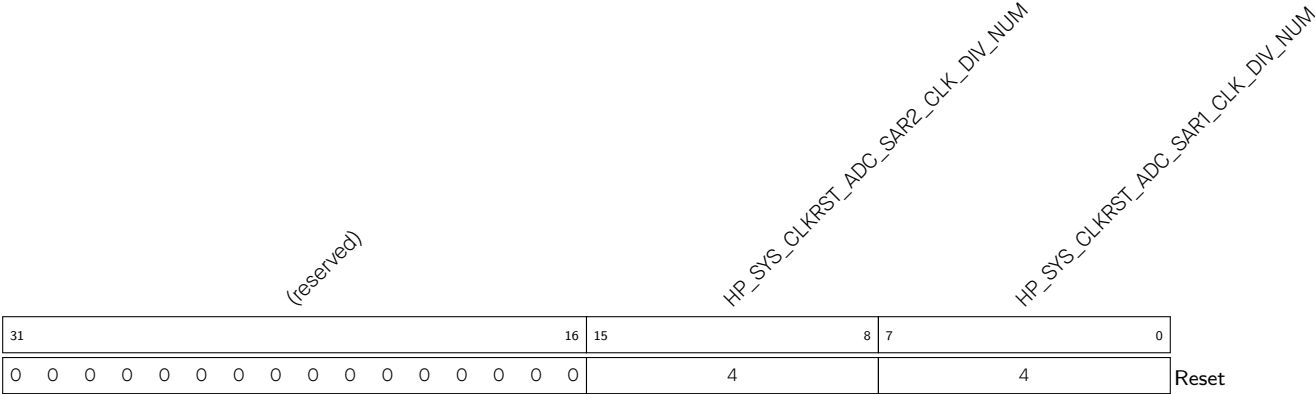
HP_SYS_CLKRST_ADC_CLK_DIV_NUM Configures the integer part of the HPADC_CLK clock divisor.

(R/W)

HP_SYS_CLKRST_ADC_CLK_DIV_NUMERATOR Configures the numerator of the divisor’s fractional part for HPADC_CLK. (R/W)

HP_SYS_CLKRST_ADC_CLK_DIV_DENOMINATOR Configures the denominator of the divisor’s fractional part for HPADC_CLK. (R/W)

Register 10.42. HP_SYS_CLKRST_PERI_CLK_CTRL24_REG (0x00A4)



HP_SYS_CLKRST_ADC_SAR1_CLK_DIV_NUM Configures the integer part of the clock divisor for ADC_SAR1_CLK. (R/W)

HP_SYS_CLKRST_ADC_SAR2_CLK_DIV_NUM Configures the integer part of the clock divisor for ADC_SAR2_CLK. (R/W)

Register 10.43. HP_SYS_CLKRST_PERI_CLK_CTRL25_REG (0x00A8)

(reserved)						HP_SYS_CLKRST_ISP_CLK_EN HP_SYS_CLKRST_ISP_CLK_SEL HP_SYS_CLKRST_CRYPT0_KM_CLK_EN HP_SYS_CLKRST_CRYPT0_ECDSA_CLK_EN HP_SYS_CLKRST_CRYPT0_SHA_CLK_EN HP_SYS_CLKRST_CRYPT0_SEC_CLK_EN HP_SYS_CLKRST_CRYPT0_RSA_CLK_EN HP_SYS_CLKRST_CRYPT0_HMAC_CLK_EN HP_SYS_CLKRST_CRYPT0_ECC_CLK_EN HP_SYS_CLKRST_CRYPT0_DSA_CLK_EN HP_SYS_CLKRST_CRYPT0_AES_CLK_EN HP_SYS_CLKRST_CRYPT0_CLK_SRC_SEL														(reserved)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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Reset

HP_SYS_CLKRST_CRYPT0_CLK_SRC_SEL Configures the clock source for CRYPTO_CLK.

- 0: XTAL_CLK
 - 1: RC_FAST_CLK
 - 2: PLL_F240M_CLK
 - 3: PLL_F160M_CLK
- (R/W)

HP_SYS_CLKRST_CRYPT0_AES_CLK_EN Configures whether to enable AES_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_CRYPT0_DSA_CLK_EN Configures whether to enable the RSA_DS clock.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_CRYPT0_ECC_CLK_EN Configures whether to enable ECC_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_CRYPT0_HMAC_CLK_EN Configures whether to enable HMAC_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

HP_SYS_CLKRST_CRYPT0_RSA_CLK_EN Configures whether to enable RSA_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

Continued on the next page...

Register 10.43. HP_SYS_CLKRST_PERI_CLK_CTRL25_REG (0x00A8)

Continued from the previous page...

HP_SYS_CLKRST_CRYPT0_SEC_CLK_EN Configures whether to enable SEC_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_CRYPT0_SHA_CLK_EN Configures whether to enable SHA_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_CRYPT0_ECDSA_CLK_EN Configures whether to enable the ECDSA_DS clock (ECDSA_DS_CLK).

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_CRYPT0_KM_CLK_EN Configures whether to enable KM_CLK.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_ISP_CLK_SRC_SEL Configures the clock source for ISP_CLK.

0: XTAL_CLK

1: PLL_F240M_CLK

2: PLL_F160M_CLK

3: Invalid

(R/W)

HP_SYS_CLKRST_ISP_CLK_EN Configures whether to enable ISP_CLK.

0: Disable

1: Enable

(R/W)

Register 10.44. HP_SYS_CLKRST_PERI_CLK_CTRL26_REG (0x00AC)

(reserved)				HP_SYS_CLKRST_H264_CLK_DIV_NUM				HP_SYS_CLKRST_H264_CLK_EN				HP_SYS_CLKRST_H264_CLK_SRC_SEL				HP_SYS_CLKRST_IOMUX_CLK_DIV_NUM				HP_SYS_CLKRST_IOMUX_CLK_EN				HP_SYS_CLKRST_IOMUX_CLK_SRC_SEL				HP_SYS_CLKRST_ISP_CLK_DIV_NUM			
31	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	Reset		

Reset

HP_SYS_CLKRST_ISP_CLK_DIV_NUM Configures the clock divisor of ISP_CLK. (R/W)

HP_SYS_CLKRST_IOMUX_CLK_SRC_SEL Configures the clock source for IOMUX_CLK.

0: XTAL_CLK

1: PLL_F80M_CLK

(R/W)

HP_SYS_CLKRST_IOMUX_CLK_EN Configures whether to enable the IOMUX_CLK clock.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_IOMUX_CLK_DIV_NUM Configures the clock divisor of IOMUX_CLK. (R/W)

HP_SYS_CLKRST_H264_CLK_SRC_SEL Configures the clock source for H264_CLK.

0: XTAL_CLK

1: PLL_F240M_CLK

(R/W)

HP_SYS_CLKRST_H264_CLK_EN Configures whether to enable the H264_CLK clock.

0: Disable

1: Enable

(R/W)

HP_SYS_CLKRST_H264_CLK_DIV_NUM Configures the clock divisor of H264_CLK. (R/W)

Register 10.45. HP_SYS_CLKRST_CLK_FORCE_ON_CTRL0_REG (0x00B4)

(reserved)																		HP_SYS_CLKRST_GMAC_TX_CLK_FORCE_ON HP_SYS_CLKRST_SAR2_CLK_FORCE_ON HP_SYS_CLKRST_SAR1_CLK_FORCE_ON HP_SYS_CLKRST_L2MEM_MEM_CLK_FORCE_ON HP_SYS_CLKRST_L2CACHE_MEM_CLK_FORCE_ON HP_SYS_CLKRST_L1CACHE_I1_MEM_CLK_FORCE_ON HP_SYS_CLKRST_L1CACHE_I0_MEM_CLK_FORCE_ON HP_SYS_CLKRST_L1CACHE_D_MEM_CLK_FORCE_ON HP_SYS_CLKRST_TRACE_SYS_CLK_FORCE_ON HP_SYS_CLKRST_TRACE_CPU_CLK_FORCE_ON HP_SYS_CLKRST_L1CACHE_I1_CPU_CLK_FORCE_ON HP_SYS_CLKRST_L1CACHE_I0_CPU_CLK_FORCE_ON HP_SYS_CLKRST_L1CACHE_D_CPU_CLK_FORCE_ON HP_SYS_CLKRST_BUSMON_CPU_CLK_FORCE_ON HP_SYS_CLKRST_SPM_CPU_CLK_FORCE_ON HP_SYS_CLKRST_CPUICM_GATED_CLK_FORCE_ON																			
31																		18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	Reset				

HP_SYS_CLKRST_CPUICM_GATED_CLK_FORCE_ON Configures the CPUICM_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_SPM_CPU_CLK_FORCE_ON Configures the SPM_CPU_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_BUSMON_CPU_CLK_FORCE_ON Configures the BUSMON_CPU_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_L1CACHE_CPU_CLK_FORCE_ON Configures the L1CACHE_CPU_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_L1CACHE_D_CPU_CLK_FORCE_ON Configures the L1DCACHE_CPU_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

Continued on the next page...

Register 10.45. HP_SYS_CLKRST_CLK_FORCE_ON_CTRL0_REG (0x00B4)

Continued from the previous page...

HP_SYS_CLKRST_L1CACHE_IO_CPU_CLK_FORCE_ON Configures the L1I0CACHE_CPU_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_L1CACHE_I1_CPU_CLK_FORCE_ON Configures the L1I1CACHE_CPU_CLK clock gating path.

(R/W)

HP_SYS_CLKRST_TRACE_CPU_CLK_FORCE_ON Configures the TRACE_CPU_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_TRACE_SYS_CLK_FORCE_ON Configures the TRACE_SYS_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_L1CACHE_MEM_CLK_FORCE_ON Configures the L1CACHE_MEM_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_L1CACHE_D_MEM_CLK_FORCE_ON Configures the L1DCACHE_MEM_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_L1CACHE_IO_MEM_CLK_FORCE_ON Configures the L1I0CACHE_MEM_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_L1CACHE_I1_MEM_CLK_FORCE_ON Configures the L1I1CACHE_MEM_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

Continued on the next page...

Register 10.45. HP_SYS_CLKRST_CLK_FORCE_ON_CTRL0_REG (0x00B4)

Continued from the previous page...

HP_SYS_CLKRST_L2CACHE_MEM_CLK_FORCE_ON Configures the L2CACHE_MEM_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_L2MEM_MEM_CLK_FORCE_ON Configures the L2MEM_MEM_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_SAR1_CLK_FORCE_ON Configures the SAR1_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

HP_SYS_CLKRST_SAR2_CLK_FORCE_ON Configures the SAR2_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

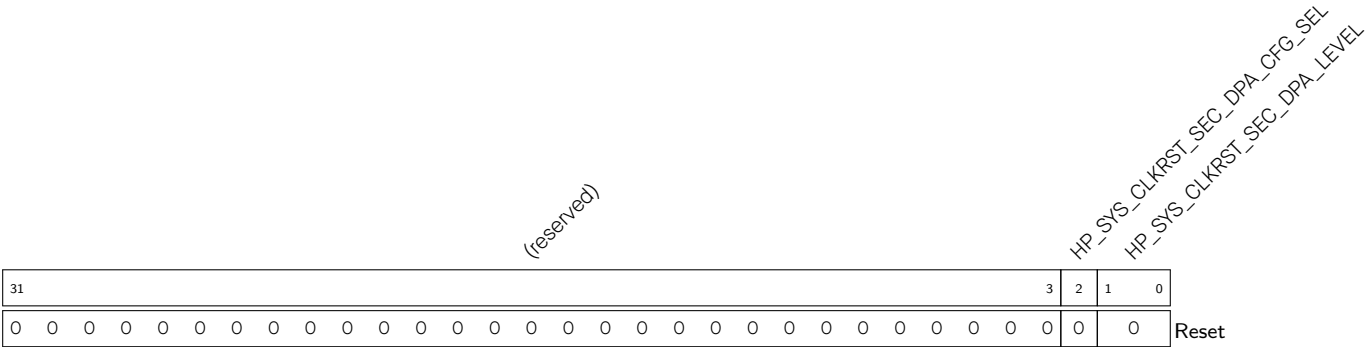
HP_SYS_CLKRST_GMAC_TX_CLK_FORCE_ON Configures the GMAC_TX_CLK clock gating path.

0: Clock gating is enabled

1: Force on the clock, bypassing the clock gating

(R/W)

Register 10.46. HP_SYS_CLKRST_DPA_CTRL0_REG (0x00B8)



HP_SYS_CLKRST_SEC_DPA_LEVEL Configures the security level of CRYPTO_CLK against DPA attacks.

0: Low security level

1: High security level

(R/W)

HP_SYS_CLKRST_SEC_DPA_CFG_SEL Configures the source of the security level of CRYPTO_CLK against DPA attacks.

0: The security level is determined by EFUSE_SEC_DPA_LEVEL

1: The security level is determined by HP_SYS_CLKRST_SEC_DPA_LEVEL

(R/W)

Register 10.47. HP_SYS_CLKRST_ANA_PLL_CTRL0_REG (0x00BC)

(reserved)																						HP_SYS_CLKRST_MSPI_CAL_STOP HP_SYS_CLKRST_MSPI_CAL_END HP_SYS_CLKRST_SYS_PLL_CAL_STOP HP_SYS_CLKRST_SYS_PLL_CAL_END HP_SYS_CLKRST_SDIO_PLL_CAL_STOP HP_SYS_CLKRST_SDIO_PLL_CAL_END HP_SYS_CLKRST_CPU_PLL_CAL_STOP HP_SYS_CLKRST_CPU_PLL_CAL_END HP_SYS_CLKRST_PLLA_CAL_STOP HP_SYS_CLKRST_PLLA_CAL_END																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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HP_SYS_CLKRST_PLLA_CAL_END Represents whether the APLL_CLK calibration is finished.

0: Calibration in progress

1: Calibration done

(RO)

HP_SYS_CLKRST_PLLA_CAL_STOP Configures whether to start or stop the APLL_CLK calibration.

0: Start calibration

1: Stop calibration

(R/W)

HP_SYS_CLKRST_CPU_PLL_CAL_END Represents whether the CPLL_CLK calibration is finished.

0: Calibration in progress

1: Calibration done

(RO)

HP_SYS_CLKRST_CPU_PLL_CAL_STOP Configures whether to start or stop the CPLL_CLK calibration.

0: Start calibration

1: Stop calibration

(R/W)

HP_SYS_CLKRST_SDIO_PLL_CAL_END Represents whether the SDIO_PLL calibration is finished.

0: Calibration in progress

1: Calibration done

(RO)

HP_SYS_CLKRST_SDIO_PLL_CAL_STOP Configures whether to start or stop the SDIO_PLL calibration.

0: Start calibration

1: Stop calibration

(R/W)

Continued on the next page...

Register 10.47. HP_SYS_CLKRST_ANA_PLL_CTRL0_REG (0x00BC)

Continued from the previous page...

HP_SYS_CLKRST_SYS_PLL_CAL_END Represents whether the SPLL_CLK calibration is finished.

0: Calibration in progress

1: Calibration done

(RO)

HP_SYS_CLKRST_SYS_PLL_CAL_STOP Configures whether to start or stop the SPLL_CLK calibration.

0: Start calibration

1: Stop calibration

(R/W)

HP_SYS_CLKRST_MSPI_CAL_END Represents whether the MPLL_CLK calibration is finished.

0: Calibration in progress

1: Calibration done

(RO)

HP_SYS_CLKRST_MSPI_CAL_STOP Configures whether to start or stop the MPLL_CLK calibration.

0: Start calibration

1: Stop calibration

(R/W)

Register 10.48. HP_SYS_CLKRST_HP_RST_EN0_REG (0x00C0)

HP_SYS_CLKRST_RST_EN_DMA2D																															HP_SYS_CLKRST_RST_EN_JPEG																															(reserved)																															HP_SYS_CLKRST_RST_EN_CSI_BRG																															HP_SYS_CLKRST_RST_EN_CSI_HOST																															HP_SYS_CLKRST_RST_EN_DSI_BRG																															HP_SYS_CLKRST_RST_EN_DUAL_MSPI_APB																															HP_SYS_CLKRST_RST_EN_DUAL_MSPI_AXI																															HP_SYS_CLKRST_RST_EN_GDMA																															HP_SYS_CLKRST_RST_EN_SPMON																															HP_SYS_CLKRST_RST_EN_L2MEMMON																															HP_SYS_CLKRST_RST_EN_L2_MEM																															HP_SYS_CLKRST_RST_EN_L1_D_CACHE																															HP_SYS_CLKRST_RST_EN_L1_I1_CACHE																															HP_SYS_CLKRST_RST_EN_L1_IO_CACHE																															HP_SYS_CLKRST_RST_EN_HP_CACHE																															HP_SYS_CLKRST_RST_EN_HP_SPM																															(reserved)																															HP_SYS_CLKRST_RST_EN_CORETRACE1																															HP_SYS_CLKRST_RST_EN_CORE1_GLOBAL																															HP_SYS_CLKRST_RST_EN_CORE0_GLOBAL																														
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6											1	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
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HP_SYS_CLKRST_RST_EN_CORECTRL Configures whether to reset CORECTRL.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_CORE0_GLOBAL Configures whether to reset HP CPU0.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_CORE1_GLOBAL Configures whether to reset HP CPU1.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_CORETRACE0 Configures whether to reset RISC-V Trace Encoder0.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_CORETRACE1 Configures whether to reset RISC-V Trace Encoder1.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_HP_SPM Configures whether to reset SPM.

0: Release from reset

1: Reset

(R/W)

Continued on the next page...

Register 10.48. HP_SYS_CLKRST_HP_RST_ENO_REG (0x00C0)

Continued from the previous page...

HP_SYS_CLKRST_RST_EN_HP_CACHE Configures whether to reset L1 I-Cache0, L1 I-Cache1, L1 D-Cache, L2 Cache.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_L1_IO_CACHE Configures whether to reset L1 I-Cache0.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_L1_I1_CACHE Configures whether to reset L1 I-Cache1.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_L1_D_CACHE Configures whether to reset L1 D-Cache.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_L2_CACHE Configures whether to reset L2 Cache.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_L2_MEM Configures whether to reset L2 MEM.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_L2MEMMON Configures whether to reset L2 MEM monitor.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_SPMON Configures whether to reset SPM monitor.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_GDMA Configures whether to reset VDMA.

0: Release from reset

1: Reset

(R/W)

Continued on the next page...

Register 10.48. HP_SYS_CLKRST_HP_RST_ENO_REG (0x00C0)

Continued from the previous page...

HP_SYS_CLKRST_RST_EN_MSPI_AXI Configures whether to reset FLASH MSPI AXI.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_DUAL_MSPI_AXI Configures whether to reset PSRAM MSPI AXI.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_MSPI_APB Configures whether to reset FLASH MSPI APB.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_DUAL_MSPI_APB Configures whether to reset PSRAM MSPI APB.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_DSI_BRG Configures whether to reset DSI Bridge.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_CSI_HOST Configures whether to reset CSI Host.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_CSI_BRG Configures whether to reset CSI Bridge and ISP.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_JPEG Configures whether to reset JPEG.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_DMA2D Configures whether to reset 2DDMA.

0: Release from reset

1: Reset

(R/W)

Register 10.49. HP_SYS_CLKRST_HP_RST_EN1_REG (0x00C4)

HP_SYS_CLKRST_RST_EN_ETM																																HP_SYS_CLKRST_RST_EN_PONT																																HP_SYS_CLKRST_RST_EN_LEDC																																HP_SYS_CLKRST_RST_EN_CAN2																																HP_SYS_CLKRST_RST_EN_CAN1																																HP_SYS_CLKRST_RST_EN_CAN0																																HP_SYS_CLKRST_RST_EN_PWM1																																HP_SYS_CLKRST_RST_EN_PWM0																																HP_SYS_CLKRST_RST_EN_RMT																																HP_SYS_CLKRST_RST_EN_I2C0																																HP_SYS_CLKRST_RST_EN_I2C1																																HP_SYS_CLKRST_RST_EN_I3CSLV																																HP_SYS_CLKRST_RST_EN_UHCI																																HP_SYS_CLKRST_RST_EN_UART4_APB																																HP_SYS_CLKRST_RST_EN_UART3_APB																																HP_SYS_CLKRST_RST_EN_UART2_APB																																HP_SYS_CLKRST_RST_EN_UART1_APB																																HP_SYS_CLKRST_RST_EN_UART0_APB																																HP_SYS_CLKRST_RST_EN_UART3_CORE																																HP_SYS_CLKRST_RST_EN_UART2_CORE																																HP_SYS_CLKRST_RST_EN_UART1_CORE																																HP_SYS_CLKRST_RST_EN_UART0_CORE																																(reserved)																																HP_SYS_CLKRST_RST_EN_TIMERGRP1																																HP_SYS_CLKRST_RST_EN_TIMERGRP0																																HP_SYS_CLKRST_RST_EN_IOMUX																																HP_SYS_CLKRST_RST_EN_AXI_PDMA																																HP_SYS_CLKRST_RST_EN_AHB_PDMA																																HP_SYS_CLKRST_RST_EN_PPA																																																																																																																																																								
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

HP_SYS_CLKRST_RST_EN_PPA Configures whether to reset PPA.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_AHB_PDMA Configures whether to reset GDMA-AHB.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_AXI_PDMA Configures whether to reset GDMA-AXI.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_IOMUX Configures whether to reset IO MUX.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_STIMER Configures whether to reset the System Timer.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_TIMERGRP0 Configures whether to reset Timer Group 0.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_TIMERGRP1 Configures whether to reset Timer Group 1.

0: Release from reset

1: Reset

(R/W)

Continued on the next page...

Register 10.49. HP_SYS_CLKRST_HP_RST_EN1_REG (0x00C4)

Continued from the previous page...

HP_SYS_CLKRST_RST_EN_UART0_CORE Configures whether to reset UART0.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_UART1_CORE Configures whether to reset UART1.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_UART2_CORE Configures whether to reset UART2.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_UART3_CORE Configures whether to reset UART3.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_UART4_CORE Configures whether to reset UART4.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_UART0_APB Configures whether to reset UART0 APB.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_UART1_APB Configures whether to reset UART1 APB.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_UART2_APB Configures whether to reset UART2 APB.

0: Release from reset

1: Reset

(R/W)

Continued on the next page...

Register 10.49. HP_SYS_CLKRST_HP_RST_EN1_REG (0x00C4)

Continued from the previous page...

HP_SYS_CLKRST_RST_EN_UART3_APB Configures whether to reset UART3 APB.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_UART4_APB Configures whether to reset UART4 APB.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_UHCI Configures whether to reset UHCI.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_I3CMST Configures whether to reset I3C Master.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_I3CSLV Configures whether to reset I3C Slave.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_I2C1 Configures whether to reset I2C1.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_I2C0 Configures whether to reset I2C0.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_RMT Configures whether to reset RMT.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_PWM0 Configures whether to reset MCPMWO.

0: Release from reset

1: Reset

(R/W)

Continued on the next page...

Register 10.49. HP_SYS_CLKRST_HP_RST_EN1_REG (0x00C4)

Continued from the previous page...

HP_SYS_CLKRST_RST_EN_PWM1 Configures whether to reset MCPWM1.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_CAN0 Configures whether to reset TWAIO.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_CAN1 Configures whether to reset TWAI.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_CAN2 Configures whether to reset TWAI.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_LEDC Configures whether to reset LEDC.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_PCNT Configures whether to reset PCNT.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_ETM Configures whether to reset ETM.

0: Release from reset

1: Reset

(R/W)

Register 10.50. HP_SYS_CLKRST_HP_RST_EN2_REG (0x00C8)

Continued from the previous page...

HP_SYS_CLKRST_RST_EN_SPI2 Configures whether to reset SPI2.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_SPI3 Configures whether to reset SPI3.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_LCDCAM Configures whether to reset the LCD_CAM.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_ADC Configures whether to reset ADC.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_BITSRAMBLER Configures whether to reset the Bit Scrambler.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_BITSRAMBLER_RX Configures whether to reset the Bit Scrambler RX.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_BITSRAMBLER_TX Configures whether to reset the Bit Scrambler TX.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_CRYPTO Configures whether to reset the crypto modules.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_SEC Configures whether to reset SEC.

0: Release from reset

1: Reset

(R/W)

Continued on the next page...

Register 10.50. HP_SYS_CLKRST_HP_RST_EN2_REG (0x00C8)

Continued from the previous page...

HP_SYS_CLKRST_RST_EN_AES Configures whether to reset AES.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_DS Configures whether to reset the RSA_DS peripheral.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_SHA Configures whether to reset SHA.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_HMAC Configures whether to reset HMAC.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_ECDSA Configures whether to reset ECDSA_DS.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_RSA Configures whether to reset RSA.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_ECC Configures whether to reset ECC.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_KM Configures whether to reset the Key Manager.

0: Release from reset

1: Reset

(R/W)

HP_SYS_CLKRST_RST_EN_H264 Configures whether to reset H264.

0: Release from reset

1: Reset

(R/W)

Register 10.51. HP_SYS_CLKRST_HP_FORCE_NORSTO_REG (0x00CC)

HP_SYS_CLKRST_FORCE_NORST_I3CMST	HP_SYS_CLKRST_FORCE_NORST_UHCI	HP_SYS_CLKRST_FORCE_NORST_UART4	HP_SYS_CLKRST_FORCE_NORST_UART3	HP_SYS_CLKRST_FORCE_NORST_UART2	HP_SYS_CLKRST_FORCE_NORST_UART1	(reserved)	HP_SYS_CLKRST_FORCE_NORST_TIMERGRP1	HP_SYS_CLKRST_FORCE_NORST_TIMERGRP0	HP_SYS_CLKRST_FORCE_NORST_STIMER	HP_SYS_CLKRST_FORCE_NORST_IOMUX	HP_SYS_CLKRST_FORCE_NORST_AXI_PDMA	(reserved)	HP_SYS_CLKRST_FORCE_NORST_PPA	HP_SYS_CLKRST_FORCE_NORST_DMA2D	HP_SYS_CLKRST_FORCE_NORST_JPEG	HP_SYS_CLKRST_FORCE_NORST_CSI_BRG	HP_SYS_CLKRST_FORCE_NORST_CSI_HOST	HP_SYS_CLKRST_FORCE_NORST_DSI_BRG	HP_SYS_CLKRST_FORCE_NORST_DUAL_MSPI_APB	HP_SYS_CLKRST_FORCE_NORST_DUAL_MSPI_AXI	HP_SYS_CLKRST_FORCE_NORST_GDMA	HP_SYS_CLKRST_FORCE_NORST_SPMMON	HP_SYS_CLKRST_FORCE_NORST_L2MEMMON	HP_SYS_CLKRST_FORCE_NORST_CORETRACE1	HP_SYS_CLKRST_FORCE_NORST_CORE0	Reset					
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

HP_SYS_CLKRST_FORCE_NORST_CORE0 Configures whether HP CPU0 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_CORE1 Configures whether HP CPU1 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_CORETRACE0 Configures whether RISC-V Trace Encoder 0 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_CORETRACE1 Configures whether RISC-V Trace Encoder 1 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_L2MEMMON Configures whether L2 MEM monitor can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_SPMMON Configures whether SPM monitor can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

Continued on the next page...

Register 10.51. HP_SYS_CLKRST_HP_FORCE_NORSTO_REG (0x00CC)

Continued from the previous page...

HP_SYS_CLKRST_FORCE_NORST_GDMA Configures whether VDMA can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_MSPI_AXI Configures whether FLASH MSPI can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_DUAL_MSPI_AXI Configures whether PSRAM MSPI can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_MSPI_APB Configures whether FLASH MSPI APB can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_DUAL_MSPI_APB Configures whether PSRAM MSPI APB can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_DSI_BRG Configures whether DSI Bridge can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_CSI_HOST Configures whether CSI Host can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_CSI_BRG Configures whether CSI Bridge, ISP can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

Continued on the next page...

Register 10.51. HP_SYS_CLKRST_HP_FORCE_NORSTO_REG (0x00CC)

Continued from the previous page...

HP_SYS_CLKRST_FORCE_NORST_JPEG Configures whether JPEG can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_DMA2D Configures whether 2DDMA can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_PPA Configures whether PPA can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_AHB_PDMA Configures whether GDMA-AHB can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_AXI_PDMA Configures whether GDMA-AXI can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_IOMUX Configures whether IO MUX can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_STIMER Configures whether the System Timer can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_TIMERGRPO Configures whether Timer Group 0 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_TIMERGRP1 Configures whether Timer Group 1 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

Continued on the next page...

Register 10.51. HP_SYS_CLKRST_HP_FORCE_NORSTO_REG (0x00CC)

Continued from the previous page...

HP_SYS_CLKRST_FORCE_NORST_UART0 Configures whether UART0 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_UART1 Configures whether UART1 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_UART2 Configures whether UART2 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_UART3 Configures whether UART3 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_UART4 Configures whether UART4 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_UHCI Configures whether UHCI can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_I3CMST Configures whether I3C Master can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

Register 10.52. HP_SYS_CLKRST_HP_FORCE_NORST1_REG (0x00D0)

(reserved)																														
HP_SYS_CLKRST_FORCE_NORST_H264																														
HP_SYS_CLKRST_FORCE_NORST_H264																														
HP_SYS_CLKRST_FORCE_NORST_H264																														
HP_SYS_CLKRST_FORCE_NORST_BITSRAMBLER_TX																														
HP_SYS_CLKRST_FORCE_NORST_BITSRAMBLER_RX																														
HP_SYS_CLKRST_FORCE_NORST_BITSRAMBLER																														
HP_SYS_CLKRST_FORCE_NORST_ADC																														
HP_SYS_CLKRST_FORCE_NORST_LCDCAM																														
HP_SYS_CLKRST_FORCE_NORST_SPI3																														
HP_SYS_CLKRST_FORCE_NORST_SPI2																														
HP_SYS_CLKRST_FORCE_NORST_I2S2																														
HP_SYS_CLKRST_FORCE_NORST_I2S1																														
HP_SYS_CLKRST_FORCE_NORST_I2S0																														
HP_SYS_CLKRST_FORCE_NORST_PARLIO_TX																														
HP_SYS_CLKRST_FORCE_NORST_PARLIO_RX																														
HP_SYS_CLKRST_FORCE_NORST_INTRMTX																														
HP_SYS_CLKRST_FORCE_NORST_ETM																														
HP_SYS_CLKRST_FORCE_NORST_PCNT																														
HP_SYS_CLKRST_FORCE_NORST_LEDC																														
HP_SYS_CLKRST_FORCE_NORST_CAN2																														
HP_SYS_CLKRST_FORCE_NORST_CAN1																														
HP_SYS_CLKRST_FORCE_NORST_CAN0																														
HP_SYS_CLKRST_FORCE_NORST_PWM1																														
HP_SYS_CLKRST_FORCE_NORST_PWM0																														
HP_SYS_CLKRST_FORCE_NORST_RMT																														
HP_SYS_CLKRST_FORCE_NORST_I2C0																														
HP_SYS_CLKRST_FORCE_NORST_I2C1																														
HP_SYS_CLKRST_FORCE_NORST_I3CSLV																														

31		27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

HP_SYS_CLKRST_FORCE_NORST_I3CSLV Configures whether I3C Slave can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_I2C1 Configures whether I2C1 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_I2CO Configures whether I2CO can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP SYS CLRST FORCE NORST RMT Configures whether RMT can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_PWM0 Configures whether MCPWM0 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP SYS CLRST FORCE NORST PWM1 Configures whether MCPWM1 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP SYS CLRST FORCE NORST CANO Configures whether TWAIO can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

Register 10.52. HP_SYS_CLKRST_HP_FORCE_NORST1_REG (0x00D0)

Continued from the previous page...

HP_SYS_CLKRST_FORCE_NORST_CAN1 Configures whether TWAI1 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_CAN2 Configures whether TWAI2 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_LEDC Configures whether LEDC can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_PCNT Configures whether PCNT can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_ETM Configures whether ETM can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_INTRMTX Configures whether the interrupt matrix can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_PARLIO Configures whether PARLIO can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_PARLIO_RX Configures whether PARLIO RX can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_PARLIO_TX Configures whether PARLIO TX can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

Continued on the next page...

Register 10.52. HP_SYS_CLKRST_HP_FORCE_NORST1_REG (0x00D0)

Continued from the previous page...

HP_SYS_CLKRST_FORCE_NORST_I2S0 Configures whether I2S0 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_I2S1 Configures whether I2S1 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_I2S2 Configures whether I2S2 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_SPI2 Configures whether SPI2 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_SPI3 Configures whether SPI3 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_LCDCAM Configures whether the LCD_CAM can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_ADC Configures whether ADC can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

Continued on the next page...

Register 10.52. HP_SYS_CLKRST_HP_FORCE_NORST1_REG (0x00D0)

Continued from the previous page...

HP_SYS_CLKRST_FORCE_NORST_BITSRAMBLER Configures whether the Bit Scrambler can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_BITSRAMBLER_RX Configures whether the Bit Scrambler RX can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

HP_SYS_CLKRST_FORCE_NORST_BITSRAMBLER_TX Configures whether the Bit Scrambler TX can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

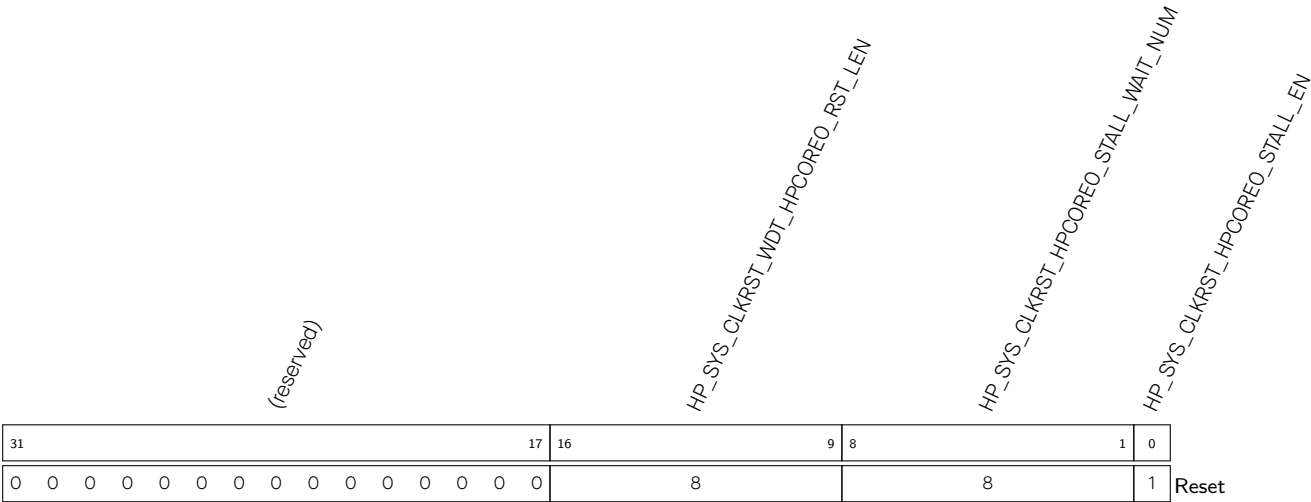
HP_SYS_CLKRST_FORCE_NORST_H264 Configures whether H264 can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

Register 10.53. HP_SYS_CLKRST_HPWDT_COREO_RST_CTRL0_REG (0x00D4)



HP_SYS_CLKRST_HPCOREO_STALL_EN Configures the behavior when MWDT triggers HP CPU0 reset.

0: Immediately reset HP CPU0

1: Stall HP CPU0 first

(R/W)

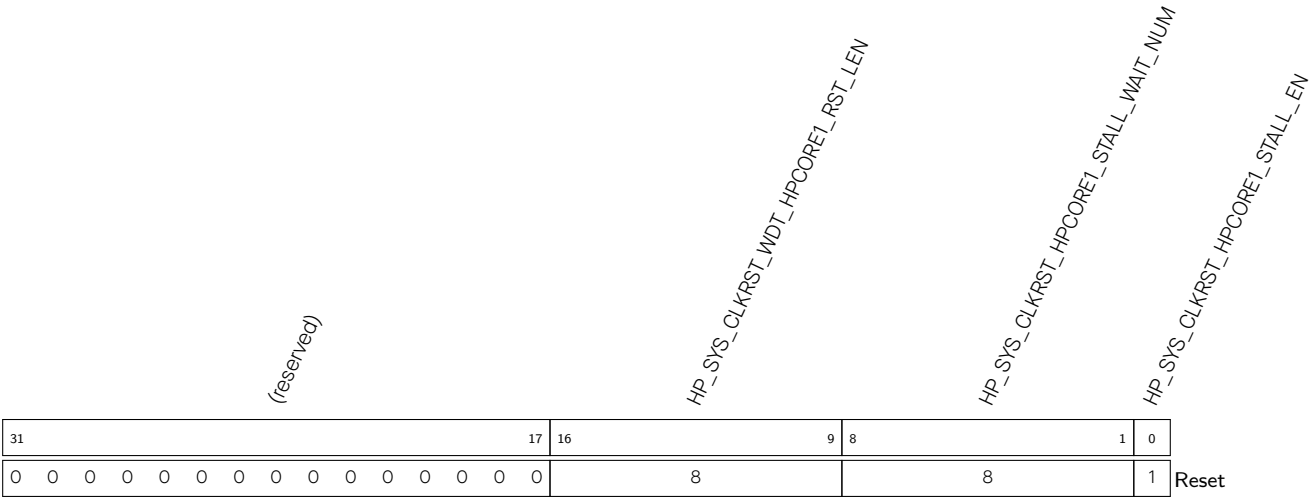
HP_SYS_CLKRST_HPCOREO_STALL_WAIT_NUM Configures the duration of HP CPU0 stall when MWDT triggers HP CPU0 reset and HP_SYS_CLKRST_HPCOREO_STALL_EN is set to 1.

Measurement unit: Clock cycles. (R/W)

HP_SYS_CLKRST_WDT_HPCOREO_RST_LEN Configures the duration of the reset when MWDT triggers HP CPU0 reset.

Measurement unit: Clock cycles. (R/W)

Register 10.54. HP_SYS_CLKRST_HPWDT_CORE1_RST_CTRL0_REG (0x00D8)



HP_SYS_CLKRST_HPCORE1_STALL_EN Configures the behavior when MWDT triggers HP CPU1 reset.

- 0: Immediately reset HP CPU1
- 1: Stall HP CPU1 first

(R/W)

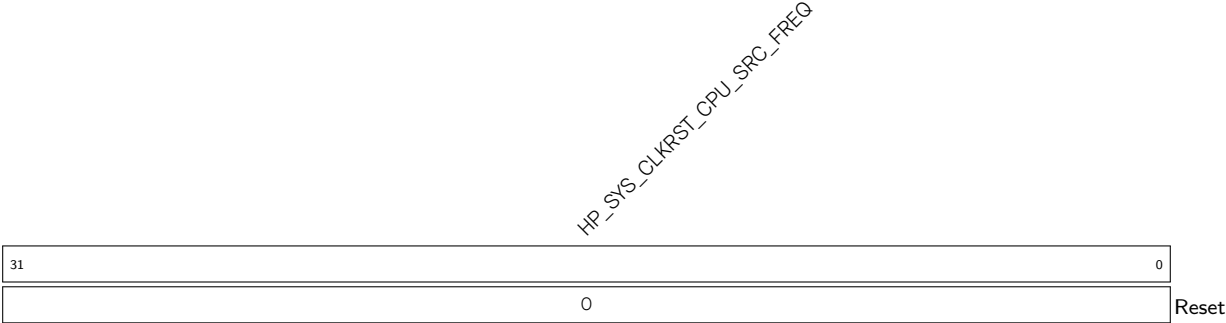
HP_SYS_CLKRST_HPCORE1_STALL_WAIT_NUM Configures the duration of HP CPU1 stall when MWDT triggers HP CPU1 reset and HP_SYS_CLKRST_HPCORE1_STALL_EN is set to 1.

Measurement unit: Clock cycles. (R/W)

HP_SYS_CLKRST_WDT_HPCORE1_RST_LEN Configures the duration of the reset when MWDT triggers HP CPU1 reset.

Measurement unit: Clock cycles. (R/W)

Register 10.55. HP_SYS_CLKRST_CPU_SRC_FREQ0_REG (0x00DC)

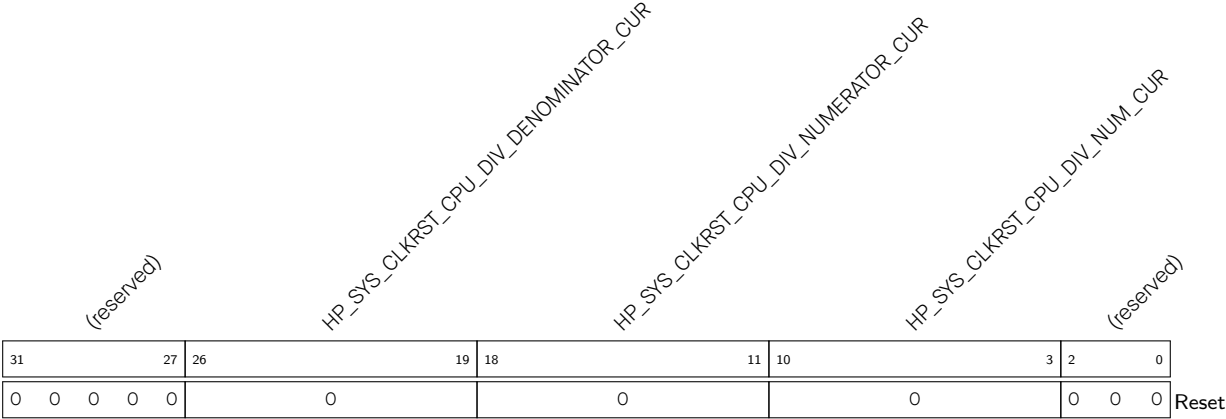


HP_SYS_CLKRST_CPU_SRC_FREQ Represents the CPU source clock frequency.

- 80: RC_FAST_CLK
- 160: XTAL_CLK
- 1600: CPLL_CLK

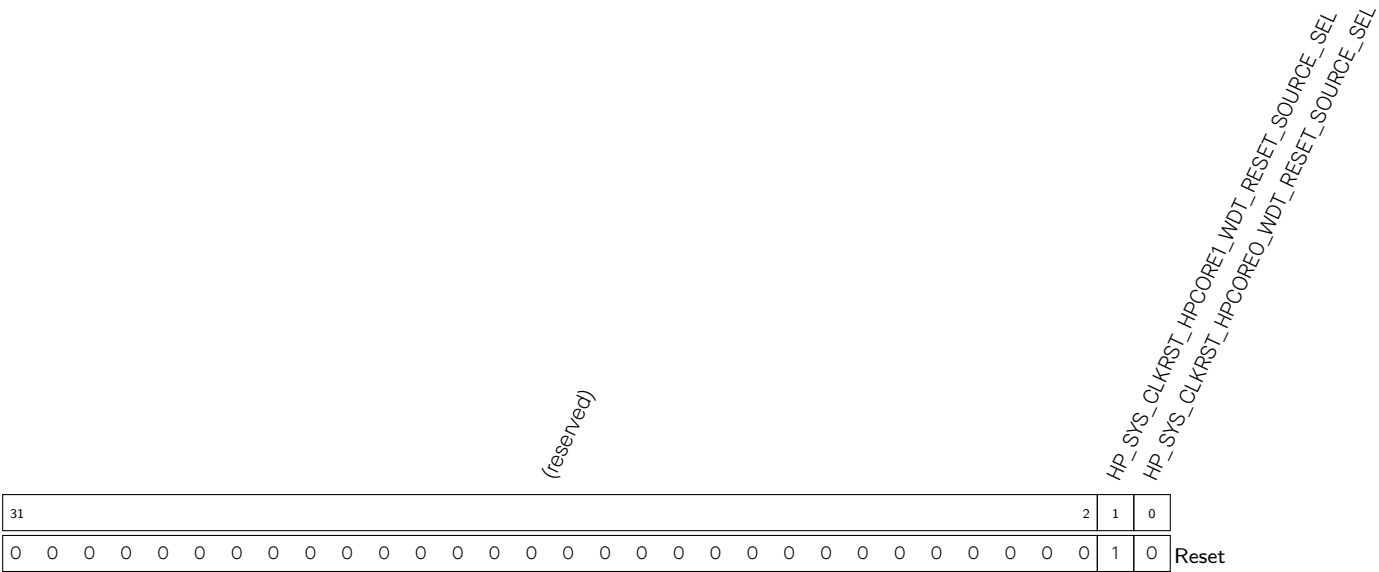
(RO)

Register 10.56. HP_SYS_CLKRST_CPU_CLK_STATUS0_REG (0x00E0)



- HP_SYS_CLKRST_CPU_DIV_NUM_CUR Represents the integer part of the current CPU_CLK clock divisor. (RO)
- HP_SYS_CLKRST_CPU_DIV_NUMERATOR_CUR Represents the current numerator of the divisor's fractional part for the CPU_CLK. (RO)
- HP_SYS_CLKRST_CPU_DIV_DENOMINATOR_CUR Represents the current denominator of the divisor's fractional part for the CPU_CLK. (RO)

Register 10.57. HP_SYS_CLKRST_HPCORE_WDT_RESET_SOURCEO_REG (0x00EC)



HP_SYS_CLKRST_HPCORE0_WDT_RESET_SOURCE_SEL Configures the MWDT used to trigger the HP CPU0 reset.

0: MWDTO

1: MWDT1

(R/W)

HP_SYS_CLKRST_HPCORE1_WDT_RESET_SOURCE_SEL Configures the MWDT used to trigger the HP CPU1 reset.

0: MWDTO

1: MWDT1

(R/W)

Register 10.58. HP_SYS_CLKRST_CPU_WAITI_CTRL0_REG (0x00F4)

(reserved)																															HP_SYS_CLKRST_REG_CORE1_WAITI_ICG_EN HP_SYS_CLKRST_REG_CORE0_WAITI_ICG_EN		
31																															2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	Reset

HP_SYS_CLKRST_REG_CORE0_WAITI_ICG_EN Configures whether CPU Core0’s WAITI signal is used to control the clock gate. 0: Do not use the WAITI signal to control the clock gate. 1: Use the WAITI signal to control the clock gate. If this bit is set for both Core0 and Core1, the related clock is gated only when both Core0’s and Core1’s WAITI signals are active. (R/W)

HP_SYS_CLKRST_REG_CORE1_WAITI_ICG_EN Configures whether CPU Core1’s WAITI signal is used to control the clock gate. 0: Do not use the WAITI signal to control the clock gate. 1: Use the WAITI signal to control the clock gate. If this bit is set for both Core0 and Core1, the related clock is gated only when both Core0’s and Core1’s WAITI signals are active. (R/W)

10.5.2 LP Always on Clock and Reset (LP_CLKRST) Registers

The addresses in this section are relative to LP_CLKRST base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.
For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

Register 10.59. LP_CLKRST_LP_CLK_CONF_REG (0x0000)

(reserved)																				LP_CLKRST_ANA_SEL_REF_PLL8M												LP_CLKRST_LP_PERI_DIV_NUM												LP_CLKRST_FAST_CLK_SEL												LP_CLKRST_SLOW_CLK_SEL											
31																				11	10	9					4	3	2	1	0																																				
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																				0	0x0								0x1	0x0	Reset																																				

LP_CLKRST_SLOW_CLK_SEL Configures the clock source of LP_SLOW_CLK.

- 0: RC_SLOW_CLK
 - 1: XTAL32K_CLK
 - 2: Invalid
 - 3: OSC_SLOW_CLK
- (R/W)

LP_CLKRST_FAST_CLK_SEL Configures the clock source of LP_FAST_CLK.

- 0: RC_FAST_CLK
 - 1: XTAL_CLK
 - 2: PLL_LP_CLK
 - 3: Invalid
- (R/W)

LP_CLKRST_LP_PERI_DIV_NUM Configures the clock divisor for LP_PERI_CLK. (R/W)

LP_CLKRST_ANA_SEL_REF_PLL8M Configures the reference clock of PLL_LP_CLK.

- 0: Invalid
 - 1: XTAL32K_CLK
- (R/W)

Register 10.60. LP_CLKRST_LP_CLK_EN_REG (0x0008)

Diagram of the 32-bit Reset Reason Register (RSTRST). The register is divided into two main sections:

- Reset Reason (Bits 31-25):**
 - 31: LP_CLKRST_FOSC_CLK_FORCE_ON
 - 30: LP_CLKRST_XTAL_CLK_FORCE_ON
 - 29: LP_CLKRST_PLL0M_CLK_FORCE_ON
 - 28: LP_CLKRST_ETM_EVENT_TICK_EN
 - 27: LP_CLKRST_CK_EN_LP_RAM
 - 26: LP_CLKRST_LP_RTC_XTAL_FORCE_ON
 - 25: (reserved)
- Reserved (Bits 24-0):** All bits are 0.

A 'Reset' signal is shown as an arrow pointing to the register.

LP_CLKRST_LP_RTC_XTAL_FORCE_ON Configures whether to force on the XTAL_CLK control of LP CALL.

0: Controlled by hardware

1: Force on, bypassing hardware control

(R/W)

LP_CLKRST_CK_EN_LP_RAM Configure whether to enable the clock of LP RAM.

0: Disable

1: Enable

(R/W)

LP_CLKRST_ETM_EVENT_TICK_EN Configures whether to enable the ETM event timer.

0: Disable

1: Enable

(R/W)

LP_CLKRST_PLL8M_CLK_FORCE_ON Configures whether to force on PLL_LP_CLK as LP_FAST_CLK source.

0: Controlled by hardware

1: Force on, bypassing hardware control

(R/W)

LP_CLKRST_XTAL_CLK_FORCE_ON Configures whether to force on XTAL_CLK as LP_FAST_CLK source.

0: Controlled by hardware

1: Force on, bypassing hardware control

(R/W)

LP_CLKRST_FOSC_CLK_FORCE_ON Configures whether to force on RC_FAST_CLK as LP FAST CLK source.

0: Controlled by hardware

1: Force on, bypassing hardware control

(R/W)

Register 10.61. LP_CLKRST_LP_RST_EN_REG (0x000C)

Diagram illustrating the Reset Vector field structure (bits 31 to 0). The field is divided into two main sections:

- Reset Vector (bits 31 to 24):** This section contains 8 groups of 3 bits each, labeled as follows:
 - LP_CLKRST_RST_EN_LP_RAM
 - LP_CLKRST_RST_EN_LP_AONEFUSEREG
 - LP_CLKRST_RST_EN_LP_MAILBOX
 - LP_CLKRST_RST_EN_LP_RTC
 - LP_CLKRST_RST_EN_LP_TIMER
 - LP_CLKRST_RST_EN_LP_WDT
 - LP_CLKRST_RST_EN_LP_ANAPERI
 - LP_CLKRST_RST_EN_LP_HUK
- Reset Mode (bits 23 to 0):** This section is labeled (reserved).

LP CLKRST RST EN LP HUK Configures whether to reset LP HUK.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_RST_EN_LP_ANAPERI Configures whether to reset LP analog peripherals.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_RST_EN_LP_WDT Configures whether to reset RWDT.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_RST_EN_LP_TIMER Configures whether to reset LP timer.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_RST_EN_LP_RTC Configures whether to reset the LP clock calibration module.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_RST_EN_LP_MAILBOX Configures whether to reset LP Mailbox.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_RST_EN_LP_AONEFUSEREG Configures whether to reset LP eFuse.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_RST_EN_LP_RAM Configures whether to reset LP RAM.

0: Release from reset

1: Reset

(R/W)

Register 10.62. LP_CLKRST_RESET_CAUSE_REG (0x0010)

LP_CLKRST_HPCORE1_RESET_FLAG_CLR																															LP_CLKRST_HPCORE1_RESET_CAUSE_CLR																															LP_CLKRST_HPCORE0_RESET_FLAG_CLR																															LP_CLKRST_HPCORE0_RESET_CAUSE_CLR																															LP_CLKRST_LPCORE_RESET_FLAG_CLR																															LP_CLKRST_LPCORE_RESET_CAUSE_CLR																															(reserved)																															LP_CLKRST_HPCORE1_RESET_FLAG																															LP_CLKRST_HPCORE1_RESET_CAUSE																															LP_CLKRST_HPCORE0_RESET_FLAG																															LP_CLKRST_HPCORE0_RESET_CAUSE																															LP_CLKRST_LPCORE_RESET_FLAG																															LP_CLKRST_LPCORE_RESET_CAUSE																														
31	30	29	28	27	26	25	24					21	20	19					14	13	12					7	6	5					0																																																																																																																																																																																																																																																																																																																																																																																	
0	0	0	0	0	0	1	0	0	0	0	0		0x0				0		0x0				0		0x0				0		0x0				Reset																																																																																																																																																																																																																																																																																																																																																																															

Reset

LP_CLKRST_LPCORE_RESET_CAUSE Represents the reset source of LP CPU.

- 0x01: Chip reset
- 0x09: PMU LP Peripheral reset
- 0x0A: PMU LP CPU reset
- 0x0F: Brown-out system reset
- 0x10: RWDT system reset
- 0x12: Super watchdog reset
- 0x13: Power glitch reset
- 0x14: Software LP CPU reset
- (RO)

LP_CLKRST_LPCORE_RESET_FLAG Represents the reset flag of LP CPU.

- 0: Not reset
- 1: Reset
- (RO)

Continued on the next page...

Register 10.62. LP_CLKRST_RESET_CAUSE_REG (0x0010)

Continued from the previous page...

LP_CLKRST_HPCORE0_RESET_CAUSE Represents the reset source of HP CPU0.

0x01: Chip reset
0x03: Software system reset
0x05: PMU core reset
0x07: MWDT core reset
0x09: RWDT core reset
0x0B: MWDT CPU reset
0x0C: Software CPU reset
0x0D: RWDT CPU reset
0x0F: Brown-out system reset
0x10: RWDT system reset
0x12: Super watchdog reset
0x13: Power glitch reset
0x14: eFuse reset
0x16: USB (JTAG) reset
0x17: USB (UART) reset
0x18: JTAG CPU reset
0x1A: Lockup reset
0x1B: PSDET reset
(RO)

LP_CLKRST_HPCORE0_RESET_FLAG Represents the reset flag of HP CPU0.

0: Not reset
1: Reset
(RO)

Continued on the next page...

Register 10.62. LP_CLKRST_RESET_CAUSE_REG (0x0010)

Continued from the previous page...

LP_CLKRST_HPCORE1_RESET_CAUSE Represents the reset source of HP CPU1.

0x01: Chip reset
0x03: Software system reset
0x05: PMU core reset
0x07: MWDTCPU core reset
0x09: RWDTCPU core reset
0x0B: MWDTCPU reset
0x0C: Software CPU reset
0x0D: RWDTCPU reset
0x0F: Brown-out system reset
0x10: RWDTCPU system reset
0x12: Super watchdog reset
0x13: Power glitch reset
0x14: eFuse reset
0x16: USB (JTAG) reset
0x17: USB (UART) reset
0x18: JTAG CPU reset
0x1A: Lockup reset
(RO)

LP_CLKRST_HPCORE1_RESET_FLAG Represents the reset flag of HP CPU1.

0: Not reset
1: Reset
(RO)

LP_CLKRST_LPCORE_RESET_CAUSE_PMU_LP_CPU_MASK Configures whether to mask the reset source code 0x0A of PMU LP CPU reset.

0: Not mask
1: Mask
(R/W)

Continued on the next page...

Register 10.62. LP_CLKRST_RESET_CAUSE_REG (0x0010)

Continued from the previous page...

LP_CLKRST_LPCORE_RESET_CAUSE_CLR Write 1 to clear LP_CLKRST_LPCORE_RESET_CAUSE.
(WT)

LP_CLKRST_LPCORE_RESET_FLAG_CLR Write 1 to clear LP_CLKRST_LPCORE_RESET_FLAG. (WT)

LP_CLKRST_HPCORE0_RESET_CAUSE_CLR Write 1 to clear LP_CLKRST_HPCORE0_RESET_CAUSE.
(WT)

LP_CLKRST_HPCORE0_RESET_FLAG_CLR Write 1 to clear LP_CLKRST_HPCORE0_RESET_FLAG.
(WT)

LP_CLKRST_HPCORE1_RESET_CAUSE_CLR Write 1 to clear LP_CLKRST_HPCORE1_RESET_CAUSE.
(WT)

LP_CLKRST_HPCORE1_RESET_FLAG_CLR Write 1 to clear LP_CLKRST_HPCORE1_RESET_FLAG.
(WT)

Register 10.63. LP_CLKRST_HPCPU_RESET_CTRL0_REG (0x0014)

LP_CLKRST_HPCORE1_STAT_VECTOR_SEL				LP_CLKRST_HPCORE1_OCD_HALT_ON_RESET				LP_CLKRST_HPCORE1_SW_RESET				LP_CLKRST_HPCORE1_STALL_WAIT				LP_CLKRST_HPCORE1_RESET_EN				LP_CLKRST_LP_WDT_HPCORE1_RESET_EN				LP_CLKRST_LP_WDT_HPCORE1_RESET_LENGTH				LP_CLKRST_HPCORE0_STAT_VECTOR_SEL				LP_CLKRST_HPCORE0_OCD_HALT_ON_RESET				LP_CLKRST_HPCORE0_SW_RESET				LP_CLKRST_HPCORE0_STALL_WAIT				LP_CLKRST_LP_WDT_HPCORE0_RESET_EN				LP_CLKRST_LP_WDT_HPCORE0_RESET_LENGTH				LP_CLKRST_HPCORE0_LOCKUP_RESET_EN			
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																								
1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	Reset																							

Reset

LP_CLKRST_HPCORE0_LOCKUP_RESET_EN Configures whether to enable Lockup to trigger HP CPU0 reset.

0: Disable

1: Enable

(R/W)

LP_CLKRST_LP_WDT_HPCORE0_RESET_LENGTH Configures the duration of the reset when RWDT and software trigger HP CPU0 reset.

Measurement unit: Clock cycles. (R/W)

LP_CLKRST_LP_WDT_HPCORE0_RESET_EN Configures whether to enable RWDT to trigger HP CPU0 reset.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HPCORE0_STALL_WAIT Configures the time HP CPU0 stalls when RWDT and software trigger HP CPU0 reset.

Measurement unit: Clock cycles. (R/W)

LP_CLKRST_HPCORE0_STALL_EN Configures the behavior when RWDT and software trigger HP CPU0 reset.

0: Immediately reset HP CPU0

1: Stall HP CPU0 first

(R/W)

LP_CLKRST_HPCORE0_SW_RESET Write 1 to trigger HP CPU0 software reset. (WT)

LP_CLKRST_HPCORE0_OCD_HALT_ON_RESET Configures whether HP CPU0 halts at the first instruction after released from reset.

0: No halt

1: Halt

(R/W)

Continued on the next page...

Register 10.63. LP_CLKRST_HPCPU_RESET_CTRL0_REG (0x0014)

Continued from the previous page...

LP_CLKRST_HPCORE0_STAT_VECTOR_SEL Configures the address from which the HP CPU0 starts.

0: From LP SPM RAM (0x50108000)

1: From HP SPM ROM (0x4FC00000)

(R/W)

LP_CLKRST_HPCORE1_LOCKUP_RESET_EN Configures whether to enable Lockup to trigger HP CPU1 reset.

0: Disable

1: Enable

(R/W)

LP_CLKRST_LP_WDT_HPCORE1_RESET_LENGTH Configures the duration of the reset when RWDT and software trigger HP CPU1 reset.

Measurement unit: Clock cycles. (R/W)

LP_CLKRST_LP_WDT_HPCORE1_RESET_EN Configures whether to enable RWDT to trigger HP CPU1 reset.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HPCORE1_STALL_WAIT Configures the time HP CPU1 stalls when RWDT and software trigger HP CPU1 reset.

Measurement unit: Clock cycles. (R/W)

LP_CLKRST_HPCORE1_STALL_EN Configures the behavior when RWDT and software trigger HP CPU1 reset.

0: Immediately reset HP CPU1

1: Stall HP CPU1 first

(R/W)

LP_CLKRST_HPCORE1_SW_RESET Write 1 to trigger HP CPU1 software reset. (WT)

LP_CLKRST_HPCORE1_OCD_HALT_ON_RESET Configures whether HP CPU1 halts at the first instruction after released from reset.

0: No halt

1: Halt

(R/W)

LP_CLKRST_HPCORE1_STAT_VECTOR_SEL Configures the address from which the HP CPU1 starts.

0: From LP SPM RAM (0x50108000)

1: From HP SPM ROM (0x4FC00000)

(R/W)

Register 10.66. LP_CLKRST_XTAL32K_REG (0x0030)

LP_CLKRST_DAC_XTAL32K			LP_CLKRST_DBUF_XTAL32K			LP_CLKRST_DGM_XTAL32K			LP_CLKRST_DRES_XTAL32K			(reserved)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
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LP_CLKRST_DRES_XTAL32K Configures the bias resistor of XTAL32K_CLK.
0: Turn off internal resistor, use external resistor
1-3: Internal bias resistor value
(R/W)

LP_CLKRST_DGM_XTAL32K Configures the gm of XTAL32K_CLK. (R/W)

LP_CLKRST_DBUF_XTAL32K Configures the buffer of XTAL32K_CLK.
0: Single-ended buffer
1: Differential buffer
(R/W)

LP_CLKRST_DAC_XTAL32K Configures the bias current DAC of XTAL32K_CLK. (R/W)

Register 10.67. LP_CLKRST_HP_CLK_CTRL_REG (0x0040)

(reserved) LP_CLKRST_HP_MPLL_500M_CLK_EN LP_CLKRST_HP_SPLL_480M_CLK_EN LP_CLKRST_HP_CPLL_400M_CLK_EN LP_CLKRST_HP_XTAL_40M_CLK_EN LP_CLKRST_HP_FOSC_20M_CLK_EN LP_CLKRST_HP_SDIO_PLLO_CLK_EN LP_CLKRST_HP_SDIO_PLL1_CLK_EN LP_CLKRST_HP_AUDIO_PLL2_CLK_EN (reserved) LP_CLKRST_HP_PLL_8M_CLK_EN LP_CLKRST_HP_SOSC_150K_CLK_EN LP_CLKRST_HP_XTAL_32K_CLK_EN LP_CLKRST_HP_PAD_EMAC_TXRX_CLK_EN LP_CLKRST_HP_PAD_EMAC_RX_CLK_EN LP_CLKRST_HP_PAD_I2S0_MCLK_EN LP_CLKRST_HP_PAD_I2S1_MCLK_EN LP_CLKRST_HP_PAD_I2S2_MCLK_EN LP_CLKRST_HP_PAD_UART0_SLP_CLK_EN LP_CLKRST_HP_PAD_UART1_SLP_CLK_EN LP_CLKRST_HP_PAD_UART2_SLP_CLK_EN LP_CLKRST_HP_PAD_UART3_SLP_CLK_EN LP_CLKRST_HP_PAD_UART4_SLP_CLK_EN LP_CLKRST_HP_PAD_PARLIO_RX_CLK_EN LP_CLKRST_HP_PAD_PARLIO_TX_CLK_EN LP_CLKRST_HP_ROOT_CLK_SRC_SEL																															
31	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	Reset	

LP_CLKRST_HP_ROOT_CLK_SRC_SEL Configures the clock source of ROOT_CLK.

- 0: XTAL_CLK
 - 1: CPLL_CLK (360 MHz)
 - 2: RC_FAST_CLK
 - 3: Invalid
- (R/W)

LP_CLKRST_HP_ROOT_CLK_EN Configures whether to enable ROOT_CLK.

- 0: Disable
 - 1: Enable
- (R/W)

LP_CLKRST_HP_PAD_PARLIO_TX_CLK_EN Configures whether to enable PAD_PARLIO_TX_CLK from IO pin.

- 0: Disable
 - 1: Enable
- (R/W)

LP_CLKRST_HP_PAD_PARLIO_RX_CLK_EN Configures whether to enable PAD_PARLIO_RX_CLK from IO pin.

- 0: Disable
 - 1: Enable
- (R/W)

LP_CLKRST_HP_PAD_UART4_SLP_CLK_EN Configures whether to enable PAD_UART4_SLP_CLK from IO pin.

- 0: Disable
 - 1: Enable
- (R/W)

LP_CLKRST_HP_PAD_UART3_SLP_CLK_EN Configures whether to enable PAD_UART3_SLP_CLK from IO pin.

- 0: Disable
 - 1: Enable
- (R/W)

Continued on the next page...

Register 10.67. LP_CLKRST_HP_CLK_CTRL_REG (0x0040)

Continued from the previous page...

LP_CLKRST_HP_PAD_UART2_SLP_CLK_EN Configures whether to enable PAD_UART2_SLP_CLK from IO pin.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_PAD_UART1_SLP_CLK_EN Configures whether to enable PAD_UART1_SLP_CLK from IO pin.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_PAD_UART0_SLP_CLK_EN Configures whether to enable PAD_UART0_SLP_CLK from IO pin.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_PAD_I2S2_MCLK_EN Configures whether to enable PAD_I2S2_MCLK from IO pin.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_PAD_I2S1_MCLK_EN Configures whether to enable PAD_I2S1_MCLK from IO pin.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_PAD_I2S0_MCLK_EN Configures whether to enable PAD_I2S0_MCLK from IO pin.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_PAD_ETH_TX_CLK_EN Configures whether to enable PAD_ETH_TX_CLK from IO pin.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.67. LP_CLKRST_HP_CLK_CTRL_REG (0x0040)

Continued from the previous page...

LP_CLKRST_HP_PAD_EMAC_RX_CLK_EN Configures whether to enable PAD_EMAC_TX_CLK from IO pin.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_PAD_EMAC_TXRX_CLK_EN Configures whether to enable PAD_EMAC_TXRX_CLK from IO pin.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_XTAL_32K_CLK_EN Configures whether to enable XTAL32K_CLK.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_SOSC_150K_CLK_EN Configures whether to enable RC_SLOW_CLK.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_PLL_8M_CLK_EN Configures whether to enable PLL_LP_CLK.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_AUDIO_PLL_CLK_EN Configures whether to enable APLL_CLK.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_SDIO_PLL2_CLK_EN Configures whether to enable SDIO_PLL2_CLK.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.67. LP_CLKRST_HP_CLK_CTRL_REG (0x0040)

Continued from the previous page...

LP_CLKRST_HP_SDIO_PLL1_CLK_EN Configures whether to enable SDIO_PLL1_CLK.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_SDIO_PLLO_CLK_EN Configures whether to enable SDIO_PLLO_CLK.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_FOSC_20M_CLK_EN Configures whether to enable RC_FAST_CLK.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_XTAL_40M_CLK_EN Configures whether to enable XTAL_CLK.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_CPLL_400M_CLK_EN Configures whether to enable CPLL_CLK.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_SPLL_480M_CLK_EN Configures whether to enable SPLL_CLK.

0: Disable

1: Enable

(R/W)

LP_CLKRST_HP_MPLL_500M_CLK_EN Configures whether to enable MPLL_CLK.

0: Disable

1: Enable

(R/W)

Register 10.68. LP_CLKRST_HP_USB_CLKRST_CTRL0_REG (0x0044)

(reserved)		LP_CLKRST_USB_I2M_DIV_NUM		LP_CLKRST_USB_25M_DIV_NUM		LP_CLKRST_USB_48M_DIV_NUM		LP_CLKRST_USB_DEVICE_OTG11_48M_CLK_EN		LP_CLKRST_USB_OTG11_48M_CLK_EN		LP_CLKRST_USB_OTG11_BK_SYS_CLK_EN		LP_CLKRST_USB_OTG11_SLEEP_MODE		LP_CLKRST_USB_OTG20_BK_SYS_CLK_EN		LP_CLKRST_USB_OTG20_SLEEP_MODE		
31	30	29	22	21	14	13	6	5	4	3	2	1	0							
0	0	39			19			9			1	1	1	0	1	0	Reset			

LP_CLKRST_USB_OTG20_SLEEP_MODE Not used. (R/W)

LP CLKRST USB OTG20 BK SYS CLK EN Not used. (R/W)

LP_CLKRST_USB_OTG11_SLEEP_MODE Not used. (R/W)

LP_CLKRST_USB_OTG11_BK_SYS_CLK_EN Not used. (R/W)

LP_CLKRST_USB_OTG11_48M_CLK_EN Configures whether to enable the Full-Speed USB 2.0 OTG PHY clock.
0: Disable
1: Enable
(R/W)

LP_CLKRST_USB_DEVICE_48M_CLK_EN Configures whether to enable the USB Serial/JTAG PHY clock.
0: Disable
1: Enable
(R/W)

LP_CLKRST_USB_48M_DIV_NUM Configures the division number from USB 480M to 25M. (R/W)

LP_CLKRST_USB_25M_DIV_NUM Configures the division number from USB 500M to 25M. (R/W)

LP_CLKRST_USB_12M_DIV_NUM Configures the division number from USB 480M to 12M. (R/W)

Register 10.69. LP_CLKRST_HP_USB_CLKRST_CTRL1_REG (0x0048)

LP_CLKRST_RST_EN_USB_OTG20_ADP Configures whether to reset High-Speed USB 2.0 OTG ADP.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_RST_EN_USB_OTG20_PHY Configures whether to reset High-Speed USB 2.0 OTG PHY.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_RST_EN_USB_OTG20 Configures whether to reset High-Speed USB 2.0 OTG.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_RST_EN_USB_OTG11 Configures whether to reset Full-Speed USB 2.0 OTG.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_RST_EN_USB_DEVICE Configures whether to reset USB Serial/JTAG.

0: Release from reset

1: Reset

(R/W)

Continued on the next page...

Register 10.69. LP_CLKRST_HP_USB_CLKRST_CTRL1_REG (0x0048)

Continued from the previous page...

LP_CLKRST_USB_OTG20_PHYREF_CLK_SRC_SEL Configures the clock source of High-Speed USB 2.0 OTG HS PHY.

- 0: PLL_F12M_CLK
 - 1: PLL_F25M_CLK
 - 2: PAD_USB_HSPHY_REFCLK
 - 3: Invalid
- (R/W)

LP_CLKRST_USB_OTG20_PHYREF_CLK_EN Configures whether to enable High-Speed USB 2.0 OTG HS PHY REFCLK.

- 0: Disable
 - 1: Enable
- (R/W)

LP_CLKRST_USB_OTG20_ULPI_CLK_EN Configures whether to enable High-Speed USB 2.0 OTG ULPI clock.

- 0: Disable
 - 1: Enable
- (R/W)

Register 10.70. LP_CLKRST_HP_SDMC_MAC_RST_CTRL_REG (0x004C)

Diagram of the LP_CLKRST_FORCE_NORST_EN_MAC register. The register is 32 bits wide, with bits 31, 30, 29, and 28 labeled as LP_CLKRST_FORCE_NORST_EN_MAC, LP_CLKRST_FORCE_NORST_EN_MAC, LP_CLKRST_FORCE_NORST_EN_MAC, and LP_CLKRST_FORCE_NORST_EN_MAC respectively. Bit 27 is labeled as (reserved). The register is reset to 0.

LP_CLKRST_RST_EN_SDMMC Configures whether to reset SDMMC.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_FORCE_NORST_SDMC Configures whether SDMMC can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

LP_CLKRST_RST_EN_EMAC Configures whether to reset EMAC.

0: Release from reset

1: Reset

(R/W)

LP_CLKRST_FORCE_NORST_ETH Configures whether ETH can be reset.

0: Can be reset

1: Force not to be reset

(R/W)

Register 10.71. LP_CLKRST_DATE_REG (0x03FC)

Diagram of the LP_CLKRST_CLK_EN register. The register is 32 bits wide. Bit 31 is labeled LP_CLKRST_CLK_EN. Bits 30 to 0 are labeled (reserved).

LP_CLKRST_CLK_EN Configures resister clock gating.

0: Support clock only when application writes registers

1: Force on clock gating for registers

(R/W)

10.5.3 LP Peripheral Clock and Reset (LPPERI) Registers

The addresses in this section are relative to LPPERI base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 10.72. LPPERI_CLK_EN_REG (0x0000)

[illegible]

LPPERI_CK_EN_RNG Configures whether to enable LP RNG clock.

0: Disable

1: Enable

(R/W)

LPperi_ck_en_lp_tsens Configures whether to enable LP temperature sensor clock.

0: Disable

1: Enable

(R/W)

LPPERI_CK_EN_LP_PMS Configures whether to enable LP PMS clock.

0: Disable

1: Enable

(R/W)

LPPERI_CK_EN_LP_EFUSE Configures whether to enable LP eFuse clock.

0: Disable

1: Enable

(R/W)

LPPERI_CK_EN_LP_IOMUX Configures whether to enable LP IO MUX clock.

0: Disable

1: Enable

(R/W)

LPPERI CK EN LP TOUCH Configures whether to enable LP touch sensor clock.

0: Disable

1: Enable

(R/W)

LPPERI_CK_EN_LP_SPI Configures whether to enable LP SPI clock.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 10.72. LPPERI_CLK_EN_REG (0x0000)

Continued from the previous page...

LPPERI_CK_EN_LP_ADC Configures whether to enable LP ADC clock.

0: Disable

1: Enable

(R/W)

LPPERI_CK_EN_LP_I2S_RX Configures whether to enable LP I2S RX clock.

0: Disable

1: Enable

(R/W)

LPPERI_CK_EN_LP_I2S Configures whether to enable LP I2S clock.

0: Disable

1: Enable

(R/W)

LPPERI_CK_EN_LP_I2CMST Configures whether to enable LP I2C master clock.

0: Disable

1: Enable

(R/W)

LPPERI_CK_EN_LP_I2C Configures whether to enable LP I2C clock.

0: Disable

1: Enable

(R/W)

LPPERI_CK_EN_LP_UART Configures whether to enable LP UART clock.

0: Disable

1: Enable

(R/W)

LPPERI_CK_EN_LP_INTR Configures whether to enable LP interrupt matrix clock.

0: Disable

1: Enable

(R/W)

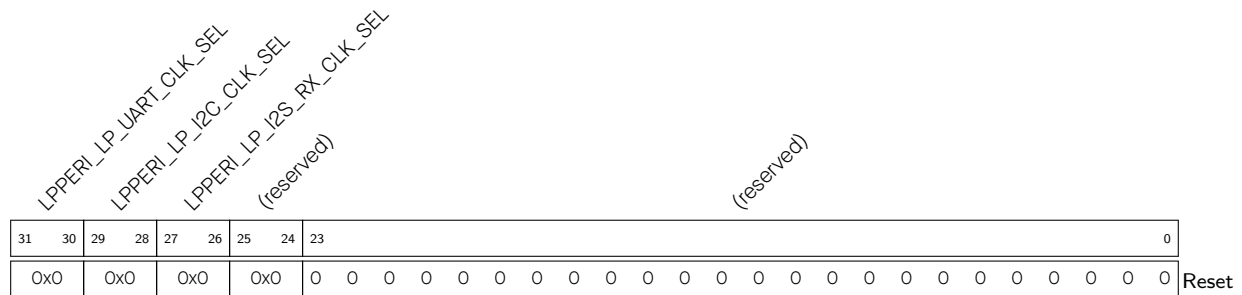
LPPERI_CK_EN_LP_CORE Configures whether to force on LP CPU clock.

0: Controlled by hardware

1: Force on, bypassing hardware control

(R/W)

Register 10.73. LPPERI_CORE_CLK_SEL_REG (0x0004)



LPPERI_LP_I2S_RX_CLK_SEL Configures LP I2S RX clock source.

- 0: LP_FAST_CLK
1: XTAL_D2_CLK
2: PLL_LP_CLK
3: Invalid
(R/W)

LPPERI_LP_I2C_CLK_SEL Configures LP I2C clock source.

- 0: LP_FAST_CLK
1: XTAL_D2_CLK
2: PLL_LP_CLK
3: Invalid
(R/W)

LPPERI_LP_UART_CLK_SEL Configures LP UART clock source.

- 0: RC_FAST_CLK
1: XTAL_D2_CLK
2: PLL_LP_CLK
3: Invalid
(R/W)

Register 10.74. LPPERI_RESET_EN_REG (0x0008)

LPPERI_RST_EN_LP_CORE LPPERI_RST_EN_LP_ROM LPPERI_RST_EN_LP_INTR LPPERI_RST_EN_LP_UART LPPERI_RST_EN_LP_I2C LPPERI_RST_EN_LP_I2CMST LPPERI_RST_EN_LP_I2S LPPERI_RST_EN_LP_ADC LPPERI_RST_EN_LP_SPI LPPERI_RST_EN_LP_TOUCH LPPERI_RST_EN_LP_IOMUX LPPERI_RST_EN_LP_EFUSE LPPERI_RST_EN_LP_PMS LPPERI_RST_EN_LP_TSENS (reserved)																																	
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17																		0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

LPPERI_RST_EN_LP_TSENS Configures whether to reset LP temperature sensor.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_PMS Configures whether to reset LP PMS.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_EFUSE Configures whether to reset LP eFuse.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_IOMUX Configures whether to reset LP IO MUX.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_TOUCH Configures whether to reset LP touch sensor.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_SPI Configures whether to reset LP SPI.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_ADC Configures whether to reset LP ADC.

0: Release from reset

1: Reset

(R/W)

Continued on the next page...

Register 10.74. LPPERI_RESET_EN_REG (0x0008)

Continued from the previous page...

LPPERI_RST_EN_LP_I2S Configures whether to reset LP I2S.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_I2CMST Configures whether to reset LP I2C master.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_I2C Configures whether to reset LP I2C.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_UART Configures whether to reset LP UART.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_INTR Configures whether to reset LP interrupt matrix.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_ROM Configures whether to reset LP ROM.

0: Release from reset

1: Reset

(R/W)

LPPERI_RST_EN_LP_CORE Writing 1 triggers LP CPU software reset. (WT)

UPPER_LPCORE_DBGM_UNAVAILABLE

(reserved)

Reset

(R/W)

Register 10.76. LPPERI_MEM_CTRL_REG (0x0028)

LPPERI_LP_UART_MEM_FORCE_PU																															LPPERI_LP_UART_WAKEUP_FLAG_CLR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
LPPERI_LP_UART_MEM_FORCE_PD																															LPPERI_LP_UART_WAKEUP_FLAG																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
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LPPERI_LP_UART_WAKEUP_FLAG_CLR Write 1 to clear LPPERI_LP_UART_WAKEUP_FLAG. (WT)

LPPERI_LP_UART_WAKEUP_FLAG Represents whether an LP UART wakeup has occurred.

0: No LP UART wakeup occurred

1: LP UART wakeup occurred

(R/WTC/SS)

LPPERI_LP_UART_WAKEUP_EN Configures whether to enable the LP UART wakeup function.

0: Enable

1: Disable

(R/W)

LPPERI_LP_UART_MEM_FORCE_PD Configures whether to force the LP UART memory to power down.

0: Do not force power down

1: Force power down

(R/W)

LPPERI_LP_UART_MEM_FORCE_PU Configures whether to force the LP UART memory to power up.

0: Do not force power up

1: Force power up

(R/W)

Register 10.77. LPPERI_ADC_CTRL_REG (0x002C)

LPPERI_LPADC_SAR1_DIV_NUM								LPPERI_LPADC_SAR2_DIV_NUM								LPPERI_LPADC_FUNC_DIV_NUM								LPPERI_SAR1_CLK_FORCE_ON LPPERI_SAR2_CLK_FORCE_ON (reserved)													
31								24	23								16	15								8	7	6	5								0
0x4								0x4								0x4								0	0	0	0	0	0	0	0	0	0	Reset			

LPPERI_SAR2_CLK_FORCE_ON Configures whether to force on the LP ADC SAR1 clock.

0: Controlled by hardware

1: Force on, bypassing hardware control

(R/W)

LPPERI_SAR1_CLK_FORCE_ON Configures whether to force on the LP ADC SAR2 clock.

0: Controlled by hardware

1: Force on, bypassing hardware control

(R/W)

LPPERI_LPADC_FUNC_DIV_NUM Configures the LP ADC function clock divisor. (R/W)

LPPERI_LPADC_SAR2_DIV_NUM Configures the LP ADC SAR1 clock divisor.(R/W)

LPPERI_LPADC_SAR1_DIV_NUM Configures the LP ADC SAR2 clock divisor.(R/W)

Register 10.78. LPPERI_LP_I2S_RXCLK_DIV_NUM_REG (0x0030)

LPPER1_LP_I2S_RX_CLKM_DIV_NUM																								(reserved)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
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LPPERI_LP_I2S_RX_CLKM_DIV_NUM Configures the integer part of the LP I2S RX clock divisor.

(R/W)

Register 10.79. LPPERI_LP_I2S_RXCLK_DIV_XYZ_REG (0x0034)

LPPERI_LP_I2S_RX_CLKM_DIV_X									
LPPERI_LP_I2S_RX_CLKM_DIV_Y									
LPPERI_LP_I2S_RX_CLKM_DIV_Z									
LPPERI_LP_I2S_RX_CLKM_DIV_YN1 (reserved)									
31	23	22	14	13	5	4	3	0	
0x0		0x1		0x0		0	0	0	0

Reset

LPPERI_LP_I2S_RX_CLKM_DIV_YN1 Configures the yn1 coefficient of the LP I2S RX clock divisor. (R/W)

LPPERI_LP_I2S_RX_CLKM_DIV_Z Configures the z coefficient of the LP I2S RX clock divisor. (R/W)

LPPERI_LP_I2S_RX_CLKM_DIV_Y Configures the y coefficient of the LP I2S RX clock divisor. (R/W)

LPPERI_LP_I2S_RX_CLKM_DIV_X Configures the x coefficient of the LP I2S RX clock divisor. (R/W)

Register 10.80. LPPERI_DATE_REG (0x03FC)

LPPERI_CLK_EN																													
(reserved)																													
31	30																												0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

LPPERI_CLK_EN Configures register clock gating.
 0: Support clock only when application writes registers
 1: Force on clock gating for registers
 (R/W)